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Projection Pursuit mit Distanzkorrelation und Distanzvarianz

Explizite Riemannsche Gradienten auf der Einheitssphäre und der Stiefel-Mannigfaltigkeit

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Projection Pursuit with Distance Correlation and Distance Variance

Closed-Form Riemannian Gradients on the Unit Sphere and the Stiefel Manifold

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Maximilian Scheiblauer

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Wien, 27. August 2026

Maximilian Scheiblauer

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Kurzfassung

Lineare Dimensionsreduktion wird zumeist von Kriterien zweiter Ordnung gesteuert: die Methode der kleinsten Quadrate liest eine Kovarianz zwischen Zielgröße und Prädiktoren, die Hauptkomponentenanalyse eine unter den Prädiktoren. Ist die Abhängigkeit nichtlinear und symmetrisch, oder liegt die interessante Struktur einer Projektion in ihrer Form statt in ihrer Varianz, so können solche Kriterien die Richtung vollständig verfehlen.

Distanzbasierte Abhängigkeitsmaße besitzen diesen blinden Fleck nicht. Die vorliegende Arbeit behandelt das numerische Problem, zwei von ihnen über Projektionen zu maximieren: die quadrierte Distanzkorrelation im überwachten Fall und die Distanzvarianz, ein Streuungsmaß, im unüberwachten Fall. Keine der beiden Maximierungen besitzt eine geschlossene Lösung, und beide Zielfunktionen sind nicht konkav.

Der methodische Beitrag ist ein Riemannscher Gradient in geschlossener Form für beide Indizes auf der Einheitssphäre und auf der Stiefel-Mannigfaltigkeit, eine Ableitung, welche die bestehende Literatur zur Distanzkovarianz vermeidet. Die Suche wird hierdurch zu einer Optimierung erster Ordnung auf einer Mannigfaltigkeit, deren Schritt weniger kostet als eine finite Differenz. Die Richtigkeit des Gradienten wird an zentralen Differenzen und an zwei ableitungsfreien Verfahren geprüft.

Es folgen eine überwachte Simulation, eine unüberwachte Simulation und eine Fallstudie zu Überrenditen von Anleihen. Zwei Grenzen werden festgehalten und nicht behoben. Die Zielfunktion addiert Einzelbewertungen der Richtungen und ist hierdurch blind für Abhängigkeit zwischen ihnen. Auch zu nennen ist, dass keine lineare Deflation nichtlineares Signal entfernt, sodass die sequentielle Extraktion jenseits der ersten Richtung keine unbedenkliche Variante besitzt.

Abstract

Linear dimension reduction is most often steered by criteria of second order: ordinary least squares reads a covariance between the response and the predictors, principal component analysis one among the predictors. Where the dependence is nonlinear and symmetric, or where the interesting structure of a projection lies in its shape rather than in its variance, such criteria can miss the direction entirely.

Distance-based measures of dependence do not carry this blind spot. This thesis treats the numerical problem of maximising two of them over projections: the squared distance correlation in the supervised case, and the distance variance, a dispersion index, in the unsupervised one. Neither maximisation has a closed-form solution, and both objectives are non-concave.

The methodological contribution is a closed-form Riemannian gradient for both indices on the unit sphere and on the Stiefel manifold, a derivative the existing distance-covariance literature avoids. Hereby the search becomes a first-order optimisation on a manifold, at a cost per step below that of a finite difference. The gradient is examined against central differences and against two derivative-free solvers.

A supervised simulation, an unsupervised simulation and a case study on excess bond returns follow. Two limitations are recorded rather than resolved. The objective adds per-direction scores and is hereby blind to dependence between the directions it selects. Also to mention is, that no linear deflation removes nonlinear signal, so sequential extraction has no sound variant beyond the first direction.

Contents

CHAPTER 1

Introduction

The methods Ordinary Least Squares (OLS) and principal component analysis (PCA) belong to the most frequently applied procedures for setting variables in relation to one another and for reducing the dimension of a data set. Both of them search for a direction in the predictor space, and both determine this direction on the basis of the second moments of the data. Least squares uses here the covariance between the predictors and the target variable, whereas the principal component analysis uses exclusively the covariance of the predictors. The second moments are a cheap summary of a sample, and for many problems they are sufficient. Nevertheless, the two following examples show cases in which this is not sufficient, and they build the point of departure of the present work.

1.1 Two examples

The first example is a regression problem with a single predictor. Let X be uniformly distributed on $[-1, 1]$ and $Y = X^2$, whereby no noise is added, so that the response variable is determined completely through the predictor. If one fits this sample by means of the method of least squares, there results the nearly horizontal line in the left part of Figure 1.1. The estimated slope lies close to zero and the sample correlation amounts to -0.03. The fit is not led astray through noise, since no noise is present, and it also does not improve with a larger quantity of data. If one repeats the experiment with $n = 10{,}000$, the correlation lies closer to zero than at $n = 500$ rather than further from it; the decline is not monotone in n, but no sample size lifts the correlation away from zero. A larger sample makes the linear summary thus more confident regarding a relationship which it does not recognise: between $n = 100$ and $n = 10{,}000$ the absolute correlation falls from 0.15 to 0.01. What is missing here is not a better fit, but a criterion which registers a dependence of a form which the covariance is not able to see. Section 1.2 names the criterion used in this work and returns to this sample.

The reason for this failure is visible in the criterion itself. Let $X \in \mathbb{R}^p$ be a centred predictor vector and $Y \in \mathbb{R}$ a centred response. Least squares seeks the coefficient vector β whose linear combination $X^\top \beta$ predicts Y with the smallest mean squared error,

$$\beta_{\text{OLS}} = \arg\min_{\beta} \ \mathbb{E}\big[(Y - X^\top \beta)^2\big], \tag{1.1}$$

which in the sample is solved in closed form by the normal equations, $\beta_{\text{OLS}} = \text{Cov}(X)^{-1} \text{Cov}(X, Y)$. Expanding the square and using that X and Y are centred gives $\mathbb{E}\big[(Y - X^\top \beta)^2\big] = \text{Var}(Y) - 2\,\beta^\top \text{Cov}(X, Y) + \beta^\top \text{Cov}(X)\,\beta$. The first summand does not involve β, so the minimisation is identical with

$$\max_{\beta} \ \big[\, 2\,\beta^\top \text{Cov}(X, Y) - \beta^\top \text{Cov}(X)\,\beta \,\big]. \tag{1.2}$$

Written in this form, the only feature of the joint distribution which the method is able to exploit is the second-order quantity $\text{Cov}(X, Y)$. That is the correct criterion when the dependence is linear, and it makes a cheap and natural baseline; under a correctly specified linear model with uncorrelated errors of equal variance, the Gauss–Markov theorem states that β_{OLS} possesses the smallest variance among all linear unbiased estimators (Rao, 1973, Ch. 4). Nevertheless, it is at the same time the limitation. Wherever the relationship is a *symmetric* nonlinearity the covariance vanishes although Y is fully determined by X, and (1.2) then contains no usable signal at all. For $Y = X^2$ with X symmetric about the origin, $\text{Cov}(X, Y) = \mathbb{E}[X^3] = 0$, since the right half of the parabola and the left half contribute equal and opposite amounts. Therefore the remedy pursued in this work does not consist in fitting a better curve. It consists in replacing the *measure* of association which the search optimises.

The second example contains no response variable at all. Two latent variables with the same variance are generated, one Gaussian and one bimodal, and afterwards rotated by 35° into the observed plane. Since both components exhibit the same variance, the covariance matrix of the observed data is approximately a multiple of the identity matrix, with two eigenvalues which differ by 0.051, so that with respect to the dispersion no direction in the plane stands out. Therefore, for the principal component analysis there is no eigenvalue gap available on which it could fall back. The direction determined by it is fixed through the respective sampling fluctuation, which resolves the nearly balanced ratio of the eigenvalues, and it lies 59° away from the bimodal axis. In the right part of Figure 1.1, the bimodal axis is visible, and along this direction the data separate into two groups.

The variance is not the only summary of a projected sample which could be maximised here. A dispersion measure which reacts to the shape of a distribution and not only to its scale would keep the role which the variance plays in the principal component analysis, and would at the same time respond to features such as the two groups above. Exactly this is proposed by Nordhausen and Radojičić (2026), namely a distance-based dispersion measure which is used as a projection index in place of the variance.

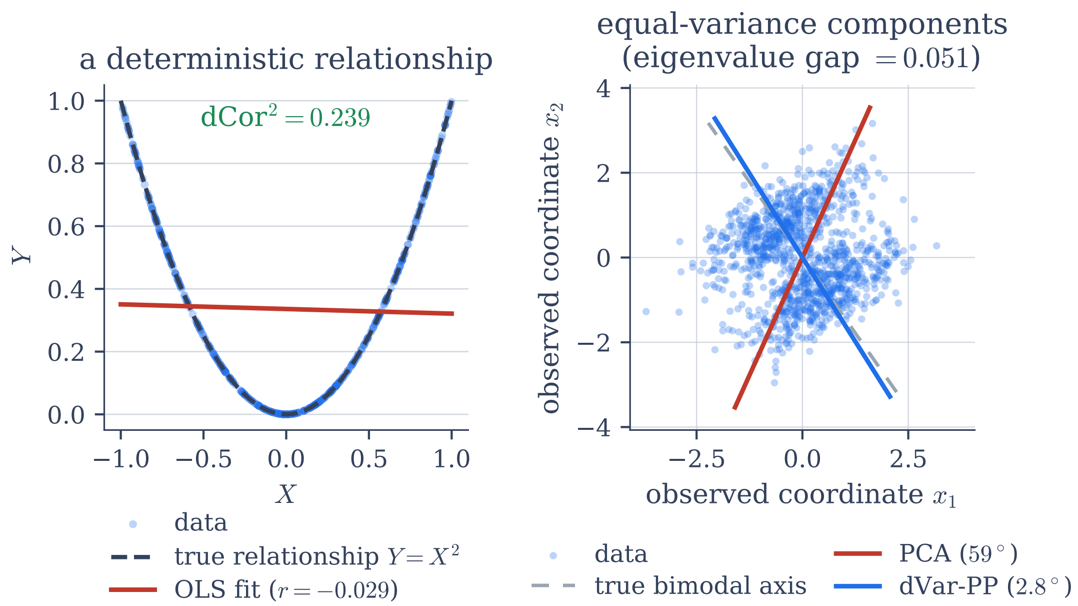

Figure 1.1: The two introductory examples. *Left:* $Y = X^2$ with $X \sim \text{Uniform}(-1, 1)$ and no noise; the least-squares fit (red) is nearly flat although the response is determined exactly by the predictor. *Right:* two latent components of equal variance, one Gaussian and one bimodal and rotated. The eigenvalues differ by 0.051, and the principal direction (red) misses the bimodal axis (grey dashed) by 59°. The distance-based quantities which are annotated in the two panels, the squared distance correlation on the left and the distance-variance direction (blue) on the right, are defined in Section 1.2 and discussed there.

Neither of the two examples represents a failure of the estimation or of the implementation. In both cases the quantity which is optimised by the respective method simply does not capture the structure that is present. Both examples show, that a criterion which is sensitive to a larger part of the joint distribution carries usable information in situations in which a criterion of second order carries none. By construction, both examples are extreme. What such a criterion costs, and whether it still pays outside such extremes, is what this work examines.

1.2 Distance correlation and the cost of maximising it

The criteria used here are the distance correlation and the distance variance (Székely et al., 2007; Székely and Rizzo, 2009). The ordinary covariance asks whether Y tends to lie above its mean when X lies above its mean, and it averages these signed deviations. The distance-based measures compare instead *pairs* of observations. They ask whether two observations which are far distant from one another in X tend also to be far distant from one another in Y. Since they work with distances instead of with signed deviations from a mean, they react to patterns which remain unconsidered under a signed averaging. On the parabola sample of Section 1.1, where the absolute correlation falls to 0.01 as the sample grows, the distance correlation takes on the squared, bias-corrected scale defined in Chapter 3, on which it is reported throughout the whole work, the value $\mathrm{dCor}^2 = 0.24$, and it varies between 0.238 and 0.245 from $n = 100$ to $n = 10{,}000$.

The distance correlation possesses a property which the ordinary correlation lacks. It is zero exactly then when the two variables are independent. From a correlation of zero one can conclude only, that no linear relationship is present. In this the distance correlation does not stand alone. The rank statistic of Hoeffding (1948) and the kernel dependence measures of Fukumizu et al. (2009) characterise independence in the same manner, and Sejdinovic et al. (2013) show that the distance-based and the kernel-based family are two descriptions of one construction. What recommends the distance correlation for the present purpose is therefore not the property itself, which it shares with these measures, but the circumstance that it is a smooth function of the sample which can be differentiated in closed form with respect to a projection direction, as Chapter 3 sets out.

The distance variance is the same construction, applied to a single variable against itself. Like the ordinary variance, it grows with the dispersion of a distribution; once this scale is fixed to unit variance, however, it reacts to the shape of the distribution. It is here important to name the direction of this reaction, because it is more narrowly framed than non-Gaussianity in general. It rises when the mass of the distribution moves away from the centre, and it falls when the mass concentrates there; it thus distinguishes a Gaussian distribution from the bimodal and multimodal alternatives for which the projection pursuit usually searches, while heavy distribution tails shift it in the opposite direction. Section 2.1 sets out the resulting ordering of distributions, which the unsupervised method inherits. On the rotated sample above, which the principal component analysis misses by 59°, the index reconstructs the bimodal axis with a deviation of 2.8°.

That non-Gaussianity is the customary target of a projection index rests upon two grounds. The first is that it marks what is rare. Under general conditions, most low-dimensional projections of a high-dimensional point cloud are approximately normal (Diaconis and Freedman, 1984), so that a projection which is visibly not normal reflects a genuine structure and not the generic behaviour which any arbitrary direction exhibits. The second ground is the identifiability. In the linear latent-variable model of the independent component analysis the mixing is recoverable precisely because at most one latent component may be Gaussian (Hyvärinen and Oja, 2000), so that a search for non-Gaussian projections is a search for the latent components themselves. The index of the present work stands beside this tradition rather than within it. It is a dispersion measure, it orders the distributions according to how far their mass lies from the centre, and the Gaussian distribution falls in the middle of this ordering and not at one of its ends.

For neither of the two maximisations does a closed-form solution exist. Least squares reduces to the normal equations, and principal component analysis to an eigenvalue decomposition. By contrast, in the present case the direction must be searched for. The search space is the set of the unit-length directions in the p-dimensional space, and this set is described by $p-1$ free angles. Evaluating the criterion on a grid is not a practicable option, already for small p. For $p = 20$, a grid with ten points per angle demands evaluations in the order of magnitude of 10^{19}, and ten points per angle amount to a spacing of 18° between neighbouring grid directions, 36° for the single angle which runs over the full circle. Therefore even such an effort would yield no more than a coarse solution. The single evaluations are not cheap either. Each one of them contains a sum over all pairs of observations and therefore grows quadratically with the sample size. When a derivative is approximated through central differences, two evaluations per coordinate become necessary, through which this effort is multiplied by $2p$. In addition, the criterion is not concave, and it is estimated from a finite sample. For this reason, a single local search is not sufficient and several starting points are needed.

The emphasis of the present work lies upon these algorithmic and numerical questions, and not upon the theory of the measures themselves; their statistical properties are established in the literature and are assumed here as given. Both criteria are differentiated in closed form, and the costs of this lie in the same order of magnitude as a single evaluation of the criterion.

1.3 Scope of the study

The work implements an optimiser for both criteria and investigates afterwards its behaviour.[1] The investigation is conducted on simulated data, where the true direction is known and the recovery of this direction can be measured directly, as well as on a published benchmark for real data, where this is not possible. The contributions are the following.

- A closed-form gradient for both criteria on the sphere and on the Stiefel manifold.
- A simulation study of the supervised method, which shows under which conditions it recovers the true direction and where difficulties arise at the extraction of further directions.
- A systematic study of the distance variance as a projection index, in comparison with the principal component analysis and FastICA. The study separates the scale contribution of the index from its shape contribution and names the settings in which the index is of advantage, as well as those in which it is not. The advantage which it establishes is one over the variance-based criteria; against the classical shape-based ones the index performs similarly. That a distance-based dispersion measure is used as the projection index in place of the variance is the proposal of Nordhausen and Radojičić (2026), made in their discussion of the robust distance

[1] Code and data to reproduce every result in this thesis are available at `https://github.com/maxscheiblauer/Dcor-SDR-Dvar-PP`.

covariance (Leyder et al., 2026); the present work implements this proposal and evaluates it, on simulated data in Chapter 5 and on real data in Chapter 6.

- An evaluation of both methods on a published benchmark for excess returns of bonds.

Chapter 2 introduces the distance-based measures, the two lines of research from which the methods descend, the classical baselines against which they are compared, and those elements of manifold optimisation which are used later. In Chapter 3 the shared machinery is described: the derivation of the gradient, the optimiser, the strategies for the extraction of more than one direction, a robust variant of the index, and the computational limits of the whole construction. The supervised method is treated in Chapter 4. This chapter begins with the recovery of a single direction and treats then the difficulties which appear as soon as further directions are asked for. In the same manner the unsupervised method is treated in Chapter 5. Included there are the factor-model setting, in which the principal component analysis is optimal by construction, the setting of equal variance, in which the distance-based index possesses an advantage, and a sweep over outlier contamination, in which the index deteriorates more slowly than the criteria built on squared deviations. In Chapter 6 both methods are applied to the bond-return benchmark. At last, Chapter 7 discusses the simulated results and the real-data results together, states the limitations and the questions which remain open, and concludes.

CHAPTER 2

Background

This chapter presents the tools upon which the two proposed methods build. It treats only that material which is used in the later chapters and is for this reason no overview of dimension reduction. The presentation is constructed around the distance-based dependence (Section 2.1). The supervised method stands in the tradition of the sufficient dimension reduction (Section 2.2), while the unsupervised method draws on the projection pursuit and on the independent component analysis (Section 2.3). Both are assessed against the classical baselines of second order (Section 2.4). Also to mention are the robustness considerations (Section 2.5) and the framework of the manifold optimisation (Section 2.6), upon which the algorithmic development in Chapter 3 is dependent.

2.1 Distance covariance, correlation, and variance

The classical measure for the connection between two random vectors is the Pearson covariance, which captures only *linear* co-movement and can therefore be exactly zero for variables which are strongly, but nonlinearly, dependent. The introductory example makes this concrete: for $Y = X^2$ with X symmetric around the origin it holds that $\mathrm{Cov}(X, Y) = \mathbb{E}[X^3] = 0$, since X^3 is an odd function of a symmetric variable, although Y is a deterministic function of X (Section 1.1). The distance covariance, introduced by Székely et al. (2007), removes this blind spot.

Let us assume that we observe a sample $(U_1, V_1), \ldots, (U_n, V_n)$ with $U_i \in \mathbb{R}^{d_U}$ and $V_i \in \mathbb{R}^{d_V}$. For every pair of observations i, j one forms the two distances

$$a_{ij} = \|U_i - U_j\|, \qquad b_{ij} = \|V_i - V_j\|, \tag{2.1}$$

that is, the distance between the two points in the U-space and the V-space.

The question which the distance covariance poses is whether pairs which lie far apart in the U-space also tend to lie far apart in the V-space. A naive correlation of the a_{ij}

and b_{ij} does not answer this. Since raw distances are non-negative and carry information at the marginal level, the comparison is distorted. Therefore each distance matrix is subjected to *double centring*, in that from every entry the mean $\bar{a}_i$ of its row and the mean $\bar{a}_j$ of its column are subtracted and the overall mean $\bar{a}$ is added again,

$$A_{ij} = a_{ij} - \bar{a}_i - \bar{a}_j + \bar{a}, \qquad B_{ij} = b_{ij} - \bar{b}_i - \bar{b}_j + \bar{b}. \tag{2.2}$$

The (squared) empirical distance covariance is then simply the mean product of these centred distances,

$$\widehat{\mathrm{dCov}}^2(U, V) = \frac{1}{n^2} \sum_{i,j=1}^{n} A_{ij}\, B_{ij}. \tag{2.3}$$

If one reads (2.3) literally, the expression is large and positive when large centred U-distances coincide with large centred V-distances, that is to say, when the geometry of the two point clouds moves jointly. It averages itself to zero when the two distance patterns are independent of one another. Nothing in this construction gives preference to linear structure. Therefore it reacts to dependence of any form whatsoever.

The distance variance is the same quantity, computed against itself,

$$\widehat{\mathrm{dVar}}^2(U) = \widehat{\mathrm{dCov}}^2(U, U) = \frac{1}{n^2} \sum_{i,j=1}^{n} A_{ij}^2, \tag{2.4}$$

the mean squared centred distance, a measure for how variable the pairwise distances within U are. Both quantities are defined here in squared form, and dCov and dVar without an exponent denote their non-negative roots. The root of the distance variance is called the *distance standard deviation* by Edelmann et al. (2020), a name that fits a quantity which scales like a standard deviation. This work uses dVar for both and distinguishes them by the exponent. If one normalises the distance covariance with the two distance variances, one obtains the distance correlation

$$\mathrm{dCor}(U, V) = \frac{\mathrm{dCov}(U, V)}{\sqrt{\mathrm{dVar}(U)\ \mathrm{dVar}(V)}} \in [0, 1], \tag{2.5}$$

a number which takes over the role of an absolute correlation coefficient, however for arbitrary and not only linear dependence. Other than the Pearson correlation it possesses no sign. Every quantity in (2.5) is a mean of non-negative terms, so that the range of values is $[0, 1]$ instead of $[-1, 1]$. Also the value 0 carries a meaning which does not belong to a Pearson correlation of 0, as (2.8) below confirms. Sample values of the supervised index are given throughout this work on the squared, bias-corrected scale dCor^2, which is laid out in Section 3.2 and which is the quantity that the supervised optimiser maximises. Since the squaring is monotone on $[0, 1]$, no ordering is changed. The unsupervised index carries the opposite convention, and its values, those of Table 2.1 included, are the unsquared dVar of the doubly centred estimator, for the reason set out below.

The original characterisation. Székely et al. (2007) give a more abstract, but equivalent population definition. They define the distance covariance by way of the characteristic functions φ_U, φ_V and the joint one $\varphi_{U,V}$ as a weighted L^2-distance between the joint characteristic function and the product of the marginal functions,

$$\mathrm{dCov}^2(U,V) = \int \left|\varphi_{U,V}(t,s) - \varphi_U(t)\,\varphi_V(s)\right|^2 w(t,s)\,\mathrm{d}t\,\mathrm{d}s, \tag{2.6}$$

and $\mathrm{dVar}^2(U) = \mathrm{dCov}^2(U,U)$. Of importance is the weight w, which is not integrable at the origin. Precisely this leads to the agreement of both definitions, since it is, up to a constant, the only choice for which the integral is finite for all distributions with first moment and which remains unchanged under rotations and scalings of the data. That finiteness is the one distributional requirement of the construction and it holds throughout this work: dCov, and with it dVar and dCor, is defined for $\mathbb{E}\|U\| < \infty$ and $\mathbb{E}\|V\| < \infty$ and for no wider class (Székely et al., 2007). No second moment is asked for, and no density, no continuity and no parametric form at all; a marginal without a finite mean, a Cauchy among them, is outside the definition. The definition is worth bearing in mind, since it makes transparent, that the distance covariance vanishes exactly then when independence is present.

The integral possesses an equivalent representation in ordinary moments of pairwise differences, entirely without a characteristic function. If one writes (U', V') and (U'', V'') for independent copies of (U, V), then it holds that

$$\mathrm{dCov}^2(U,V) = \mathbb{E}\|U-U'\|\,\|V-V'\| + \mathbb{E}\|U-U'\|\;\mathbb{E}\|V-V'\| - 2\,\mathbb{E}\|U-U'\|\,\|V-V''\|, \tag{2.7}$$

an identity of Székely et al. (2007), discussed in these terms by Raymaekers and Rousseeuw (2025). Both sides are population quantities. Since one obtains exactly the doubly centred average, if every expectation is replaced through the mean over the observed pairs and the terms are collected together, the sample formula (2.3) is the plug-in estimator of the right-hand side and not perhaps a second form of it. Therefore the quantity is estimable, without ever evaluating a characteristic function.

Regarding the replacement of the Pearson correlation in this work, it is the characterisation of independence which makes (2.5) the right one. The integral (2.6) vanishes exactly then when the joint characteristic function factorises into the product of the marginal functions. Therefore it holds that

$$\mathrm{dCor}(U,V) = 0 \quad \Longleftrightarrow \quad U \perp\!\!\!\perp V. \tag{2.8}$$

Of the Pearson correlation only the weaker implication is guaranteed, that independence enforces a correlation of zero, not its converse. The equivalence in (2.8) is exactly that which the supervised method needs in order to recognise the structure which escapes the ordinary least-squares estimation.

Distance variance and shape. The diagonal case, the distance variance $\mathrm{dVar}^2(U) = \mathrm{dCov}^2(U,U)$ of a single univariate projection, is the index which drives the unsupervised method. Through the *Brownian distance covariance*, Székely and Rizzo (2009) give it a clear interpretation. Take two independent standard Brownian motions W and W' on the line, drawn independently of the data. With them the observations are relabelled, the variable U through the centred value $W(U) - \mathbb{E}[W(U) \mid W]$ of the first path at the point U and V through the corresponding value of the second path. For each draw of the pair of paths, an ordinary covariance between the two relabelled variables is obtained, and Székely and Rizzo (2009) show, that the average of that covariance over the randomness of the paths is exactly $\mathrm{dCov}^2(U,V)$. Of importance is what a Brownian path does to a variable. Its increments over disjoint intervals are independent and its squared increment over an interval has expectation equal to the length of that interval, $\mathbb{E}\big[(W(u) - W(u'))^2\big] = |u - u'|$. Hereby $W(U)$ registers the whole arrangement of the mass of U along the line and not its first two moments alone, and the construction sees more than second-order structure. If W is replaced by the identity map, the ordinary covariance is obtained again.

On the diagonal this is what ties the index to the spread of a single distribution. There $\mathrm{dVar}^2(U)$ is the averaged variance of the relabelled variable, and since Brownian variance accumulates in proportion to the separation $|u - u'|$ of the two arguments, that quantity is large when the values of U typically lie far from one another and small when they cluster. At first the index measures the dispersion, before it measures anything else, exactly as a standard deviation does, since the distance variance is scale-equivariant, $\mathrm{dVar}(aU) = |a|\,\mathrm{dVar}(U)$, and a doubling of a variable doubles its distance variance. Two distributions of equal variance can differ nevertheless in how far their values typically lie from one another, so that the index continues to move once the scale is fixed. Precisely this continued change at held dispersion is what the unsupervised method exploits. The ordering which is thereby produced is worth being formulated precisely, since one easily assumes the wrong one. The index rises when the mass of the distribution moves away from its centre, and it falls when the mass concentrates itself in one peak. The values are listed in Table 2.1. A two-point variable, whose entire mass lies at the extremes, attains the maximum, and the multimodal mixtures and the uniform case follow. The two-component mixture lies at 0.754 ahead of the three- and four-component ones at 0.745 and 0.737, since splitting the same total mass into more components moves some of it back towards the centre. Below the Gaussian distribution lie the skewed as well as the peaked, heavy-tailed cases.

Separating the components of one mixture makes the same point along a single axis. Here the two component means are moved apart, while the sample is standardised back to unit variance after every draw. Hereby the index rises monotonically, from 0.631 when the means coincide, through 0.666 at two within-component standard deviations and 0.754 at four, to 0.905 at twelve, approaching the two-point limit of 1.000 as the two components become well separated points. Since a two-component mixture with coincident means and equal component variances is a Gaussian distribution, that first value estimates the same population quantity as the Gaussian row of Table 2.1, and the gap between 0.631 and 0.635 is sampling noise, the spread of the index across the ten draws being 0.003 in either case. Only how far the mass sits from the centre changes along that sweep, while the variance and the number of modes stay as they are.

Regarding the measures for non-Gaussianity customarily used in this literature (Section 2.3), these assess a Gaussian distribution as the lowest and rise in every direction away from it. The distance variance places the Gaussian distribution thus in the *middle* of its ordering, which makes the latter the narrower one. The maximisation of the distance variance searches for directions whose projected mass lies away from the centre, and Chapter 5 works out what this ordering achieves and what it does not achieve.

The classical summary that moves along the same axis is the kurtosis, which at fixed variance is read as peakedness, small for a distribution whose mass sits at the extremes and large for one concentrated in a single peak with heavy tails. The distance variance runs the other way along that axis. How far the correspondence holds is shown by the excess kurtosis reported beside it in Table 2.1. Down to the Gaussian the two orderings are exact mirror images, since the six distributions from the two-point variable to the Gaussian appear in the same sequence whether they are sorted by decreasing distance variance or by increasing excess kurtosis. Below the Gaussian they part company. The distance variance ranks the centred χ^2_3 above the centred exponential and both above the Laplace, whereas the excess kurtosis places the Laplace at 2.79 below both of them. Furthermore it separates the t_3 and the lognormal by an amount that its own sampling variation swamps, and these are the two rows without a usable fourth moment. The correspondence is therefore not an identity, and on heavy tails the distance variance is the better behaved of the two, since it is built from first moments of pairwise distances and needs no fourth moment to exist. For the purpose of naming what the index reacts to, it behaves at fixed variance like a measure of anti-peakedness. Among the light-tailed shapes the two statistics agree on the ordering exactly.

Estimation and computation. The sample formulas (2.3) and (2.4) are the basic estimators. Therefore the matrix of the pairwise distances is formed, is doubly centred, and products of the centred entries are averaged. This double centring (2.2) is the original convention, and it carries a bias of finite samples. The cause lies in the diagonal. A distance matrix possesses $a_{ii} = 0$ on its diagonal, but the row, column and overall means which are subtracted in (2.2) average over all n entries, self-distances included. The centring therefore removes somewhat less than the mean distance level which it should remove, and a positive remainder of the order $1/n$ survives on average. Under independence the consequence is exact. The population value $\mathrm{dCov}^2(U, V)$ is zero, but the doubly centred estimator (2.3) is strictly positive in every finite sample. The bias vanishes in the limit, is however not negligible at the sample sizes used here, n in the order of magnitude of some hundreds.

The *U-centring* of Székely and Rizzo (2014) removes the bias exactly. It divides the row and column sums through $n - 2$ instead of through n, the overall sum through

Table 2.1: Distance variance of unit-variance distributions, averaged over ten independent samples of size $n = 4000$, with the sample excess kurtosis of the same draws beside it. At fixed variance the index orders distributions by how far their mass lies from the centre. The Gaussian sits in the middle of that ordering rather than at one end, which is why the index is not a measure of non-Gaussianity. The three mixtures place their components four within-component standard deviations apart. The two entries marked † have no finite fourth moment or a very heavy one, so their sample excess kurtosis does not settle: across the ten draws it varies by ± 309 for t_3 and ± 33 for the lognormal, against below 0.3 in every other row.

Distribution (unit variance)	dVar	excess kurtosis
two-point ± 1	1.000	-2.00
balanced bimodal mixture	0.754	-1.28
trimodal mixture	0.745	-1.26
four-component mixture	0.737	-1.23
uniform	0.729	-1.19
Gaussian $\mathcal{N}(0,1)$	0.635	-0.04
centred χ^2_3	0.593	4.22
centred exponential	0.575	6.76
Laplace	0.541	2.79
Student t_3	0.465	$129^\dagger$
standardised lognormal	0.431	$57.8^\dagger$

$(n-1)(n-2)$, and it sets the diagonal of the centred matrix to zero by hand instead of letting the centring determine it. The raw self-distances a_{ii} are zero to begin with, so nothing is removed from the matrix itself; what the three adjustments remove is the contribution of those n zeros to the row, column and overall averages, and with it the positive $1/n$ remainder described above. With these three adjustments the estimator becomes a genuine U-statistic: a mean of a fixed symmetric kernel over all distinct subsets of the sample of fixed size, here four. Since every term of this mean is itself an unbiased estimate of the same population quantity and no observation is ever paired with itself, the mean is exactly unbiased and not merely approximately. Here it is unbiased for dCov^2 as soon as $n \geq 4$ holds, which is the size of the kernel. The corrected denominators named above stay positive already from $n = 3$ onwards; what carries the same bound as the kernel is the factor $n(n-3)$ which the normalisation to dCor^2 in Chapter 3 brings with it.

This work retains both conventions, one for each method, because the two methods place opposite requirements on the estimator. The supervised method measures a projection against a fixed target quantity and should not, near independence, see structure which is not present; it uses therefore the unbiased U-centred form, whose vanishing expectation under independence is exactly the property which a dependence measure needs. The unsupervised method, by contrast, *maximises* the distance variance, and there the biased, doubly centred form is the natural choice. It is a genuine squared norm, $\mathrm{dVar}^2 = \|A\|_F^2/n^2$ with A as the doubly centred matrix, and therefore non-negative by construction. The U-centred distance variance carries no such guarantee and can become negative in finite samples. This changes nothing about which direction is the best one, since the taking of a square root does not reorder values; it changes, however, whether the search can move at all, because the derivative of a square root has the root in the denominator and does therefore not exist there, where the quantity beneath it is zero or negative. The exact expressions as well as the gradients derived from them are given in Chapter 3. A naive evaluation costs $O(n^2)$, since it builds up the dense $n \times n$ distance matrix, but the *value* alone can be computed in $O(n \log n)$ with the AVL-tree algorithm of Huo and Székely (2016), refined by Chaudhuri and Hu (2019) and implemented in the `dcor` package (Ramos-Carreño and Torrecilla, 2023). That fast path serves the evaluation only; Section 3.3 shows why the gradient cannot use it, and what that costs.

2.2 The supervised lineage: sufficient dimension reduction

The supervised method stands in the tradition of *sufficient dimension reduction* (SDR). The underlying question is one of regression. A response Y is to be brought into relation with a p-dimensional predictor X, whereby p is large enough that the direct estimation of the regression function becomes difficult. What SDR seeks is to replace X by a small number of linear combinations $B^\top X$, with $B \in \mathbb{R}^{p\times d}$ and $d \ll p$, and this in such a way that *no information about Y gets lost.* Formally one searches for a matrix B such that

$$Y \perp\!\!\!\perp X \mid B^\top X, \tag{2.9}$$

which means, that once the d projected coordinates $B^\top X$ are known, the remaining directions of X are irrelevant for Y. The column space spanned by such a matrix B is called a *dimension-reducing subspace.* Subspaces of this kind are not unique as the whole of $\mathbb{R}^p$ fulfils the condition trivially. Therefore the object of actual interest is the smallest such subspace. Under mild conditions the intersection of all dimension-reducing subspaces is itself again a dimension-reducing subspace; after Cook (1998) it is then designated the *central subspace* $\mathcal{S}_{Y|X}$, and its dimension d is the structural dimension of the regression. The goal of SDR consists in estimating this subspace from the data, and in the ideal case also d.

About the shape of the regression function f in $Y = f(B^\top X) + \varepsilon$, condition (2.9) makes no statement at all. SDR estimates those *directions* which carry the dependence, without laying itself down on a model for how the response depends upon them. This distinguishes it from ordinary regression. At the same time it is the property which makes a dependence measure into the natural instrument, for the defining condition (2.9) is itself a statement about conditional independence.

Considered historically, the predominant strategy estimates the central subspace over *inverse moments*, i.e. over characteristics of the conditional distribution of X given Y, such as the conditional expectation $\mathbb{E}[X \mid Y]$ or the conditional covariance, which under suitable conditions are guaranteed either to lie in the central subspace or to disclose it. Such methods are computationally cheap, since they can be led back to an eigendecomposition. Unfortunately, this speed is paid for with distributional assumptions on X, typically that its distribution is elliptically symmetric, and beyond that they may remain blind towards certain symmetric dependence structures.

A different strategy would be to search directly for those directions B which *maximise a dependence measure between Y and $B^\top X$* instead. Because of the independence characterisation (2.8), the distance covariance is a natural measure of this kind. That the maximisation identifies the right object is nevertheless not self-evident and has to be proven. Sheng and Yin (2016) show, that under mild conditions on the predictors the maximiser of the population $\mathrm{dCov}(Y, B^\top X)$ over the Stiefel manifold spans the central subspace, and that the sample maximiser is consistent for it; neither a link function nor a nonparametric estimate of the regression function enters those conditions, and the elliptical symmetry which the inverse-moment methods require is not among them either. What the measure does ask of the data is the finite first moment of Section 2.1, which the elliptical families satisfy in any case, so the comparison is one of shape assumptions and not of none against some. The supervised method follows this line. Its direct predecessors are Sheng and Yin (2013), who estimate a single-index direction through maximisation of $\mathrm{dCov}(Y, X\beta)$, and Sheng and Yin (2016), who extend the construction to a multidimensional subspace. Their population objective is, up to the choice of the optimiser, the supervised objective of Chapter 4. Where the present work differs from them is in the computation. They optimise with sequential quadratic programming on the Stiefel manifold, while here a closed-form Riemannian gradient is derived and passed over to a customary adaptive optimiser. That the gradient plays a role is no mere conjecture. Wu and Chen (2021) report, that standard manifold solvers come to a standstill in distance-covariance based SDR when they are run without analytical derivatives, and they propose majorise-minimise procedures in order to compensate for this. The analytical gradient of Chapter 3 responds directly to this difficulty.

2.3 The unsupervised lineage: projection pursuit and ICA

Where no response is available which could guide the reduction, the canonical instrument for the detection of low-dimensional structure in high-dimensional data is *principal component analysis* (PCA) (Jolliffe, 2002). Among all d-dimensional linear projections, the projection onto the leading principal components minimises the mean squared reconstruction error and, equivalently to this, preserves the maximally possible variance. Hereby it is optimal in the Karhunen–Loève sense. If the interesting structure in the data *really is* its second-order scatter then no linear method whatsoever can improve upon PCA.

The premise of projection pursuit is, that this is often *not* the case. A direction may carry very little variance and nevertheless disclose the most about the data. The standard example is given by two well-separated clusters whose separation lies along an axis of

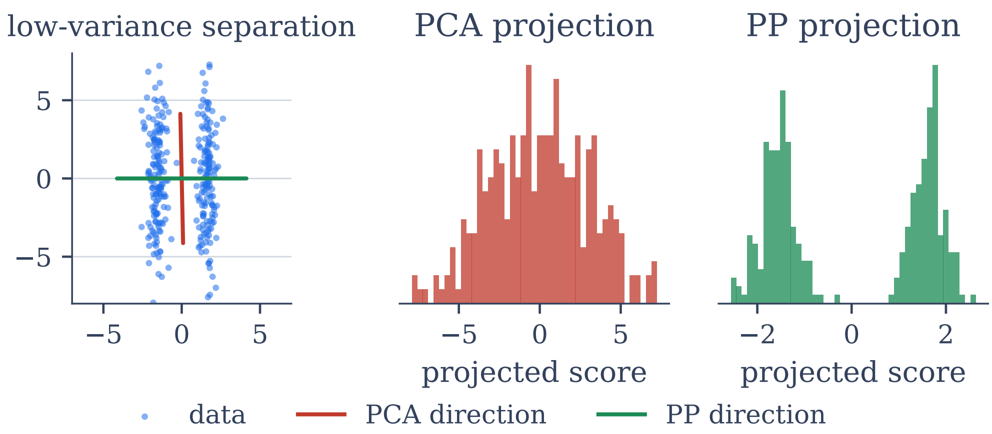

Figure 2.1: Why an interestingness index need not be variance. *Left:* two clusters whose separation runs along the axis of *small* spread, while the large spread is within each cluster. *Centre:* projected onto the leading principal component, the clusters fall on top of one another and the projection looks unimodal. *Right:* projected onto the direction a projection-pursuit index selects, the two groups separate. PCA is not failing at its own task; the structure does not lie where variance is largest.

small variance, which PCA discards in favour of the directions of high variance *within* each cluster. Figure 2.1 shows this situation. For non-elliptical data the same point is made systematically clear by invariant coordinate selection (Tyler et al., 2009), which finds such directions by comparing two scatter matrices instead of maximising a single one. *Projection pursuit*, introduced by Friedman and Tukey (1974) and systematised by Huber (1985), formalises the search for directions of this kind in that the variance is replaced by an *interestingness index*, a functional of the one-dimensional projected distribution which is then maximised over the projection directions. In nearly the whole of the literature, this index is a measure of *non-Gaussianity*, whereby a projection is the more interesting, the more strongly its distribution deviates from a normal distribution.

The reason why non-normality is the correct notion of interestingness is made precise by Diaconis and Freedman (1984). Under general conditions, *most* low-dimensional projections of a high-dimensional point cloud are approximately normally distributed. A projection which deviates markedly from normality is therefore rare, and rare means informative. It reflects genuine structure and not the generic central-limit behaviour that an arbitrary direction exhibits in any case. Concretely, indices such as the absolute kurtosis or the negentropy preferred by Huber (1985) are often used to single out projections of this kind, though they do not all do so in the same way. The negentropy is a genuine divergence from the normal distribution of equal variance and is minimised there. The absolute kurtosis is a single standardised moment, which departs from its Gaussian value when the projection is heavy-tailed or flat and multimodal but is not a distance from normality. A distribution far from Gaussian can carry a kurtosis close to that of the Gaussian. Maximising either pulls out those directions along which the data are heavy-tailed, skewed or multimodal. Radojičić et al. (2020) set out how indices of this kind order distributions and how this ordering connects non-normality with independent component analysis; the comparison given there is the point of reference for the index employed in the present work, which orders distributions differently.

The same state of affairs carries a warning with it, which is sharpened by Bickel et al. (2018). In high dimensions and outside of a favourable sample-size regime, already the empirical fluctuations of pure Gaussian noise suffice in order to produce projections which *look* non-normal, so that no projection pursuit procedure can reliably distinguish genuine structure from this artefact.

In this sense the unsupervised method of the present work is a projection pursuit method, though with one restriction. Its interestingness index is the *distance variance* of the one-dimensional projection, and the ordering that index imposes on distributions is the one set out in Section 2.1. That ordering is directed towards one side of the normal distribution rather than away from it in every direction. Against PCA it can nevertheless achieve what PCA cannot, though not through directions of small variance: the distance variance is scale-equivariant, so a direction of small spread receives a small value from it as well. The gain lies with directions of *equal* variance, where the covariance matrix has no eigenvalue gap left for PCA to rank by while the pairwise-distance profiles of the projections still differ (Section 5.4.2). The justification given above and the high-dimensional limit continue to hold; what does not continue to hold is the equating of the index with non-normality, and Chapter 5 reports where this difference becomes visible.

When several such directions are searched for simultaneously and the index is non-normality, projection pursuit becomes *independent component analysis* (ICA). The FastICA family of Hyvärinen and Oja (2000) maximises contrast functions (kurtosis, log-cosh and the like), which are disguised projection indices, after a prewhitening of the data. One conceptual distinction within this family is of importance for the present work. Contrast-function ICA measures the *marginal* non-normality of each extracted direction separately, and the distance variance index of this work judges each extracted direction on its own in the same way. The distance covariance ICA of Matteson and Tsay (2017), on the contrary, minimises the sum of the *pairwise* distance covariances between the extracted components and aims thereby at true mutual independence instead of marginal non-normality. Section 3.1 takes up that distinction, which bounds what either method of this work is able to see.

There is a second and more specific condition on an index used for this purpose. Radojičić et al. (2020) show, that in the independent component model a projection index recovers the independent components when it is a squared dispersion measure that is sub-additive or super-additive, in the sense that the measure applied to a sum of independent variables is bounded by the corresponding combination of the measures of the summands. Certain functions of the entropy and of the kurtosis satisfy this and are squared dispersion measures in the required sense. The distance variance is a sub-additive dispersion measure (Radojičić et al., 2020), but it is the unsquared one. The quantity that enters

(3.1) is dVar and not dVar^2, for the reason of differentiability given in Section 2.1. What that difference costs the index in its ability to separate independent components is not settled here, and it is one of the two structural restrictions that Chapter 7 names.

A closely related strand, projection pursuit regression (Friedman and Stuetzle, 1981), introduced the idea of residual deflation which the supervised method takes over and which proves itself insufficient in Section 4.4; the difficulty of the non-convex landscapes resulting from this was already stated by Friedman (1987).

2.4 Classical baselines

The proposed methods are judged against the customary second-order procedures. *Principal component analysis* (PCA), introduced in Section 2.3, is the unsupervised baseline throughout. *Ordinary least squares* (OLS) is the baseline for the supervised method. It maximises the covariance with the response and is therefore blind towards symmetric nonlinear dependence (Section 1.1).

Two further baselines require an explanation on account of implementation details. FastICA is a projection pursuit method in its own right, maximising a contrast function over the whitened directions, which is not the criterion of PCA. It is run here with full whitening, that is on all p principal components rather than on a leading subset of them, so that the contrast function and not the variance ordering decides which directions are returned. Since whitening equalises the variances before the contrast function is evaluated, the criterion has nothing left to read in a model whose signal direction is marked by its variance alone; FastICA is accordingly reported in Section 5.4.2, whose construction carries the structure in the shape instead, and where $p = k = 2$ leaves nothing to discard. The *partial least squares* method (PLS) (Wold et al., 2001) is the supervised linear counterpart for the chapter on real data. It extracts directions of maximal covariance between predictors and response and is thus the natural linear method which one has to surpass if it is claimed that *nonlinear* dependence on the response brings an added value.

2.5 Robustness of distance-based measures

Even in their simple form the distance-based methods are more robust than the second-order baseline procedures against which they are compared; Table 5.6 in Section 5.4.5 measures the difference. OLS and PCA rest on *squared* deviations from a mean; a single far-lying observation therefore enters with its squared displacement and may dominate the fit, respectively the leading eigenvector. Distance covariance and distance variance proceed differently. They aggregate *pairwise absolute distances* $\|z_i - z_j\|$, and these grow only linearly in the displacement of an observation, so that the replacement of squared second-order moments by pairwise distances already damps the influence of extreme points.

One might conclude from this, that a distance-based method is thereby robust. Leyder et al. (2026) clarify this question by distinguishing three quantities. The *influence function* describes the effect of an infinitesimal contamination at one point. Its finite-sample counterpart is the *sensitivity curve*, which measures the change caused by the addition of one observation to a sample of size n. Separate from both is the *breakdown point*, the smallest fraction of contaminated observations which is able to drive the estimator towards an uninformative limiting value.

For the Euclidean exponent employed here, distance covariance, distance variance and distance correlation all possess a *bounded* influence function. However, it is not a redescending one, so that a contaminating point, no matter how far outside it may lie, still contributes with nearly its maximal weight; only for exponents below one does the influence function redescend (Leyder et al., 2026). This is a statement at the population level. It does not carry over to finite samples: the sensitivity curve is unbounded, and if one observation of a fixed sample is shifted far enough outwards, the index tends to infinity. The breakdown point is correspondingly zero. Both facts become reconciled only in the limit, in which the sensitivity curve converges towards the influence function; and at the sample sizes used here the finite-sample behaviour does make itself noticeable.

The remedy proposed by these authors consists in a bounded, *redescending* transformation, which Leyder et al. (2026) call the *biloop*, after the two ellipses that make up its image. It standardises every coordinate by means of its median and its median absolute deviation and then wraps the standardised value onto those ellipses, so that extreme values are folded back towards the centre. Since the transformed sample lies within a bounded region, an observation situated far outside is pulled back among the remaining ones before the index registers it, so that the finite-sample sensitivity is removed and the influence of a point decays towards zero with growing distance instead of saturating. All three quantities therefore improve at once, since the index computed on a bounded transform carries a redescending influence function and a breakdown value strictly larger than zero, which the plain index does not attain (Leyder et al., 2026). The unsupervised method takes over this transformation for its robust variant (Section 5.4.5). The same authors extend the construction into a complete robust independent component analysis (Leyder et al., 2025), the natural multivariate successor of the univariate transformation employed here, which keeps the objective of Matteson and Tsay (2017) described above and replaces the plain index inside it by the bounded one; that is the same substitution the robust variant makes on the projection pursuit side, and it is noted as future work. Bounding the influence costs something in return, in differentiability and in accuracy on clean data; Section 3.6 gives the transformation in closed form and Chapter 5 measures both.

2.6 Optimisation on matrix manifolds

Both methods reduce to one and the same type of optimisation problem: the maximisation of a smooth objective function $J(B)$ over matrices $B \in \mathbb{R}^{p\times k}$ whose columns are orthonormal, $B^\top B = I_k$.

The constraint deserves a remark, for it is imposed on the two sides for different reasons. Both objective functions depend on B only through the scores XB, but they answer differently to a rescaling of a column. The supervised objective does not change at all, since the distance correlation (2.5) divides by the distance variances of its own arguments, so that without a normalisation the maximisation possesses a continuum of equivalent solutions along which a gradient procedure drifts. The unsupervised index is scale-equivariant instead, $\mathrm{dVar}(aU) = |a|\,\mathrm{dVar}(U)$, so that stretching a column raises the objective without bound and no maximiser exists at all. Therefore the normalisation is needed in the one case in order to single out one point of a continuum, and in the other in order to make the problem well posed at all. An orthogonal change of basis within the same span is not neutral for either objective. Both are sums of indices of the individual columns XB_j, so a rotation $B \mapsto BQ$ with $Q^\top Q = I_k$ leaves the span untouched while mixing the columns, and with them every summand. What is optimised here is therefore an orthonormal *frame*, not a subspace.

In the case of independent component analysis the normalisation follows from the prewhitening, after which the unmixing matrix is orthogonal by construction, while in projection pursuit it forms part of the search itself and holds the k directions fixed as a frame instead of as k copies of one single maximiser. In sufficient dimension reduction, by contrast, the target is a *subspace*, the central subspace $\mathcal{S}_{Y|X}$, which fixes no basis at all; here the target fixes no basis at all, and $B^\top B = I_k$ would merely select a representative of it if the objective depended on the span alone. The objective employed here does not: it scores each column separately, so the constraint does restrict, and the frame which the optimisation returns is one particular basis of the estimated subspace. The constraint does not, however, deliver uncorrelated projections, a gap to which Section 3.1 returns. In principal component analysis this gap does not arise. Since the directions are eigenvectors of the covariance matrix, orthogonal weights produce uncorrelated scores there by construction, which an arbitrary orthonormal frame does not.

The constraint has the consequence that ordinary gradient ascent cannot be applied directly: a Euclidean step $B + \eta\,\nabla J$ almost never satisfies $B^\top B = I_k$ anew, so that the iterate leaves the feasible set immediately.

The set of such orthonormal frames is a smooth surface inside $\mathbb{R}^{p\times k}$, the *Stiefel manifold*

$$\mathrm{St}(p,k) = \{\, B \in \mathbb{R}^{p\times k} : B^\top B = I_k \,\}, \tag{2.10}$$

which for $k = 1$ is simply the unit sphere in $\mathbb{R}^p$. *Riemannian optimisation* denotes the discipline of carrying out gradient-based optimisation while remaining upon such a surface, and for the purposes of the present work it relies on two operations, each one of which carries a simple geometric meaning.

The first is the *projection onto the tangent space.* At a point B on the manifold, the tangent space is the set of those directions in which one may move without violating the constraint to first order: a movement along these directions alters $B^\top B$ only by an amount of second order in the step size, whence for a small step the constraint remains satisfied up to a negligible error. It is the flat plane which approximates the curved surface at B best. If one differentiates the constraint $B^\top B = I_k$ along a curve, it becomes apparent, that a tangent direction η has to fulfil the condition $B^\top \eta + \eta^\top B = 0$, so that the forbidden directions are exactly those of the form BS with symmetric S. Removing this part from a Euclidean gradient G, one obtains the tangent projection

$$P_B(G) = G - B\,\mathrm{sym}(B^\top G), \qquad \mathrm{sym}(M) = \tfrac{1}{2}(M + M^\top), \tag{2.11}$$

which on the sphere ($k = 1$) reduces to $P_w(g) = g - (w^\top g)\,w$, the ordinary orthogonalisation of g against w. What remains is the *Riemannian gradient*, the direction of steepest ascent which respects the constraint.

The second is a *retraction*. A step along a tangent direction lands slightly outside the manifold and must be mapped back onto it. A retraction is any smooth mapping $R_B(\eta)$ which accomplishes this, which fixes B for $\eta = 0$ and which agrees with the manifold to first order. The one employed here takes the thin QR factorisation of the updated matrix and retains its orthonormal factor,

$$R_B(\eta) = Q, \qquad \text{where} \qquad B + \eta = QR, \tag{2.12}$$

with $Q \in \mathrm{St}(p, k)$ and R upper triangular with positive diagonal entries $R_{ii} > 0$. After this factorisation it is called the QR retraction, in which Q collects the orthonormal directions and R the coefficients expressing the original columns in them; fixing the signs of the diagonal of R renders the factorisation, and therewith the mapping, unique and smooth (Absil et al., 2008). A Cayley-transform retraction (Li et al., 2020) is the usual alternative.

By means of these two operations every Euclidean first-order optimiser carries over to the manifold: one computes the Euclidean gradient, projects it onto the tangent space, performs an adaptive step, and retracts. The optimiser employed throughout is the Riemannian version of Adam (Bécigneul and Ganea, 2019), which transfers the adaptive momentum mechanism of Euclidean Adam (Kingma and Ba, 2015) to the Stiefel manifold by way of precisely these operations. The general theory of optimisation on matrix manifolds is developed by Edelman et al. (1998), Absil et al. (2008) and Boumal (2023); here only the two operations named above are required, and the explicit projections, the retraction and the step-size schedule appear in Chapter 3.

CHAPTER 3

Algorithmic Foundations

Both methods of the present work rest upon a similar principle. Each of them maximises a distance-based dependence index over the directions of a projection, subject to an orthonormality condition. What changes from one method to the other is the index itself: distance correlation with a target quantity for the supervised method (Chapter 4), and distance variance of the projection for the unsupervised one (Chapter 5).

This chapter develops the machinery which both methods make use of in common, so that the method chapters may concentrate upon the results. Three parts of it are new. Section 3.2 derives the analytic gradient of both indices in closed form; this is the component that renders the search practicable, and the literature on distance-covariance-based dimension reduction has so far circumvented the derivation instead of employing it. In Section 3.1, a diversity penalty is added which combines an existing independence criterion with a supervised objective function. Section 3.6 takes a robust transformation which was proposed for the estimation of dependence and employs it instead as a projection index, quantifying what the boundedness costs on uncontaminated data. The remainder is composed of established building blocks: Riemannian Adam, applied as published (Section 3.4); the deflation strategies and the joint alternative to them, whose failure modes are analysed here (Section 3.5); and the cost of the gradient together with its verification, which set the practical limits (Section 3.3).

3.1 The unified objective

Let $X \in \mathbb{R}^{n\times p}$ be the (centred) data matrix with one observation per row, and let $B \in \mathbb{R}^{p\times k}$ be a matrix carrying k projection directions in its columns. Both methods maximise one and the same object: a sum of column-wise indices, maximised over the

Stiefel manifold of orthonormal frames,

$$\max_{B \in \mathrm{St}(p,k)} \sum_{j=1}^{k} I(XB_j), \qquad \mathrm{St}(p,k) = \{B \in \mathbb{R}^{p\times k} : B^\top B = I_k\}. \tag{3.1}$$

Let us begin with a single column $B_j \in \mathbb{R}^p$, a unit vector which selects one direction in the predictor space. The product XB_j is then an n-vector: its i-th entry is the projection of the i-th observation onto this direction, so that XB_j constitutes the one-dimensional *score* of the entire sample, viewed through one single linear lens. For an optimal frame B, every column B_j is that direction whose lens reveals most of what the method is searching for.

What "the most" means is fixed by the index $I(\cdot)$. It takes the n projected values XB_j and returns one single number which measures how interesting this projection is. In the supervised case $I(\cdot) = \mathrm{dCor}^2(Y,\cdot)$, the squared distance correlation between the projection and the target quantity, which is large whenever the projection preserves a strong (possibly nonlinear) dependence on Y. In the unsupervised case $I(\cdot) = \mathrm{dVar}(\cdot)$, the distance variance of the projection, whose ordering of shapes at fixed variance is the one given in Section 2.1. Both indices were defined in Chapter 2; here they are simply scalar values assigned to a one-dimensional projection. For $k = 1$ the constraint reduces to the unit sphere S^{p-1}, and the objective function reads literally as "find the most interesting direction".

The sum over j turns k separate one-dimensional scores into one single scalar which an optimiser is able to maximise, and the orthonormality condition $B \in \mathrm{St}(p,k)$ holds the columns fixed as a genuine frame: k different axes instead of k copies of the same maximiser. It suggests itself to read (3.1) as a search for k directions each of which is, on its own, as interesting as possible. This reading, however, holds only for a greedy search which first fixes the first direction, then the best one orthogonal to it, and so forth. PCA makes the very same distinction: only its first component maximises the variance without restriction, whereas every later one is subject to orthogonality with respect to those already chosen. To solve (3.1) for all columns simultaneously is a different problem, whose solution need not be the greedy one. Both ways are employed here, sequential extraction with deflation (Section 3.5) being the greedy one, and Section 4.4.5 compares the two.

The orthonormality condition, however, controls the directions only as vectors in $\mathbb{R}^p$; about the projections which they induce in $\mathbb{R}^n$ it says nothing. Two columns may satisfy $B_i^\top B_j = 0$ exactly and nevertheless yield projected scores XB_i and XB_j which are strongly correlated, indeed nearly identical, as soon as the data is not isotropic, since X stretches the geometry of the weight space unevenly (this is precisely the gap which Section 3.5 addresses). Orthogonal *weights*, in other words, do not guarantee diverse *projections*, and a frame of orthonormal columns may silently collapse onto one single interesting direction which is then discovered k times over. PCA does not have this problem; the difference is that its directions are eigenvectors of the covariance matrix, so that the orthogonal weights deliver uncorrelated scores by construction. An orthonormal frame chosen by another criterion carries no such guarantee. In the supervised case we therefore add, optionally, a diversity penalty which takes hold where the redundancy actually resides, namely at the projections themselves,

$$\max_{B \in \mathrm{St}(p,k)} \sum_{j=1}^{k} \mathrm{dCor}^2(Y, XB_j) \;-\; \lambda \sum_{i<j} \mathrm{dCor}^2(XB_i, XB_j), \tag{3.2}$$

which penalises every pair of projected scores that are themselves dependent on one another. The coefficient λ weighs the raw column-wise dependence on the target quantity against the redundancy between the columns; its effect is examined in Section 4.4.5.

The penalty term as such is not new: to sum the pairwise distance covariances between the recovered components is precisely the objective of distance-covariance-based independent component analysis (Matteson and Tsay, 2017), where it is minimised alone in order to drive the components towards mutual independence. What is employed here is that quantity as a regulariser for a supervised objective criterion, the two being weighed against one another by λ instead of one of them being optimised on its own. This combination appears not yet to have been investigated for dimension reduction by means of distance correlation.

One structural feature of (3.1) reappears in both methods as a limitation and is for that reason worth mentioning at this point. The objective criterion is a *sum of marginal* indices, one per column. It never measures the *joint* dependence of the complete recovered projection XB. This marginal structure has consequences on both sides. For the supervised method it means that neither the joint nor the sequential optimisation is able to detect dependence which resides in the projection as a whole and not in its individual columns (Section 4.4.5); for the unsupervised method it permits the index to overlook higher-order dependencies between the factors (Section 5.4.6). Chapter 7 brings both together as one single phenomenon.

3.2 The analytical gradient

The objective criterion (3.1) is a sum over the columns of B, and each summand depends only on its own column. The gradient with respect to B therefore assembles column-wise from the gradients of the individual directions; the constraint $B^\top B = I_k$ is handled by the optimiser via projection or retraction and does not affect the derivation below. It therefore suffices to determine, for a single fixed direction $w \in \mathbb{R}^p$, the derivative $\partial I(Xw)/\partial w$.

Both indices are built from the pairwise distances of the projected sample. These distances contain an absolute value, and this is the only non-smooth component of the entire construction. It fails to be differentiable exactly where two projected scores coincide. Everywhere else the objective criterion is differentiable in w, and the derivative is available in closed form.

Structure of the section.

1. **Notation** (§1): distance matrix D, sign matrix S, the inner products used.
2. **The chain rule** (§2, Theorem 1): the structure shared by both gradients. It shows that the entire dependence on the choice of index is contained in a single matrix, the *upstream gradient* $G = \partial g/\partial D$.
3. **Centring operators** (§3): both centring operations – the double centring of the distance variance and the U-centring of the distance correlation – are linear, self-adjoint, idempotent projections. That property alone yields each of the three upstream gradients that are needed, without differentiating through the centring sums. In particular, this is where it becomes clear why the numerator of the distance correlation is *linear* in the raw distances, and why its denominator is formally the same computation as that of the distance variance.
4. **Distance variance** (§4) and **distance correlation** (§5) as two instances of the same argument.
5. **Degenerate points** (§6): the choice of search direction where the supervised index is undefined.

The derivation is elementary, but no closed form for either gradient is available in the literature, which is why the steps are carried out in full rather than merely cited.

1. Notation. Let $X \in \mathbb{R}^{n\times p}$ be the data matrix, $w \in \mathbb{R}^p$ the projection direction, and

$$z = Xw \in \mathbb{R}^n$$

the vector of n projected scores. The objective criterion does not depend directly on w: w determines the scores z, the scores determine the pairwise distances, and the criterion is evaluated from these. The derivative therefore follows the chain

$$w \longrightarrow z \longrightarrow D \longrightarrow g.$$

Two matrices describe the geometry of the projected sample:

$$D_{ij} = |z_i - z_j|, \qquad S_{ij} = \operatorname{sign}(z_i - z_j). \tag{3.3}$$

D contains the distance between every pair of projected scores; S records only their ordering. We have:

- D is symmetric with zero diagonal,
- S is **antisymmetric**, $S_{ij} = -S_{ji}$, with zero diagonal.

Furthermore, $\odot$ denotes the element-wise (Hadamard) product, $\mathbf{1} \in \mathbb{R}^n$ the vector of ones, and we use two inner products on matrices:

$$\langle M, N \rangle_F = \sum_{i,j} M_{ij} N_{ij}, \qquad \langle M, N \rangle_0 = \sum_{i \neq j} M_{ij} N_{ij}. \tag{3.4}$$

The second ignores the diagonal and is the one in which the U-centring shows its structure.

2. The chain rule through the pairwise distances. We first derive the chain in general, since the same argument carries both indices. Let g be a scalar function of the distance matrix D. We collect its derivatives with respect to the distances in the matrix

$$G_{ij} = \frac{\partial g}{\partial D_{ij}}, \tag{3.5}$$

which we call the **upstream gradient**. This matrix is the only component that differs between the two indices.

Theorem 1 (Chain rule). *If g is differentiable with symmetric upstream gradient G, then at every point with pairwise distinct scores*

$$\frac{\partial g}{\partial z} = 2\,(G \odot S)\,\mathbf{1}, \qquad \frac{\partial g}{\partial w} = X^\top \frac{\partial g}{\partial z} = 2\,X^\top (G \odot S)\,\mathbf{1}. \tag{3.6}$$

Proof. The only non-smooth operation is the absolute value; it is differentiable everywhere the two scores do not coincide, with

$$\frac{\partial |z_i - z_j|}{\partial z_k} = \operatorname{sign}(z_i - z_j)\,(\delta_{ik} - \delta_{jk}) = S_{ij}\,(\delta_{ik} - \delta_{jk}). \tag{3.7}$$

The Kronecker deltas express that the distance D_{ij} depends only on its two variables. Substituting into the chain rule gives

$$\frac{\partial g}{\partial z_k} = \sum_{i,j} \frac{\partial g}{\partial D_{ij}} \frac{\partial D_{ij}}{\partial z_k} = \sum_{i,j} G_{ij} S_{ij} (\delta_{ik} - \delta_{jk}) = \underbrace{\sum_j G_{kj} S_{kj}}_{k\text{-th row sum}} - \underbrace{\sum_i G_{ik} S_{ik}}_{k\text{-th column sum}}. \tag{3.8}$$

Both sums concern the same matrix $G \odot S$. Since G is symmetric by assumption and S is antisymmetric, $G \odot S$ is **antisymmetric**. An antisymmetric matrix has column sums equal to the negative of its row sums; the two terms in (3.8) therefore add up to twice the row sum, which gives the first equation in (3.6). The second follows immediately from the linear relation $z = Xw$, whose Jacobian is X. □

This reduces the entire problem to a single question: **what is G?**

3. Centring operators. Neither index evaluates the raw distances directly; both use a centred version. The two centrings are the *double centring*

$$\mathcal{C}_H(D) = HDH, \qquad H = I_n - \tfrac{1}{n}\mathbf{1}\mathbf{1}^\top,$$

used by the unsupervised index with the Frobenius inner product $\langle\cdot,\cdot\rangle_F$, and the *$U$-centring*

$$\begin{aligned}
\mathcal{C}_U(D)_{ij} &= D_{ij} - \frac{1}{n-2}\sum_l D_{il} - \frac{1}{n-2}\sum_l D_{lj} + \frac{1}{(n-1)(n-2)}\sum_{l,m} D_{lm} \quad (i \neq j), \\
\mathcal{C}_U(D)_{ii} &= 0,
\end{aligned} \tag{3.9}$$

used by the supervised index with $\langle\cdot,\cdot\rangle_0$. Both are orthogonal projections on their respective matrix spaces, which for $\mathcal{C}_H$ is immediate from $H = H^\top$ and $H^2 = H$, and for $\mathcal{C}_U$ is established in Székely and Rizzo (2014). As projections they are linear, self-adjoint, and idempotent, so differentiating through the centring sums is unnecessary: the projection can be moved across the inner product.

For the unsupervised index this gives, with $A = \mathcal{C}_H(D)$,

$$\begin{aligned}
\mathrm{dVar}^2 &= \tfrac{1}{n^2}\langle A, A\rangle_F, \\
\mathrm{d}\langle A, A\rangle_F &= 2\langle A, \mathcal{C}_H(\mathrm{d}D)\rangle_F = 2\langle \mathcal{C}_H(A), \mathrm{d}D\rangle_F = 2\langle A, \mathrm{d}D\rangle_F, \\
G &= \tfrac{2}{n^2}\, A.
\end{aligned} \tag{3.10}$$

For the supervised numerator, with $\tilde{A} = \mathcal{C}_U(D)$ and $\tilde{B} = \mathcal{C}_U(D^Y)$ fixed and already centred,

$$\begin{aligned}
s_{xy} &= \tfrac{1}{n(n-3)}\langle \tilde{A}, \tilde{B}\rangle_0, \\
\langle \tilde{A}, \tilde{B}\rangle_0 &= \langle \mathcal{C}_U(D), \tilde{B}\rangle_0 = \langle D, \mathcal{C}_U(\tilde{B})\rangle_0 = \langle D, \tilde{B}\rangle_0, \\
G &= \tfrac{1}{n(n-3)}\, \tilde{B}.
\end{aligned} \tag{3.11}$$

For the supervised denominator, the same calculation as (3.10) with $\mathcal{C}_U$ and $\langle\cdot,\cdot\rangle_0$:

$$\begin{aligned}
s_{xx} &= \tfrac{1}{n(n-3)}\langle \tilde{A}, \tilde{A}\rangle_0, \\
G &= \tfrac{2}{n(n-3)}\, \tilde{A}.
\end{aligned} \tag{3.12}$$

All three G are symmetric, so Theorem 1 applies in each case.

4. Distance variance (unsupervised). The unsupervised index uses the biased double centring $\mathcal{C}_H$ rather than $\mathcal{C}_U$, for the reason given in Section 2.1: it keeps the quantity under the square root non-negative, and the derivative of that root is what the search needs.

Substituting $G = \frac{2}{n^2}A$ from (3.10) into Theorem 1 gives

$$\frac{\partial\, \mathrm{dVar}^2}{\partial w} = \frac{4}{n^2}\, X^\top (A \odot S)\, \mathbf{1}. \tag{3.13}$$

The optimiser maximises $\mathrm{dVar} = \sqrt{\mathrm{dVar}^2}$, so the final gradient picks up one more chain-rule factor:

$$\frac{\partial\,\mathrm{dVar}}{\partial w} = \frac{1}{2\,\mathrm{dVar}}\,\frac{\partial\,\mathrm{dVar}^2}{\partial w} = \frac{2}{n^2\,\mathrm{dVar}}\,X^\top (A \odot S)\,\mathbf{1}. \tag{3.14}$$

5. Distance correlation (supervised). The supervised index maximises the squared distance correlation

$$\mathrm{dCor}^2(z, Y) = \frac{s_{xy}}{\sqrt{s_{xx}\,s_{yy}}}, \tag{3.15}$$

with s_{xy}, s_{xx}, s_{yy} as defined in (3.11)–(3.12). Since $\tilde{B}$ and s_{yy} depend only on Y, not on w, they are computed once. Substituting the upstream gradients from (3.11) and (3.12) into Theorem 1 gives

$$\frac{\partial s_{xy}}{\partial w} = \frac{2}{n(n-3)}\,X^\top (\tilde{B} \odot S)\,\mathbf{1}, \qquad \frac{\partial s_{xx}}{\partial w} = \frac{4}{n(n-3)}\,X^\top (\tilde{A} \odot S)\,\mathbf{1}. \tag{3.16}$$

With s_{yy} constant, the product and power rules yield

$$\frac{\partial\,\mathrm{dCor}^2}{\partial w} = \mathrm{dCor}^2 \left(\frac{1}{s_{xy}}\frac{\partial s_{xy}}{\partial w} - \frac{1}{2\,s_{xx}}\frac{\partial s_{xx}}{\partial w} \right). \tag{3.17}$$

All three upstream gradients (A, $\tilde{B}$, $\tilde{A}$) enter through the same contraction $X^\top(\cdot \odot S)\mathbf{1}$.

6. Degenerate points. The unbiased s_{xy} can become negative in finite samples. Wherever any of s_{xx}, s_{xy}, s_{yy} is non-positive, (3.17) is undefined and (3.15) has no value. The index is set to zero at such points.

A zero gradient is the obvious candidate, and it is unusable. Zero is a fixed point of the update: a column that enters the degenerate region would remain frozen there for the rest of the run. The optimiser therefore returns the ascent direction of the numerator alone,

$$\frac{\partial s_{xy}}{\partial w} = \frac{2}{n(n-3)}\,X^\top (\tilde{B} \odot S)\,\mathbf{1},$$

which increases the quantity whose sign determines whether the objective is defined, leading back into the region where optimisation is possible. Its magnitude is irrelevant; Adam normalises the step.

The region is entered often enough that this is a design decision. Across all configurations in the supervised projection-pursuit chapter with ten data seeds, a median of 1.5% of gradient calls land in the degenerate region, rising to a median of 5.7% $[0.0, 22.5]$ for the worst configuration, the noiseless `sine` response at $p = 5$.

3.3 Cost and verification

Both gradients cost $O(n^2)$ for the sign-weighted distance matrix and its row sums, plus $O(np)$ for the contraction $X^\top(\cdot)$; this is the same order as a single evaluation of the index. A central-difference gradient requires $2p$ evaluations, one per coordinate of w, so its cost is $O(pn^2)$. At $p = 50$ this is the difference between seconds and minutes per optimisation run.

The quadratic term is not avoidable, and this is where the $O(n \log n)$ fast path of Section 2.1 stops being available. That algorithm attains its speed by never forming an $n \times n$ object at all, condensing the pairwise differences into rank-based aggregates instead, whereas the chain rule above needs the sign matrix S and the centred distance matrix explicitly. The gradient therefore builds dense $n \times n$ matrices at every call and is $O(n^2)$ in time and in memory, which places a practical ceiling near $n = 2{,}000$ for interactive use: at $n = 2{,}000$ and $p = 20$ a single five-restart fit of the gradient search takes a median of 32 s. Beyond it one would have to accept a subsampled or randomised gradient, a low-rank approximation of the distance matrix, or fewer restarts, none of which are used here; Chapter 7 returns to the limit as future work.

Numerical verification. The analytical gradient is compared with a central difference of the index in its squared form, dCor^2 and dVar^2. The unsupervised gradient the optimiser uses, (3.14), differs from the checked one by the chain-rule factor of the square root alone. Figure 3.1 varies the step size h over eight orders of magnitude. A correct derivative must produce a V-shape, where truncation error falls as h^2 on the right, floating-point cancellation dominates on the left, and the minimum near $h = 3 \times 10^{-6}$ reaches about 10^{-10}. Both indices trace this V exactly; a gradient that was close but not exact would flatten out at its own error level instead. The relative error $\|\nabla_{\text{analytic}} - \nabla_{\text{numeric}}\| / \|\nabla_{\text{analytic}}\|$, minimised over h and taken as a median over ten data seeds, stays below 10^{-9} in all configurations (Table 3.1). In 12 of the 120 seed–configuration combinations, every one of them on the supervised index, the check fails with a relative error of exactly 1; these are the degenerate points of §6, where the substitute direction is deliberately not the derivative and a difference check must disagree.

Structural verification. The gradient optimiser is cross-checked against two derivative-free solvers started from the same initial directions with twenty times the iteration budget: Nelder–Mead (Nelder and Mead, 1965), which maintains a simplex and reflects its worst vertex, and COBYLA (Powell, 1994), which fits local linear models within a trust region. Neither solver has knowledge of the Stiefel manifold. If the gradient formula contained a sign error or a wrong factor, the derivative-free solvers would converge to a different direction. Table 3.2 reports the results. COBYLA matches or exceeds the gradient solution in all 36 combinations of index, sample size, dimension and seed, agreeing on the direction to within 1.7° in most cases. Nelder–Mead agrees at $p = 5$ and deteriorates to $\approx 45°$ at $p = 50$, which is the well-known failure of simplex methods in high dimension and no property of the indices.

One configuration ($n = 200$, $p = 50$) is worth noting: at one of the three seeds COBYLA attains a higher objective at a direction 15.4° away, against 5.7° and 2.5° at the other two. Two approximately equal maxima are the readiest explanation; that is the non-convexity which the random restarts of the optimiser are designed to handle.

Table 3.1: Relative error of the analytical gradient against a central difference, minimised over step sizes $h \in [10^{-8}, 10^{-3}]$. Each entry is the median over ten data seeds; the twelve degenerate combinations discussed in the text are excluded. Data are the `cubic` single-index configuration at $\sigma_\varepsilon = 0.5$.

n	p	dCor^2 gradient	dVar^2 gradient
200	5	5.0×10^{-11}	1.6×10^{-11}
200	20	1.3×10^{-10}	3.2×10^{-11}
200	50	1.4×10^{-10}	8.1×10^{-11}
500	5	5.2×10^{-11}	7.3×10^{-11}
500	20	5.8×10^{-11}	1.5×10^{-10}
500	50	1.6×10^{-10}	4.6×10^{-10}

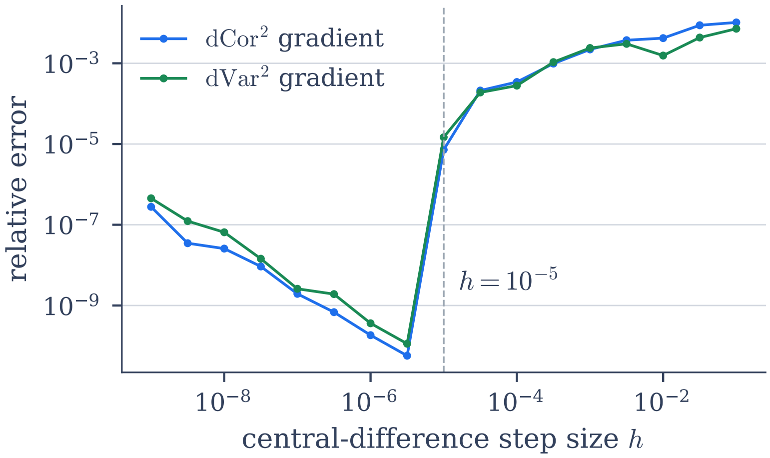

Figure 3.1: Relative error between the analytical gradient and a central difference as the step size h varies, at $n = 200$, $p = 5$. Both indices trace the same V: truncation-limited to the right, cancellation-limited to the left, with a minimum near $h = 3 \times 10^{-6}$ of about 10^{-10}. A finite difference therefore does not descend to machine precision, which an exact derivative does.

3.4 Riemannian optimisation on the sphere and Stiefel manifold

The constrained problem (3.1) is solved with Riemannian Adam (Bécigneul and Ganea, 2019) on the Stiefel manifold, running the projection-step-retraction iteration of Section 2.6

Table 3.2: The gradient optimiser against Nelder–Mead and COBYLA on the same objective from the same starting directions, three restarts each. Reported are the attained objective, the angle to the gradient solution, and the wall-clock cost as a multiple of the gradient optimiser's, each a median over three data seeds. The derivative-free solvers are given twenty times the iteration budget. Supervised data are the `cubic` configuration at $\sigma_\varepsilon = 0.5$, unsupervised data the factor model of Chapter 5 with a bimodal latent factor.

		gradient		Nelder–Mead		COBYLA	
n	p	value	cost	angle	cost	angle	cost
supervised, dCor^2							
200	5	0.6516	1×	0.09°	35×	0.09°	9×
200	20	0.6744	1×	6.06°	35×	0.37°	54×
200	50	0.6742	1×	45.24°	17×	5.71°	171×
500	5	0.6442	1×	0.08°	37×	0.08°	10×
500	20	0.6719	1×	3.74°	159×	0.22°	34×
500	50	0.6739	1×	41.23°	106×	1.68°	112×
unsupervised, dVar							
200	5	1.5382	1×	0.12°	117×	0.12°	3×
200	20	1.5755	1×	14.05°	1454×	0.31°	24×
200	50	1.6705	1×	18.58°	668×	0.49°	167×
500	5	1.5540	1×	0.11°	9×	0.10°	1×
500	20	1.5476	1×	5.53°	322×	0.30°	6×
500	50	1.6137	1×	24.27°	1043×	0.41°	19×

with the tangent projection (2.11) and the QR retraction (2.12). Figure 3.2 shows one iteration on the sphere, and Algorithm 3.1 states the single-direction procedure.

Method-specific choices. The two methods differ in three respects. First, the supervised optimiser uses $G - B\,(B^\top G)$ where the unsupervised one uses (2.11) exactly. It removes the entire column-space component of G and not merely its symmetric part, discarding the remainder $B\,\mathrm{skew}(B^\top G)$ with $\mathrm{skew}(M) = \frac{1}{2}(M - M^\top)$. What it retains is the horizontal part of the tangent space, along which the span of B changes; the discarded part rotates the frame inside the span already held. The iterates hence stay on $\mathrm{St}(p, k)$ and the ascent is a legitimate one, but it is an ascent over subspaces: the supervised optimiser moves the estimated subspace and leaves the basis within it at the orthonormal frame drawn at initialisation. For an objective invariant under $B \mapsto BQ$ this would be the exact Riemannian gradient. The objective (3.1) is not invariant under it, since it scores each column on its own, as Section 2.6 sets out, so the omission costs attainable objective value and the search over frames falls to the random restarts described below. The supervised accuracies reported in Chapters 4 and 6 do not depend on the frame, being principal angles between the estimated and the true subspace; the attained objective

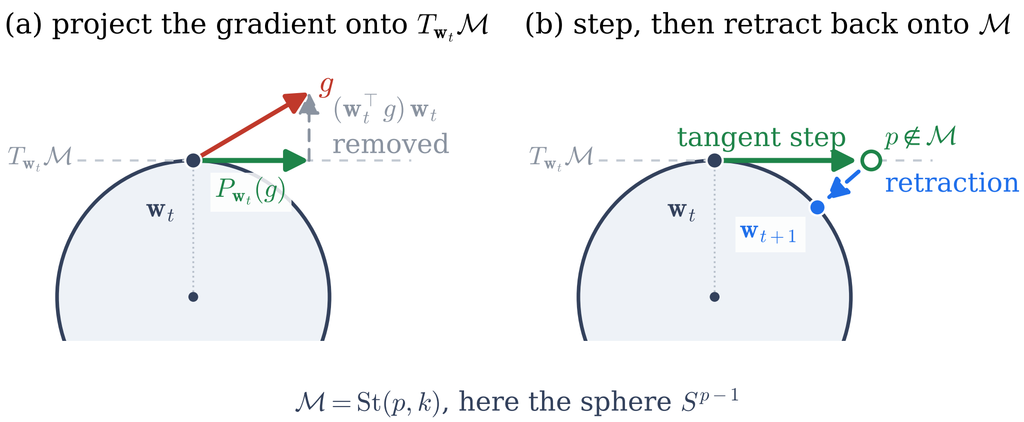

Figure 3.2: One iteration of the Riemannian optimiser on the sphere S^{p-1}. *(a)* The Euclidean gradient g is projected onto the tangent space $T_{\mathbf{w}_t}\mathcal{M}$, removing the radial component. *(b)* A step along the tangent direction lands off the manifold; the retraction maps it back by renormalisation. On the Stiefel manifold the renormalisation becomes a QR retraction and the projection becomes (2.11).

values do, and are the values the restricted search reaches. Second, the supervised optimiser cools the learning rate on a cosine schedule from $\mathrm{lr} = 0.05$ to $\mathrm{lr}_{\min} = 0.005$; the unsupervised one holds it fixed at 0.05. Third, the Adam step is norm-clipped to one in the supervised procedure, the tangent gradient in the unsupervised one. In both, Adam uses $\beta_1 = 0.9$, $\beta_2 = 0.999$, $\epsilon = 10^{-8}$, and convergence is declared when the index changes by less than 10^{-6} after a warm-up phase (twenty iterations supervised, ten unsupervised). Moment estimates are carried across iterates by re-projection rather than explicit vector transport, following Bécigneul and Ganea (2019).

Restarts. The objective is non-convex, so the optimiser is started from several random initial points (three to twenty, depending on the experiment) and the best result is retained. The spread of the attained objective across restarts serves as a diagnostic: a small spread indicates a reproducible, likely global solution; a large spread indicates a multi-modal landscape. Seeding the first restart from sliced inverse regression was tested and found, over ten data seeds, to improve the median angle by more than one degree in three of forty supervised configurations and to worsen it by more than a degree in two, the median gain across the grid being 0.00°; SIR reads the inverse conditional mean $\mathbb{E}[X \mid Y]$ and is blind to symmetric responses, which is exactly where the distance-correlation objective is not blind. The restarts are therefore left random.

Algorithm 3.1: Distance-index projection pursuit (single direction).

Input: data $X \in \mathbb{R}^{n \times p}$; index $I \in \{\mathrm{dCor}^2(Y, \cdot), \mathrm{dVar}(\cdot)\}$; restarts R; iterations T
Output: unit direction $\hat{w}$ maximising $I(Xw)$

if $I = \mathrm{dCor}^2$ **then**
  cache the U-centred response matrix $\tilde{B}$ and s_{yy}
end
for $r = 1$ **to** R **do**
  initialise w on the sphere (random, or PCA/OLS for $r = 1$)
  for $t = 1$ **to** T **do**
    $z \leftarrow Xw$; form D, S and the (U- or double-) centred matrix
    compute $I(z)$ and the Euclidean gradient g `// Eq. (3.14) or (3.17)`
    $g \leftarrow P_w(g)$ `// tangent projection, Eq. (2.11)`
    Adam moment update; $w \leftarrow w + \mathrm{step}$; $w \leftarrow w/\|w\|$
    if $|I(z) - I_{\mathrm{prev}}| < 10^{-6}$ after warm-up **then break**
  end
  keep w if its objective is the best so far
end
return best $\hat{w}$; report the objective spread across restarts

3.5 Deflation strategies

In order to obtain $k > 1$ directions, two options are at one's disposal: to extract them singly one after the other, with a deflation step in between, or to optimise the entire frame jointly. Four strategies cover these options, three of them deflations. We formulate them here once; their relative behaviour is then examined method by method (Sections 4.4 and 5.4.1).

X-deflation acts in the predictor space (weight space). As soon as we possess directions which span the columns of Q, we project the predictors onto the orthogonal complement, $X \leftarrow X - XQQ^\top$. This forces the next direction to be orthogonal to the previous ones in $\mathbb{R}^p$. The two methods implement this differently. The supervised method forms Q explicitly by a QR factorisation of the directions found so far and projects anew. The unsupervised method applies the equivalent iterative rank-one deflation $X \leftarrow X - (X\hat{w})\hat{w}^\top$ after each direction. The two agree in exact arithmetic, and the unsupervised method employs the iterative form for a practical and not for a theoretical reason: it is the cheaper in-place update, without explicit Q and without a complete re-projection, and it leaves the deflated X ready so that the next search starts from the directions already found and not from new random ones; this is a warm start which fits the parallel multi-start structure of this method. Its only price is an orthogonality drift of order 10^{-3}, quantified at the end of this section, and this drift lies far below the resolution of the subspace reconstruction metrics of Chapter 5, hence exerts no measurable influence on the accuracy; if orthonormality to machine precision is genuinely desired, the joint

Stiefel optimisation delivers it. In either case the step enforces $\hat{w}_i^\top \hat{w}_j = 0$, which on non-isotropic data leaves the projections themselves correlated, as Section 3.1 sets out.

Σ-*orthogonal deflation* performs the same step in the geometry the data impose rather than in the coordinate geometry of $\mathbb{R}^p$. Writing $\hat{\Sigma}$ for the sample predictor covariance and $\hat{\Sigma}^{-1/2}$ for its symmetric inverse square root, the search runs on the whitened predictors, X-deflation is applied there, and each direction is mapped back:

$$Z = X\hat{\Sigma}^{-1/2}, \qquad \hat{\beta} = \hat{\Sigma}^{-1/2} v, \qquad \hat{\beta}_i^\top \hat{\Sigma} \hat{\beta}_j = v_i^\top v_j. \tag{3.18}$$

The identity on the right is the whole construction: orthogonality of the whitened directions v_i is precisely $\hat{\Sigma}$-orthogonality of the recovered $\hat{\beta}_i$, and $\hat{\beta}_i^\top \hat{\Sigma} \hat{\beta}_j = 0$ states that the projections $X\hat{\beta}_i$ and $X\hat{\beta}_j$ are uncorrelated; that is the property the preceding paragraph leaves unsecured. The requirement is that $\hat{\Sigma}$ have full rank, which for $n > p$ holds almost surely. The constraint is not introduced here: the distance-covariance solver of Sheng and Yin (2016) searches in whitened coordinates and therefore imposes it throughout. Only the supervised method uses this variant, and Section 4.4.3 sets out what it is worth.

Y*-residual deflation* acts in the response space, in the tradition of projection pursuit regression (Friedman and Stuetzle, 1981). We remove the linear projection of the response variable onto the direction found, $\hat{z} = X\hat{w}/\|X\hat{w}\|$, by setting $\tilde{y} = Y - (Y^\top \hat{z})\,\hat{z}$, and the next search employs $\tilde{y}$. The problem herewith is that this removes only the *linear* part of the response variable along the direction found. As Section 4.4 shows, that is the wrong instrument for a nonlinear response variable.

Joint Stiefel optimisation avoids deflation altogether. It optimises all k columns of B at once, by way of (3.1) or its regularised form (3.2), using the tangent projection (2.11) and the QR retraction of Section 3.4. It attains orthonormality to machine precision, yet it is not free of charge. Every iteration now evaluates k projected scores and their k index gradients instead of one, which raises the dominant $O(n^2)$ effort per step by roughly a factor of k, and adds on top of this the $O(pk^2)$ QR retraction as well as the $k \times k$ symmetrisation of (2.11). In measured computing time this renders the joint optimisation at $k = 4$ approximately 1.5 times slower than the sequential extraction, the median over the five data seeds of the unsupervised multi-direction runs with $[1.1, 2.2]$ across them. What the additional effort brings in is orthonormality. The retraction holds $B^\top B = I$ to 6.9×10^{-16}, whereas sequential deflation accumulates over the same four directions a drift of 1.4×10^{-3}. Better reconstruction it does not bring. In neither of the two methods is there a configuration in which one strategy reconstructs the true subspace better than the other, across the five data seeds of each comparison (Sections 4.4.5 and 5.4.4).

3.6 The robust biloop index

Distance variance is built from pairwise distances $|z_i - z_j|$, which are unbounded: a single far-lying observation inflates many distances simultaneously and may dominate

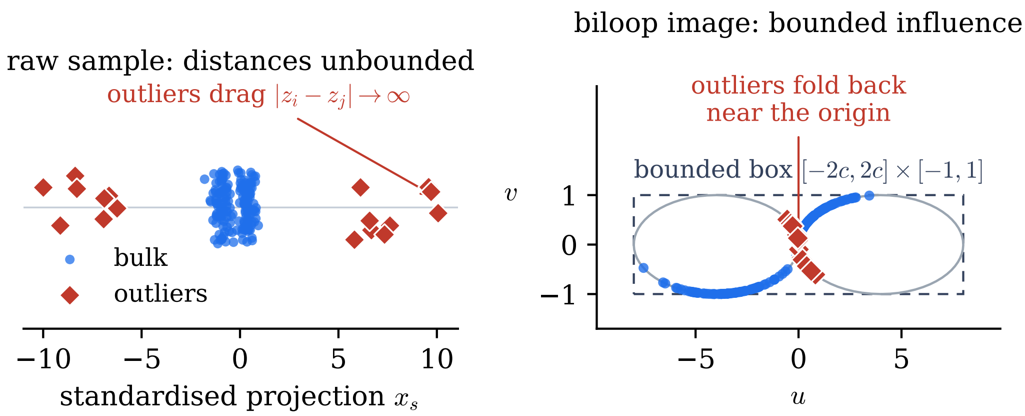

Figure 3.3: The biloop transform (3.19) on a contaminated sample. *Left:* raw standardised projection with outliers far from the bulk. *Right:* the biloop image; the bulk spreads around the two loops while extreme outliers fold back towards the origin. Hereby the sensitivity curve is bounded.

the index. The raw index already aggregates absolute rather than squared deviations and is therefore more robust than PCA or OLS, but its finite-sample sensitivity curve is unbounded and its breakdown value is zero (Leyder et al., 2026).

The robust variant replaces the raw projection with a bounded, redescending transformation before the index is computed. Following Leyder et al. (2026), the projected scores are standardised by their median and median absolute deviation, then mapped onto a bounded two-dimensional curve:

$$t = \tanh(x_s/c), \qquad (u, v) = \Big(\operatorname{sign}(x_s)\, c\,[\,1 + \cos(2\pi t \pm \pi)\,],\ \sin(2\pi t)\Big), \qquad c = 4, \tag{3.19}$$

where $\pm$ takes on the sign of x_s. As $|x_s|$ grows, $\tanh(x_s/c)$ saturates towards ± 1 and the curve carries the most extreme values back towards the origin instead of stretching them outward; the entire image lies within $[-2c, 2c] \times [-1, 1]$. The distance variance is then computed on the resulting two-dimensional points. Because the most extreme observations are folded inward, the influence of an observation decreases as it moves further away; this is the redescending behaviour that produces the robustness gains of Section 5.4.5. Since the image is bounded, the breakdown value of the index computed on it is furthermore strictly larger than zero, which the plain index does not attain (Leyder et al., 2026). Figure 3.3 shows the effect on a contaminated sample.

The transformation uses the median and absolute value and is therefore not differentiable; the analytic gradient does not apply, and the robust index falls back on central finite differences. One complete fit then costs a median of 54 times what the plain index costs, $[11, 102]$ across the twenty runs of Section 5.4.5 at $p = 30$, and it does so while running a third of the iteration cap, so the price per step is higher again. A central difference needs $2p$ evaluations of the index where the closed form needs one, and each of those evaluations is itself dearer here, the transformed points being two-dimensional. The saturation also flattens the objective surface, worsening reconstruction on clean data (Section 5.4.5), so the transformation is best reserved for the contaminated regime.

CHAPTER 4

Supervised Projection Pursuit via Distance Correlation

This chapter specifies the first of the two methods: a supervised projection pursuit that searches for the direction in predictor space whose projection is maximally dependent on a response. It is the distance-correlation specification of the optimisation machinery of Chapter 3, and the method is called *dCor-SDR* throughout. It is put to four questions in turn: the accuracy of single-direction recovery, the accuracy of recovery of several directions at once, the cost as the predictor dimension grows, and the behaviour on data whose true directions are unknown.

4.1 The objective and the estimators compared against

The projection-pursuit problem for a single direction is

$$\beta^\star = \underset{\|\beta\|=1}{\arg\max}\ \mathrm{dCor}^2(Y,\, X\beta), \tag{4.1}$$

the $k = 1$ instance of the unified objective (3.1) with $I(\cdot) = \mathrm{dCor}^2(Y, \cdot)$. The sample quantity the optimiser maximises is the U-centred, bias-corrected squared distance correlation of (3.9) and (3.15).

4.1.1 Three points of comparison

Three estimators run alongside dCor-SDR throughout the chapter. The first two are the classical baselines of the field and are described here; the third maximises the same population quantity as dCor-SDR and is set out separately in Section 4.1.2.

- **Ordinary least squares** is not a dimension-reduction method in the usual sense, but it is the most widely used statistical tool of all and serves as the reference point. Chapter 1 motivates the thesis through one of its known failure modes, a symmetric nonlinear response.

- **Sliced inverse regression** (Li, 1991) is the cheap standard baseline and the representative of the inverse-moment family. It estimates the central subspace from the *inverse* regression: the sample is sorted by the response and cut into slices, the predictor mean is taken within each slice, and the directions along which those slice means move away from the overall mean span the estimate. That is a whitening, a slicing and one eigendecomposition, which is why it costs milliseconds where the searches here cost seconds. Its blind spot is the one of least squares reached by another route: reading a *first* moment makes any dependence with a flat slice mean invisible, and under $Y = Z_1^2$ the observations with Z_1 large and positive and those with Z_1 large and negative share a response, fall in the same slice and cancel there, so every slice mean sits near the origin. Least squares is blind because $\mathrm{Cov}(X, Y) = 0$, sliced inverse regression because $\mathbb{E}[X \mid Y]$ does not move with Y.
- **The Sheng–Yin solver** (Sheng and Yin, 2016) is the closest existing method to the one developed here: it maximises distance dependence between the response and a linear projection of the predictors, which is the same population objective.

4.1.2 The Sheng–Yin solver

The comparison that runs through this chapter is a comparison against this solver, so the differences between the two are set out one at a time. There are six, and only two of them concern the search itself.

- **The population objective.** They maximise distance *covariance*, unsquared, where this chapter maximises squared distance *correlation*. Distance covariance is not scale-free, so a scale constraint on the directions has to be imposed alongside it, whereas the normalisation in (3.15) carries that constraint already.
- **The sample estimator.** Their criterion is the V-statistic, which carries the positive bias described in Section 2.1; the criterion here is the bias-corrected U-statistic. The two agree in the population and differ at finite n.
- **The coordinate system.** Their search runs on the whitened predictors $Z = (X - \bar{x})\hat{\Sigma}^{-1/2}$ and imposes its orthonormality constraint there, so the directions it returns satisfy $\hat{\beta}_i^\top \hat{\Sigma} \hat{\beta}_j = \delta_{ij}$: they are $\hat{\Sigma}$-orthogonal in the original scale rather than orthogonal as vectors.
- **Several directions at once.** For $k > 1$ the two no longer maximise the same population quantity. They score the projection as a whole, $\mathrm{dCov}(Y, B^\top X)$, and solve for all k columns simultaneously under $V^\top V = I_k$; (3.1) sums the marginal criteria $\mathrm{dCor}^2(Y, X\beta_j)$, one per column. The consequence reaches further than the choice of solver. Their objective depends on B only through $\mathrm{span}(B)$ and is unchanged

under $B \mapsto BQ$ with $Q^\top Q = I_k$, so orthonormality selects a representative of a subspace and restricts nothing, in the manner Section 2.6 describes; the objective here is not invariant under that rotation, and what it returns is one particular frame. Their procedure uses neither deflation nor sequential extraction, so it is a constrained optimisation over a frame rather than a projection pursuit in the strict sense.

- **The starting point.** They use a single start, chosen by scoring 500 orthonormalised random perturbations of an inverse-regression seed and keeping the best. The search here instead runs five independent restarts, each to convergence. Section 4.5 measures what each device buys, and Section 4.4.6 asks whether their choice of start is what puts them ahead at two directions. It is not: transplanted here it leaves recovery unchanged on one model and degrades it on another. There the ascent is led to a higher value of the sample criterion at a worse direction.

- **The optimiser.** They solve the constrained problem by sequential quadratic programming with finite-difference derivatives. This is the part the closed-form gradient of Chapter 3 replaces. A central-difference gradient costs $2p$ evaluations of the criterion where the closed form costs the equivalent of one, so the saving grows with the predictor dimension: with the objective, the manifold and the starting rule held equal, one complete fit takes 1.57 s against 34.4 s at $(n, p) = (500, 20)$, and across $p \in \{6, 20, 50, 100\}$ the ratio between the two methods widens from 2.8 to 56.5.

Neither of the two papers publishes its code. The implementation of their solver used here is a port of the MATLAB version their group distributes through the repository of Wu and Chen (2021); the code for every method and every experiment of this work is available in the repository accompanying the thesis.

4.2 Data-generating models and evaluation

4.2.1 The models

Every experiment in this chapter draws from one model class. We use the same framing as Sheng and Yin (2013): the response depends on the p predictors only through k linear combinations and on nothing else,

$$Y = g(\beta_1^\top X,\ \dots,\ \beta_k^\top X,\ \varepsilon) \tag{4.2}$$

This is the conditional-independence statement $Y \perp\!\!\!\perp X \mid B^\top X$ of (2.9) in Section 2.2. Here $B = [\beta_1, \dots, \beta_k] \in \mathbb{R}^{p\times k}$, k is the structural dimension of the central subspace $\mathcal{S} = \mathrm{col}(B)$ (Cook, 1998), and g is an arbitrary measurable function whose arguments include the noise ε. Recovering the regression means recovering $\mathcal{S}$, not any particular basis for it.

Two sub-specialisations of (4.2) matter for the experiments. In the first, the error is additive and independent of the predictors, $Y = f(B^\top X) + \sigma_\varepsilon\, \varepsilon$, $\varepsilon \sim \mathcal{N}(0,1)$: the noise level σ_ε is then a free parameter and the response surface f is identifiable. In the second, the response separates across directions and the projections are independent of one another,

$$Y = \sum_{j=1}^{k} f_j(\beta_j^\top X) + \sigma_\varepsilon\, \varepsilon, \qquad \beta_i^\top X \perp\!\!\!\perp \beta_j^\top X \;\; (i \neq j). \tag{4.3}$$

Under (4.3) a sum of per-direction scores is a faithful objective, because the response genuinely decomposes direction by direction. The configurations below are chosen to sit inside this class, on its boundary, and outside it.

Since distance correlation is invariant to orthogonal rotations of the predictors, coordinates may be rotated until the true directions are the first k canonical basis vectors, $B = [e_1, \dots, e_k]$. The informative projection is then the first k predictor coordinates, written Z_j for the j-th of them, and the remaining $p - k$ are pure noise. Recovery is then scored against the known $\mathrm{span}(e_1, \dots, e_k)$.

The configurations. Table 4.1 lists the seven response configurations used throughout the chapter's main comparison. Models A, B and C are from Sheng and Yin (2016), carried here through the restatement of Wu and Chen (2021). The direction vectors are $\beta_1, \beta_2, \beta_3$ with first six components $(1,0,0,0,0,0)$, $(0,1,0,0,0,0)$ and $(1,0.5,1,0,0,0)$ respectively, and zeros beyond; $\varepsilon, \varepsilon_1, \varepsilon_2$ are independent standard normal.

- Model **A**: $Y = (\beta_1^\top X)^2 + \beta_2^\top X + 0.1\,\varepsilon$, $k = 2$. The configuration mixes a symmetric term with a linear one in the same response.
- Model **B**: $Y = \mathrm{sign}(2\beta_1^\top X + \varepsilon_1) \log|2\beta_2^\top X + 4 + \varepsilon_2|$, $k = 2$. The response is a product of two direction-functions, not a sum, and its two errors sit inside g rather than being added to it. Here dependence lives in the interaction and not in the margins alone.
- Model **C**: $Y = \exp(\beta_3^\top X)\,\varepsilon$, $k = 1$. The conditional mean is $\mathbb{E}[Y \mid X] = 0$ identically, because the error is multiplicative and mean-zero. The whole dependence lives in the conditional variance, $\mathrm{Var}(Y \mid X) = \exp(2\beta_3^\top X)$, so no first-moment method can see it.
- **square**: $Y = Z_1^2 + \sigma_\varepsilon\, \varepsilon$, $k = 1$. Symmetric: the population covariance $\mathrm{Cov}(Z_1, Z_1^2) = 0$ vanishes although Y is fully determined by Z_1, so both OLS and SIR are structurally blind. The motivating example of Chapter 1.
- **cubic**: $Y = Z_1^3 + \sigma_\varepsilon\, \varepsilon$, $k = 1$. Monotone, so it retains a linear component and OLS remains competitive: the control case where the method has no structural advantage.

Table 4.1: The seven response configurations used throughout the chapter. Z_j denotes the j-th informative coordinate. Models A, B and C are from Sheng and Yin (2016) via the restatement of Wu and Chen (2021); the remaining four are standard test cases. *Scope* indicates whether a configuration is used in Section 4.3 (single-direction recovery), Section 4.4 (multi-direction recovery), or both.

Key	Response	k	Scope
Additive, independent indices (4.3)			
A	$Y = (\beta_1^\top X)^2 + \beta_2^\top X + 0.1\,\varepsilon$	2	both
`square`	$Y = Z_1^2 + \sigma_\varepsilon \varepsilon$	1	single
`cubic`	$Y = Z_1^3 + \sigma_\varepsilon \varepsilon$	1	single
`sum_squares`	$Y = \sum_j Z_j^2 + \sigma_\varepsilon \varepsilon$	≥ 2	multi
`product`	$Y = Z_1\, Z_2 + \sigma_\varepsilon \varepsilon$	2	multi
Error inside g, non-additive (4.2)			
B	$Y = \operatorname{sign}(2\beta_1^\top X + \varepsilon_1) \log \lvert 2\beta_2^\top X + 4 + \varepsilon_2 \rvert$	2	both
C	$Y = \exp(\beta_3^\top X)\,\varepsilon$	1	single

- **sum of squares**: $Y = \sum_{j=1}^{k} Z_j^2 + \sigma_\varepsilon\,\varepsilon,\ k \geq 2$. Rotationally symmetric in the informative subspace, so no single basis direction is identified and only the span is recoverable. It is one of the two multi-direction members of the additive class (4.3).

- **product**: $Y = Z_1\, Z_2 + \sigma_\varepsilon\,\varepsilon$, $k = 2$. The dependence lives in the interaction of two projections and in neither of them alone: $\operatorname{Cov}(Z_1, Z_1 Z_2) = 0$ when $Z_1 \perp\!\!\!\perp Z_2$, so neither OLS nor SIR can see either direction from its marginal. It is the additive-error counterpart of model B, which lets deflation behaviour be attributed to the response surface rather than to the internal errors.

The predictor distribution. The predictors are $X \sim \mathcal{N}(0, I_p)$ throughout. Nothing in the objective (4.1) requires that choice: it is a function of the pairwise distances between observations, so neither normality, nor continuity, nor the existence of a density enters its definition, and Sheng and Yin (2016) accordingly run their own models with uniform and with Poisson predictors. One moment condition does enter. The weight in (2.6) is chosen so that the integral is finite exactly for distributions with a finite first moment, and dCov is defined under $\mathbb{E}\|X\| < \infty$ and $\mathbb{E}|Y| < \infty$ (Székely et al., 2007). What is not required is a second moment, and this is the respect in which dependence-maximising methods ask less of the predictors than the inverse-moment and forward families (Sheng and Yin, 2013), which read a conditional mean and therefore need one to exist and to move. A single law is used below so that the comparison between the four estimators is not read off two grids at once.

4.2.2 Measuring recovery

Two lines through the origin, spanned by unit vectors $\hat{\beta}$ and e_1, differ by an angle whose cosine is their inner product, so the recovery angle is

$$\angle(\hat{\beta}, e_1) = \arccos|\hat{\beta}^\top e_1|, \tag{4.4}$$

taking the absolute value because $\hat{\beta}$ and $-\hat{\beta}$ define the same line and the same projection. An angle of 0° means $\hat{\beta} = \pm e_1$, that is perfect recovery, while 90° means the estimate is orthogonal to the truth and carries no signal.

For $k > 1$ the estimate and the truth are each a *subspace* i.e. a whole k-dimensional frame of directions, where the term angles needs to be clarified. A subspace is just the set of all linear combinations of a few directions: a line is a one-dimensional subspace, a plane through the origin a two-dimensional one, and the central subspace $\mathcal{S}$ a k-dimensional one. The standard way to compare them is the sequence of *principal angles* $0 \le \theta_1 \le \cdots \le \theta_k \le 90°$ between subspaces $\mathcal{U}$ and $\mathcal{V}$, defined recursively by finding the most aligned pair of directions first and then continuing:

$$\cos\theta_1 = \max_{\substack{u\in\mathcal{U},\, v\in\mathcal{V} \\ \|u\|=\|v\|=1}} u^\top v, \tag{4.5}$$

with the maximisers u_1, v_1 the first pair of *principal vectors*; each subsequent θ_m is obtained the same way, restricted to the directions orthogonal to the pairs already taken. In practice this is done via singular value decomposition: if the columns of $Q_\mathcal{U}$ and $Q_\mathcal{V}$ are orthonormal bases of the two subspaces, the singular values of $Q_\mathcal{U}^\top Q_\mathcal{V}$ are the cosines $\cos\theta_m$ (Edelman et al., 1998). Both the mean principal angle and the largest are reported (for $k = 1$ both are equal). The measure depends only on the two subspaces and not on the bases chosen for them, which matches the object being estimated, since the central subspace is defined only as a span; on the manifold of k-dimensional subspaces the principal angles are what the natural distance between two of them is built from (Absil et al., 2008), which is why they are the usual yardstick in the dimension-reduction literature.

Ordinary least squares returns a single direction whatever k is, and there is no inverse-moment extension of it to a larger subspace. Where the model has $k > 1$ its entry is therefore the one angle between the estimated line and the true plane, which is what (4.5) yields for a rank-one estimate.

Alongside the angle, the dCor^2 attained at the solution is recorded as a diagnostic, and for multi-direction runs the inter-direction $\mathrm{dCor}^2(X\hat{\beta}_i, X\hat{\beta}_j)$ as a measure of redundancy. The comparison rests on the recovery angle, which no method optimises.

4.3 Single-direction recovery

Every method in this section extracts a single direction ($k = 1$), regardless of the model's true structural dimension. For models whose central subspace has $k > 1$ (A, B and

Table 4.2: Single-direction recovery ($n = 500$, five seeds, five restarts per direction). *Angle* is the median principal angle to the true subspace in degrees. dCor^2 is the bias-corrected squared distance correlation attained at the solution (a diagnostic, not a comparison criterion). Square and cubic use $\sigma_\varepsilon = 0.5$.

	OLS		SIR		Sheng–Yin		dCor-SDR	
$p = 6$								
A (k=2)	8.9	0.404	36.1	0.225	3.4	0.277	3.9	0.313
B (k=2)	6.7	0.399	6.8	0.245	4.3	0.258	4.0	0.268
C (k=1)	67.5	0.040	5.4	0.278	3.3	0.278	4.1	0.279
square (k=1)	64.5	0.015	64.8	0.023	2.2	0.237	2.6	0.237
cubic (k=1)	2.4	0.653	4.7	0.659	2.4	0.660	3.6	0.660
$p = 20$								
A (k=2)	17.5	0.384	48.0	0.207	7.6	0.283	9.5	0.298
B (k=2)	13.1	0.432	13.5	0.256	11.3	0.259	10.7	0.265
C (k=1)	48.6	0.125	9.9	0.273	10.1	0.288	10.2	0.290
square (k=1)	61.3	0.032	79.5	0.011	5.5	0.228	6.5	0.228
cubic (k=1)	9.5	0.661	7.9	0.662	4.8	0.672	7.4	0.675

`sum_squares`) the recovered direction is scored against the true *subspace*: the principal-angle machinery of (4.5) returns the one angle between the estimated line and the true plane, which is the closest the line can come to the subspace.

Five configurations appear here: models A, B and C, `square` and `cubic`. The remaining one, `sum_squares`, is rotationally symmetric in a two-dimensional subspace, so a single direction is not the object it defines, and it enters in Section 4.4 only. Every model is run at $n = 500$ at both $p = 6$ and $p = 20$, at five data seeds, with five restarts per direction. Table 4.2 reports the median recovery angle over those seeds and the attained dCor^2.

Discussion. The blindness is a property of the response and of the predictor law together, not of the estimators alone, and it is worth naming which property. What fails on `square` is $\mathrm{Cov}(Z_1, Z_1^2) = \mathbb{E}[Z_1^3]$, and that third moment vanishes for every predictor law symmetric about its mean. The normal law used here is one such law, so the two classical estimators would be equally blind under any other symmetric choice, and would recover the direction under an asymmetric one.

Between the two distance-based estimators the margin is narrow and its sign depends on the model. At $p = 20$ dCor-SDR is ahead on model B (10.7° against 11.3°) and level on model C (10.2° against 10.1°), and behind on model A (9.5° against 7.6°), on `cubic` (7.4° against 4.8°) and on `square` (6.5° against 5.5°); at $p = 6$ the same pattern appears with every angle smaller. Comparing the two on each data set against itself separates the differences the five seeds support from those they do not. At $p = 20$ the paired difference is $+1.72°$ on model A, $+2.67°$ on `cubic` and $+0.88°$ on `square`, in each case with all five seeds on the same side of zero, so those three are differences. On model B it is $-0.66°$ over a range from $-2.09°$ to $+1.13°$ and on model C $-0.30°$ over $[-0.78°, +0.01°]$, and neither is. The largest margin anywhere in the table is under 3°, and both methods stay within a few degrees of the truth wherever the classical pair is lost. Accuracy is therefore not what separates the two searches, and Section 4.5 takes up what does.

4.4 Recovering several directions

Recovering a single direction is, in the favourable regimes above, a solved problem. The difficulty begins when the signal lives in a subspace of dimension $k > 1$ and the directions have to be extracted one at a time. Each extraction needs a way to remove the signal already found so that the next search looks elsewhere (this is *deflation*), and it is here that the supervised objective meets its first structural wall.

4.4.1 The grid and the results

When the structural dimension is $k > 1$, every method must extract a full set of directions. OLS returns a single direction regardless of k, so its entry is the angle between that line and the true plane. For SIR the full set is the leading k eigenvectors of the inverse-regression estimate. The Sheng–Yin solver searches for all k directions in its whitened coordinate system, imposing $\hat{\Sigma}$-orthogonality between them, and dCor-SDR extracts directions sequentially by X-deflation (Section 3.5): the first direction maximises $\mathrm{dCor}^2(Y, X\beta)$, then the predictors are projected orthogonally to $\hat{\beta}_1$ before the next search.

The grid differs from the single-direction one in two respects. First, only the four models with $k = 2$ appear: A, B, `sum_squares` and `product`; the three single-index configurations (`square`, `cubic`, C) cannot produce a second direction by construction. Second, recovery is scored against the full $k = 2$ subspace: the mean and maximum principal angles of (4.5) replace the single angle of (4.4). Everything else carries over from the single-direction grid: $n = 500$, $p \in \{6, 20\}$, normal predictors, five data seeds.

dCor-SDR appears in two variants: the *sequential* search extracts directions one at a time by X-deflation; the *joint* search optimises both columns simultaneously on the Stiefel manifold $\mathrm{St}(p, 2)$ with $\lambda = 0$ (Section 4.4.5). The joint search receives 20 restarts against the sequential search's 5 per direction, on stability grounds rather than a concession to either method, and even at $p = 20$ this keeps its 6.5 s per fit below the Sheng–Yin solver's 13.0 s. Table 4.3 reports the median recovery angles; each entry gives $\bar{\theta}/\theta_{\max}$, the mean and maximum principal angle.

The split between the classical and the distance-based estimators carries over from single-direction recovery without change. On `sum_squares` the classical methods return subspaces at 34.8° to 77.6° from the truth, while both distance-based methods stay below 11° at both sizes; the rotational symmetry of that response is the same obstacle as before, now spread over a plane. Model B remains the configuration where all four methods are close, and model A the one that separates the two classical estimators. Two comparisons in this table are left until the deflation machinery that produces them has been set out:

Table 4.3: Multi-direction recovery ($k = 2$) at $n = 500$, five seeds. The sequential arm uses five restarts per direction, the joint arm twenty. Each entry gives $\bar{\theta}/\theta_{\max}$ in degrees: the median (over seeds) of the mean and maximum principal angle to the true subspace. OLS returns a single direction; its entry is the angle between that direction and the true plane ($\bar{\theta} = \theta_{\max}$). Sequential dCor-SDR uses X-deflation; joint dCor-SDR optimises both columns on $\mathrm{St}(p, 2)$ at $\lambda = 0$.

	OLS	SIR	Sheng–Yin	dCor-SDR (seq.)	dCor-SDR (joint)
$p = 6$					
A	5.0 / 5.0	36.4 / 69.8	2.3 / 2.9	3.2 / 3.7	3.2 / 3.9
B	7.0 / 7.0	5.6 / 8.0	4.7 / 6.4	5.3 / 6.7	7.0 / 11.1
sum_sq.	34.8 / 34.8	63.9 / 77.2	3.8 / 5.0	4.1 / 5.4	3.9 / 5.1
product	48.7 / 48.7	60.4 / 72.3	4.9 / 7.8	5.7 / 9.6	5.2 / 9.4
$p = 20$					
A	16.4 / 16.4	47.1 / 84.8	7.2 / 8.0	8.8 / 9.8	9.4 / 11.9
B	16.2 / 16.2	14.4 / 18.3	11.0 / 14.5	12.3 / 14.7	17.9 / 22.0
sum_sq.	67.7 / 67.7	77.6 / 84.6	8.9 / 9.6	10.2 / 12.1	10.4 / 12.5
product	68.1 / 68.1	76.4 / 83.6	9.8 / 11.3	11.2 / 12.9	12.9 / 15.8

the margin between the Sheng–Yin solver and dCor-SDR, taken up in Section 4.4.6, and the two dCor-SDR variants against each other, taken up in Section 4.4.5.

4.4.2 What deflation has to achieve

What deflation must achieve is best stated through three sets of admissible direction pairs. Writing $z_1 = X\hat{\beta}_1$ and $z_2 = X\hat{\beta}_2$ for the two projections and Σ for the predictor covariance, let

$$\underbrace{\mathcal{O} = \{\hat{\beta}_1^\top \hat{\beta}_2 = 0\}}_{\text{orthogonal weights}}, \qquad \underbrace{\mathcal{U} = \{\hat{\beta}_1^\top \Sigma \hat{\beta}_2 = 0\}}_{\text{uncorrelated projections}}, \qquad \underbrace{\mathcal{I} = \{z_1 \perp\!\!\!\perp z_2\}}_{\text{independent projections}}, \qquad \mathcal{U} \supset \mathcal{I}, \tag{4.6}$$

so that independence is the most demanding of the three and uncorrelatedness the weaker consequence of it. The first set is of a different kind: $\mathcal{O}$ constrains the two *coefficient vectors* and says nothing about the data, and it coincides with $\mathcal{U}$ exactly when $\Sigma \propto I$. For any other predictor covariance the two sets merely intersect, and a pair may lie in $\mathcal{O}$ while its projections are strongly correlated. What $\hat{\beta}_1^\top \Sigma \hat{\beta}_2 = 0$ does say is that the two directions are orthogonal in the geometry the data impose, that is with respect to Σ; it does not say that the projections are independent, since a vanishing covariance constrains one second moment and leaves every higher one free.

Distance correlation cares about $\mathcal{I}$: by (2.8), two directions are redundant exactly when their projections are *dependent*. No linear deflation can deliver $\mathcal{I}$, for reasons Section 4.4.4 makes precise, so the practical question is which of the two weaker conditions a strategy should enforce, and what the shortfall costs.

Two remarks locate this against the predecessor. Sheng and Yin (2016) maximise distance *covariance* rather than distance correlation, which is not scale-free, so their formulation must impose a scale constraint on the directions separately; the normalisation in (3.15) carries that constraint already, which is one reason for preferring the correlation here. And their solver searches in whitened coordinates, so the orthonormality it imposes is $\hat{\beta}_i^\top \Sigma \hat{\beta}_j = \delta_{ij}$, membership of $\mathcal{U}$ rather than of $\mathcal{O}$. That is exactly the second strategy below, and it is borrowed from them rather than invented here.

4.4.3 Two workable strategies

Both strategies that this section recommends work in predictor space and differ only in the geometry they orthogonalise in. The first is the X-deflation of the shared machinery (Section 3.5), which makes every later direction orthogonal to its predecessors as a vector and so places the pair in $\mathcal{O}$. The second is its whitened form, introduced here.

Σ-*orthogonal deflation* orthogonalises in the geometry the data impose. Writing $\hat{\Sigma}^{-1/2}$ for the symmetric inverse square root of the sample predictor covariance, the search runs on the whitened predictors $Z = X\hat{\Sigma}^{-1/2}$, ordinary deflation is applied there, and the direction is mapped back as $\hat{\beta} = \hat{\Sigma}^{-1/2} v$. Since $\hat{\beta}_i^\top \hat{\Sigma} \hat{\beta}_j = v_i^\top v_j$, orthogonality in the whitened coordinates is precisely $\hat{\Sigma}$-orthogonality in the original ones, so the pair lies in $\mathcal{U}$: the recovered projections are uncorrelated. The one requirement is that $\hat{\Sigma}$ be of full rank, which for $n > p$ holds almost surely and which the implementation enforces by flooring the eigenvalues.

The difference between the two is invisible under isotropic predictors and is the whole story otherwise. A two-variable case makes it concrete. Take $p = 2$ with

$$\Sigma = \begin{pmatrix} 1 & 0.8 \\ 0.8 & 1 \end{pmatrix}, \qquad \hat{\beta}_1 = e_1, \quad \hat{\beta}_2 = e_2. \tag{4.7}$$

The weights are exactly orthogonal, yet the projections are $z_1 = X_1$ and $z_2 = X_2$ with $\mathrm{Cov}(z_1, z_2) = \hat{\beta}_1^\top \Sigma \hat{\beta}_2 = 0.8$: two "distinct" directions whose projections are correlated at 0.8, and hence dependent. Σ-orthogonal deflation returns instead the pair $\hat{\beta}_2 \propto \Sigma^{-1} e_2$, for which the projected covariance is zero by construction. This is the cheap fix for exactly this leak, and the experiment below measures what it is worth beyond the constructed example.

Table 4.4 runs both strategies, and the Y-residual strategy discussed in the remark below, over three `sum_squares` configurations, each with isotropic predictors and again with the AR(1) covariance $\Sigma_{ij} = 0.5^{|i-j|}$ used by Sheng and Yin (2016), at five data seeds. Two columns are reported: the mean principal angle to the true subspace, and the inter-direction redundancy $\mathrm{dCor}^2(z_1, z_2)$, which is the diagnostic the hierarchy calls for.

Three things come out of the experiment. First, with isotropic predictors the two predictor-space strategies are the same method, as they must be: their median angles agree to within 0.3° at every configuration, running between 6.5° and 7.4°, against 12.9°

Table 4.4: The three deflation strategies on the `sum_squares` response at $k > 1$, median over five data seeds. *Angle* is the mean principal angle to the true subspace in degrees, smaller being better; *redundancy* is the inter-direction dCor^2 of the recovered pair, smaller meaning the two directions carry less of the same information. Predictors carry the AR(1) covariance $\Sigma_{ij} = 0.5^{|i-j|}$, under which orthogonal weights and uncorrelated projections do not coincide; the isotropic case, under which the first two strategies coincide, is reported in the text.

	Angle (deg)			Redundancy		
Configuration	X-defl.	Σ-orth.	Y-resid.	X-defl.	Σ-orth.	Y-resid.
`sum_squares`, k=2, p=10	33.7	10.4	24.0	0.013	0.008	0.999
`sum_squares`, k=2, p=10 (noisy)	18.6	13.0	25.9	0.051	0.005	0.999
`sum_squares`, k=3, p=10	24.2	21.5	35.1	0.077	0.020	0.999

to 26.7° for Y-residual deflation. What small differences remain between the two come from the sample covariance not being exactly the identity.

Second, correlated predictors cost every strategy accuracy, and they cost X-deflation the most. At $k = 2$, $p = 10$ the median angle under X-deflation rises from 6.5° to 33.7° when the AR(1) covariance is switched on, while Σ-orthogonal deflation rises only to 10.4°; the five-seed ranges are $[8.0, 46.6]$ against $[6.9, 20.5]$, so the worse half of the X-deflation runs lies beyond anything the whitened geometry produces. The rotational symmetry of the response is what makes the configuration sensitive: when every direction in the informative plane carries the same marginal dependence, the second search is steered by the deflation and by little else, and a deflation in the wrong geometry steers it wrongly.

Third, Σ-orthogonal deflation delivers what it was constructed to deliver. Under correlated predictors its median redundancy is at or below 0.020 at every configuration, against up to 0.077 for X-deflation. That is the leak of (4.7) closing. Neither strategy reaches independence: the residual redundancy is small but not zero, which is the shortfall Section 4.4.4 shows to be unavoidable for any linear operator.

The recommendation that follows is Σ-orthogonal deflation whenever $\hat{\Sigma}$ is of full rank, which costs one eigendecomposition. The evidence behind it covers one response family at one predictor dimension, so it is a default rather than a general result; what supports it beyond these three configurations is the argument of (4.7), which does not depend on the response at all.

Y-residual deflation looks simpler and fails. A third strategy suggests itself and is worth one paragraph, because it is the one the projection-pursuit regression of Friedman and Stuetzle (1981) uses. Rather than touching the predictors, it removes from the response its least-squares projection on the direction just found, passing the residual $\tilde{y} = Y - (Y^\top u)\, u$, $u = z_1/\|z_1\|$, to the next search. The trouble is that the residual is linear while the signal is not. Removing the projection on z_1 makes $\tilde{y}$ uncorrelated with z_1, and a vanishing covariance certifies nothing about dependence: with $u \sim \text{Uniform}(-1, 1)$ and $w = u^2$ one has $\text{Cov}(u, w) = 0$ while w is a deterministic function of u. In the symmetric case the failure is total, since $\text{Cov}(Y, z_1) = 0$ already (the blindness of (1.2)), so the residual is the unchanged response and the next search returns the same direction. The measurements agree: the linear deflation check $|\,\text{corr}(\tilde{y}, z_1)| \approx 0$ passes to numerical precision, while the inter-direction dCor^2 of the recovered pair sits at 0.999 or above on all three configurations. On the criterion the objective is built from, the two recovered directions carry the same information. A nonparametric residualiser would remove nonlinear structure that a linear one cannot, which is what Friedman and Stuetzle (1981) do, at the price of the model selection a smoother brings.

4.4.4 The obstacle all three share

Every deflation strategy meets one obstacle that no amount of tuning removes, and it can be stated precisely in terms of (4.6). *All three strategies are linear operators*: a projection of X in two cases, a projection of Y in the third. A linear operator can enforce membership of $\mathcal{O}$ or of $\mathcal{U}$, both second-order conditions. But the condition the objective requires, $\mathcal{I}$, is a statement about all higher moments at once, and *no linear operation can remove higher-order dependence without knowing the functional form f.*

This can be shown directly. Take Y-residual deflation with a response of the additive form $Y = f(z_1)$. Because $z_1 = X\hat{\beta}_1$ with $X \sim \mathcal{N}(0, I_p)$ and $\|\hat{\beta}_1\| = 1$, the projection z_1 is standard Gaussian, so any square-integrable f expands in the probabilists' Hermite polynomials He_m, which are orthogonal under that law,

$$f(z_1) = \sum_{m \geq 0} a_m \, \text{He}_m(z_1), \qquad a_1 = \mathbb{E}[f(z_1)\, z_1] = \text{Cov}(Y, z_1). \tag{4.8}$$

The linear residual removes only the degree-one term (the constant goes with centring), leaving $\tilde{y} = \sum_{m \geq 2} a_m \text{He}_m(z_1)$, which is still a genuine function of z_1 whenever any a_m, $m \geq 2$, is non-zero, so $\text{dCor}(\tilde{y}, z_1) > 0$. Removing *all* the dependence would mean subtracting $\mathbb{E}[Y \mid z_1] = f(z_1)$ itself, a nonlinear function a linear operator cannot form without already knowing f. The same accounting applies to X-deflation, whose only lever is the projected covariance $\hat{\beta}_i^\top \Sigma \hat{\beta}_j$. A linear operator supplies one second-order lever per removed direction; projected independence imposes infinitely many higher-order conditions, so the tool is one order too coarse for the objective.

The mismatch is between the order of the tool, linear and second-order, and the order of the dependence distance correlation is built to detect, which is arbitrary; the particular implementations are not at fault. It surfaces in two ways: Y-residual deflation leaves projected dependence essentially intact, while X-deflation and its Σ-orthogonal variant reach a second-order condition and stop there. The natural remedy is to deflate *nonlinearly*, with an operator rich enough to reach $\mathcal{I}$; kernel-ridge residualisation is the obvious candidate and is set out as future work in Section 7.3.

A second limitation is worth naming here because it is not the same one, and Section 4.7 returns to it. Every method in this section optimises a *marginal* dependence, one that scores a single projected direction at a time, and never the dependence of the response on the recovered subspace as a whole,

$$\mathrm{dCor}^2(Y,\, XB), \qquad XB = (X\beta_1, \ldots, X\beta_k), \tag{4.9}$$

which by (2.8) vanishes exactly when the whole subspace is uninformative. Interaction structure can live in the combination of directions while being invisible to any direction alone, and the rotationally symmetric response $Y = Z_1^2 + Z_2^2$ is the clearest case: every direction in the true plane carries the same marginal dependence, so a marginal objective cannot tell one orthonormal pair of axes from another, whereas (4.9) fixes the plane at once.

4.4.5 Joint versus sequential extraction

Sequential deflation is not the only way to obtain k directions. The alternative is to optimise all of them at once on the Stiefel manifold $\mathrm{St}(p, k) = \{B \in \mathbb{R}^{p\times k} : B^\top B = I_k\}$, in words every choice of k mutually orthogonal unit directions, considered simultaneously rather than one after another, thereby maximising the summed dependence with a penalty that discourages the columns from agreeing,

$$\max_{B\in\mathrm{St}(p,k)} \sum_{j=1}^{k} \mathrm{dCor}^2\big(Y, X\beta_j\big) \;-\; \lambda \sum_{i<j} \mathrm{dCor}^2\big(X\beta_i, X\beta_j\big). \tag{4.10}$$

the supervised instance of the penalised objective (3.2). The signal term is a sum of per-column dependences; the penalty term reuses the same analytical gradient, treating one projection as the “response” of another. Optimisation is by Riemannian Adam with a QR retraction onto the manifold (Chapter 3). The Sheng–Yin solver is joint in the sense that it also solves for all k columns at once, but it carries no penalty term of this kind: its only anti-redundancy device is the hard constraint $V^\top V = I_k$ in the whitened coordinates, enforced by the sequential quadratic programme itself. The two arrangements therefore differ in what keeps the columns apart, a constraint alone there against a constraint plus an optional penalty here, and $\lambda = 0$ is the default used throughout unless a value is stated. The joint column of Table 4.3 runs it against sequential deflation on the same data, and a separate run at $p = 5$ and $p = 10$ on the `sum_squares` response repeats the comparison from the same random initial directions over five data seeds.

On recovery the experiment does not separate the two strategies where the problem is small. On `sum_squares` the medians are 6.24° sequential against 5.85° joint at $p = 5$, and 8.55° against 8.76° at $p = 10$, with the seed ranges $[3.06, 8.64]$ against $[2.94, 8.69]$ and $[8.08, 10.37]$ against $[8.35, 11.31]$ overlapping almost completely. Paired on each data set against itself, none of the four configurations puts the difference on one side of zero across the five seeds: the paired medians run from −0.45° to +0.26° and every range straddles zero. The grid of Section 4.4.1 agrees at $p = 6$, where no model separates the two by more than 1.7°. That is the expected picture at small k: a greedy method pays for its greed only once it has committed to several directions, so the deflation-based route is competitive exactly where few directions are wanted, and the configurations here stop at $k = 3$. Whether the ordering changes at larger k is not tested by this experiment.

At $p = 20$ the joint search becomes the less reliable of the two on one model. On B its median angle is 17.9° against 12.3° for sequential deflation, a paired difference of +5.02° with all five seeds on the same side of zero, $[4.22, 8.31]$. On the other three models the paired difference stays inside the seed spread, but `product` shows how the failure arrives: its paired median is +0.36° and its range reaches +31.0°, one seed on which the joint search lands somewhere else entirely. The Stiefel manifold $\mathrm{St}(20, 2)$ is a harder landscape than two successive sphere searches, and the twenty-restart budget does not always suffice for a frame of two where five per direction suffices for one at a time.

What the joint formulation does change is the redundancy between the recovered directions. Already at $\lambda = 0$ its two columns share almost no information: the inter-direction dCor^2 has median 0.000 and 0.002 and never exceeds 0.009 on the two `sum_squares` configurations, against medians of 0.071 and 0.149 for sequential deflation there. Optimising the columns together therefore separates them without any penalty being applied. The second difference is the exact orthonormality of the retraction, quantified in Section 3.5; here it costs about six times the runtime of the sequential search, 2.30 s against 0.38 s at $p = 10$.

This leaves the penalty weight λ with little to do, and the sweep confirms it. Across $\lambda \in \{0.1, 0.5, 1, 2\}$ and all four configurations, not one paired difference against $\lambda = 0$ falls on a single side of zero over the five seeds. The largest paired median is $-1.27°$, on the noisy `product` configuration at $\lambda = 2$, and its range is $[-2.47°, +0.26°]$. That is what a penalty with nothing to remove would produce: where the inter-direction dependence is already below 0.01, the term has no redundancy left to act on, and what it changes is inside the sampling noise.

Joint optimisation is therefore neither a remedy for the deflation obstacle of Section 4.4.4 nor an improvement in recovery over sequential deflation. What speaks for it is orthonormality to machine precision and directions that are close to independent without a penalty being set.

A third strategy sits between the two: extract $\hat{\beta}_1$, then $\hat{\beta}_2$ under a constraint against it, and so on, as the distance-covariance independent component analysis of Matteson and Tsay (2017) does. What distinguishes it is what the constraint means. Their target is $\mathcal{I}$ of (4.6), projections that are *independent* rather than merely orthogonal or uncorrelated, which is the condition the objective needs and the one no linear deflation reaches. It is not run separately here because (4.10) already contains it: the penalty term is precisely their ICA criterion, so the joint problem is its penalised relaxation, and the λ sweep above measures what that relaxation buys. On these configurations it buys nothing, since the redundancy it would remove is already near zero.

4.4.6 The margin against the Sheng–Yin solver

Where the single-direction comparison of Section 4.3 splits by model, the multi-direction one does not. At $p = 20$ under normal predictors their joint constrained solve is ahead on three of the four two-direction models when the two are paired on each data set against itself: by 2.44° on `sum_squares` with all five seeds on the same side of zero $[2.04, 2.69]$, by 1.55° on A $[1.20, 2.21]$ and by 1.09° on `product` $[1.00, 2.68]$. On B the two are level, the paired median being $-0.01°$ with the seeds falling on both sides, $[-1.38, 0.67]$. At $p = 6$ the margin is under 1° everywhere.

The one structural difference between the two procedures is the one named in Section 4.1.2: they solve for both columns at once under a constraint, where dCor-SDR extracts one direction and then deflates. The other candidate explanations were tested and none of them accounts for the margin. Matching their sample criterion, adopting their whitened coordinates and raising the restart budget from five to twenty each leave the margin where it was. Seeding the search from an inverse regression does worse than that, and how it fails is worth recording. At $k = 2$ and $p = 20$, with two of the five restarts spent on a sliced inverse regression and a sliced average variance estimate and the best of the five kept by attained objective, the mean principal angle on A is unchanged, 9.53° against 9.17° for random starts, while on B it degrades from 12.52° to 34.29°. The seeded arm is not failing to find a maximum. On three of the five seeds it attains a *higher* sample objective than the random arm at a much worse direction, 0.467 against 0.293 at 34.3° against 12.5° being the largest of the three, and on the other two it attains a lower objective at the same angle. Model B therefore carries a maximum of the sample criterion away from the true subspace which the inverse-regression start reaches and the random starts mostly do not, and best-of-restarts selection, which can only compare objective values, prefers it. That locates the difficulty in the criterion rather than in the search, and it is a caution against informative initialisation on this model rather than an explanation of the margin. Whitening is the only one of the four devices that moves anything at $k = 2$, and it moves it slightly: 11.49° against 12.52° on B and 9.01° against 9.17° on A. The joint Stiefel search of Section 4.4.5 is the obvious way to remove the remaining structural difference, and it does not: at $p = 20$ it is less reliable than the sequential search it would replace. The margin is small and consistent, and what causes it has not been established here.

4.5 Cost in the predictor dimension

This section measures what the closed-form gradient of Chapter 3 was built for. A finite-difference gradient has to re-evaluate the objective once per parameter, so a single constrained solve at p predictors and k directions spends $O(pk)$ objective evaluations per step, k being the number of directions solved for at once. Every timing in this section fixes $k = 1$, so the count there is $O(p)$. The closed-form gradient costs one index evaluation whatever p is, plus the $O(np)$ contraction that carries the gradient in the projected sample back to the gradient in β. The advantage should therefore grow with the predictor dimension, and should be small where the predictor dimension is small.

Two prior questions have to be settled before a cost comparison means anything, because a search that is cheaper by being worse is not cheaper. The first is whether the two searches differ in what they find when the problem is held fixed. Driving their SQP solve and Riemannian Adam on the *same* whitened problem with the *same* U-statistic objective, and comparing them on each data set against itself, the median paired difference in recovery angle is 0.00° on the single-index grid at $p = 20$ and −0.03° on model A at $(n, p) = (500, 20)$, with Adam ahead in 49% and 58% of configurations, and the attained dCor^2 agreeing to five decimal places in median on the grid. What differs is the price: 0.13 s against 0.30 s per fit at $p = 20$, and 1.57 s against 34.41 s at the larger size, a factor of twenty-two. Whatever accuracy difference remains between the two published methods therefore belongs to the sample criterion each maximises and not to the search that maximises it.

The second question is whether the two are being run at comparable budgets. Their solver spends its budget on one good starting point, 500 orthonormalised perturbations of an inverse-regression seed scored at one objective evaluation each, where the search here spends it on five independent ascents, and that difference is itself a consequence of the gradient: restarting a solve whose every step costs $O(pk)$ evaluations is expensive in a way that restarting a closed-form-gradient ascent is not. Stripping each device out and measuring it shows that the two buy the same thing. Starting their solver from a single random orthonormal point raises its median angle on the grid at $p = 20$ from 12.55° to 17.01°, and restricting this chapter's search to one start raises its median from 13.57° to 17.97°; but paired on each data set against itself neither device improves the typical fit, the median paired difference over the 219 configurations being 0.00° for their seed and 0.00° for the five restarts. What both do is prevent rare total failures: the paired *mean* differences are −4.1° and −6.2°, with worst cases of −72.5° and −74.1°, so on a few configurations a single unlucky start lands the search on the wrong side of the sphere and each device insures against that. The perturbation stage adds nothing once the seed is in place: its paired median difference is 0.00°, and at $(n, p) = (500, 20)$ the seeded and the perturbed arms agree to two decimals of a degree.

Neither the objective nor the budget, then, separates the two searches, which leaves the price of the search itself.

The two solvers cost the same at $(n, p) = (100, 6)$, 0.06 s against 0.06 s per fit, and are a factor of fourteen apart at $(500, 20)$, 14.3 s against 1.00 s. Two points give the direction of the trend without giving its shape, and $p = 20$ is where the chapter's other experiments stop. An extension in the predictor dimension supplies the curve. Model A, part (1), is run at $n = 500$ and $p \in \{6, 20, 50, 100\}$, five replicates, all four estimators; this is outside the design of Sheng and Yin (2016), whose largest predictor dimension is 20. Table 4.5 gives the numbers and Figure 4.1 puts cost and accuracy side by side.

The cost curves have the shape the derivation predicts. A finite-difference gradient must retake the differences along each of the pk parameters, so their solve grows with the

Table 4.5: Cost and accuracy against the predictor dimension, on model A part (1) at $n = 500$: median over five replicates with the range across them in brackets. *Seconds* is one complete fit, including the initialisation each method performs. *Angle* is the mean principal angle to the true plane in degrees.

	$p = 6$	$p = 20$	$p = 50$	$p = 100$
Seconds per fit				
OLS, SIR	0.000	0.000	0.001	0.002
Sheng–Yin	2.51 $[2.5, 3.1]$	13.16 $[11.2, 15.8]$	37.98 $[28.3, 76.7]$	83.92 $[70.4, 287.4]$
dCor-SDR	0.89 $[0.8, 0.9]$	1.07 $[1.0, 1.1]$	1.33 $[1.2, 1.3]$	1.49 $[1.3, 1.5]$
ratio	2.8	12.3	28.5	56.5
Mean principal angle (deg)				
OLS	6.7	16.8	31.7	41.6
SIR	35.3	43.2	52.5	57.2
Sheng–Yin	3.5 $[1.8, 4.7]$	8.1 $[6.3, 9.0]$	15.8 $[12.6, 47.9]$	58.0 $[23.3, 59.9]$
dCor-SDR	5.4 $[3.6, 6.6]$	11.0 $[9.4, 11.6]$	17.7 $[14.7, 26.1]$	48.3 $[28.3, 59.4]$

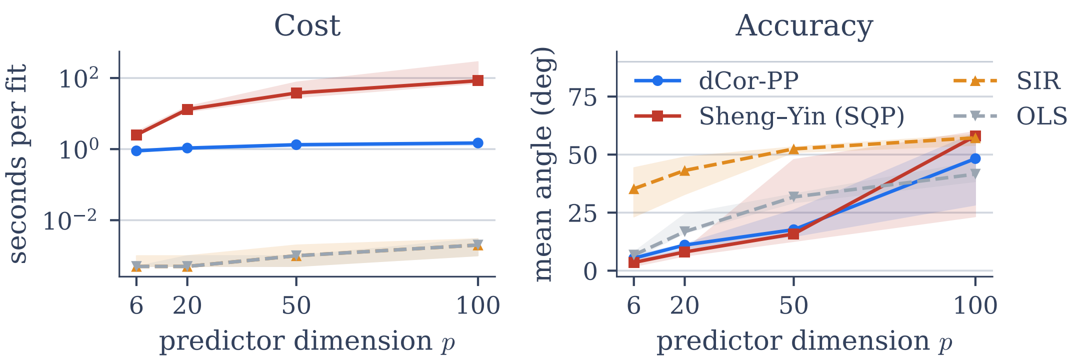

Figure 4.1: Model A part (1) at $n = 500$, five replicates; the line is the median and the band the range. *Left:* seconds per fit, on a logarithmic scale. The gradient search is nearly flat in p while the derivative-free constrained solve rises by more than an order of magnitude; the two classical estimators sit near a millisecond throughout. *Right:* mean principal angle to the true plane. Every method degrades with p at fixed n, and by $p = 100$ the two distance-based searches have joined the classical ones in failing. A two-dimensional subspace of a 100-dimensional predictor space is not recoverable from 500 observations by any of the four.

predictor dimension: 2.51 s at $p = 6$ rising to 83.9 s at $p = 100$, a factor of 33. The closed-form gradient costs one index evaluation whatever p is, and the $O(np)$ contraction that maps the gradient in the projected sample back to the gradient in β is the only p-dependent term, so the gradient search rises from 0.89 s to 1.49 s, a factor of 1.7 (Section 3.3). The ratio between the two therefore widens monotonically, from 2.8 at $p = 6$ to 56.5 at $p = 100$. The worst replicate at $p = 100$ took their solver 287 s, against 1.5 s for the gradient search. At $p = 6$, by contrast, the two are within a factor of three of each other, which is what a saving proportional to p looks like when p is small.

On accuracy their advantage does not extend above $p = 20$, and nothing replaces it. They are ahead by 1.9° and 3.0° at $p = 6$ and $p = 20$. At $p = 50$ the medians are close and the spreads are not: $[12.6, 47.9]$ against $[14.7, 26.1]$, one replicate in five losing the subspace on their side and none on this one. At $p = 100$ both fail, over ranges that overlap almost completely, and a median near 50° is not a recovered subspace under any reading. The sample is what binds there: two informative directions among 100 predictors at $n = 500$ leaves every trial projection mixing the signal with 98 noise coordinates, the mechanism already visible at $p = 20$, and it does not yield to a better search. The extension therefore establishes a cost result. The gradient makes the larger problems affordable to attempt, which is a separate matter from whether they can be solved at this sample size.

4.6 The AutoMPG data

The simulations fix the truth, which is what makes them scorable and at the same time what makes them artificial. The AutoMPG data of Quinlan (1993), which Sheng and Yin (2016) also use, supplies the opposite case: $n = 392$ cars once the six rows with a missing horsepower reading are dropped, fuel consumption in miles per gallon as the response, six standardised engine and body measurements as predictors, and no known direction to score against. With no true direction there is no angle to report, so two other quantities carry the comparison: how much dependence each single index retains, and how well the response is predicted from it, the second by five-fold cross-validation of a cubic fit, so that the comparison does not rest on the criterion one of the four methods maximises.

The ordering of the simulations reverses in Table 4.6. The gradient search returns the index with both the highest retained dependence (0.834) and the best cross-validated fit (0.862); their solver is last on both (0.784, 0.803), with OLS and SIR between, and the cost ordering of Section 4.5 holds here too at 0.43 s against 1.65 s. The four directions agree on what matters physically, weight positive and large and model year negative, and differ in how the weight loading is shared with the correlated cylinder and displacement readings. That is where the whitening in their solver shows: it spreads the loading onto `cylinders` (0.66) where the other three concentrate it on `weight`, and on predictors this correlated the redistribution costs 0.05 of cross-validated R^2.

Read down the rows, dCor-SDR returns the sparsest of the four directions. It places 0.89 on `weight` and -0.43 on `year` and leaves the other four loadings at 0.16 or below in absolute value, where SIR carries 0.26 and 0.21 on the correlated `cylinders` and

Table 4.6: A single index on the AutoMPG data ($n = 392$, $p = 6$), median over five seeds. dCor^2 is the dependence the index retains; R^2_{CV} is the five-fold cross-validated fit of a cubic in the index; *sec* is the wall-clock time one complete fit of the index takes on this data set, restarts included. Coefficients are for the standardised predictors, signed so that the weight loading is positive. The two deterministic methods have no seed spread; across the five seeds the spread of both reported values is below 0.004 for every method.

Method	dCor^2	R^2_{CV}	sec	cyl.	displ.	hp	weight	accel.	year
dCor-SDR	0.834	0.862	0.43	−0.01	0.05	0.16	0.89	0.02	−0.43
OLS	0.830	0.856	0.000	0.09	−0.12	0.00	0.89	−0.04	−0.43
SIR	0.825	0.854	0.000	0.26	0.21	0.08	0.79	0.01	−0.51
Sheng–Yin	0.784	0.803	1.65	0.66	0.17	−0.21	0.63	0.00	−0.31

`displacement` readings and the Sheng–Yin solver carries 0.66 on `cylinders`. A comparison of loadings of this kind only means something when the predictors share a scale, and here they do: all six are standardised before any of the four searches is run. Sparsity is not something the objective (4.1) asks for, so it is an observation about this data set and not a property claimed for the method. One data set is likewise not evidence of a general ordering, and a cubic in almost any sensible index of these six measurements explains four fifths of the variation; what the comparison does show is that the simulated deficit against the Sheng–Yin solver is not a property the method carries everywhere. Chapter 6 takes the real-data case up properly.

4.7 Summary

Against the two classical baselines the account is one-sided, and it is the reason the chapter exists. On the symmetric responses, `square` in one direction and `sum_squares` in two, the median recovery angle of both distance-based estimators is under 5° at $p = 6$ and under 11° at $p = 20$, where ordinary least squares and sliced inverse regression run from 35° to 80°. The reason is one property held in common: the population covariance between the response and the predictors vanishes, so both classical methods read a first moment that is not there, the one because $\mathrm{Cov}(X, Y) = 0$, the other because the inverse-regression mean is flat. Among the three models of Sheng and Yin (2016) the same split appears on model A, whose first index enters through a square, and model C separates the two classical estimators from each other rather than from the distance-based pair.

Against the solver this objective comes from, the account is mixed and is worth stating plainly. The two are level in low dimension. At $p = 20$ the Sheng–Yin solver recovers the subspace more accurately on most configurations, by 1° to 3° in median, and dCor-SDR is level on model C. What establishes the contribution is not that comparison but the matched one: run their SQP solve and Riemannian Adam on one objective and one manifold and the two are indistinguishable in accuracy, with a paired median of 0.00°, and a factor of twenty-two apart in cost at $(n, p) = (500, 20)$. The accuracy that separates the two published methods therefore lies in the sample criterion each maximises and not in the search, and the closed-form gradient is a cost contribution. It widens with the predictor dimension as the $O(pk)$ count predicts, at $k = 1$, from 2.8 times at $p = 6$ to 56.5 times at $p = 100$, and their accuracy advantage does not extend that far: the two are level at $p = 50$ with their range the wider, and at $p = 100$ two informative directions among a hundred predictors at $n = 500$ is beyond all four estimators, which is a limit of the sample rather than of the search.

Beyond one direction, two structural walls appear. The deflation problem is the deeper of the two: no linear deflation removes nonlinear signal, so Y-residual deflation collapses to repetition, and the two predictor-space strategies reach a second-order condition and stop there. Between them, orthogonalising in the Σ inner product is the better default whenever the predictor covariance is of full rank, since it costs one eigendecomposition and leaves the recovered directions measurably less redundant; the principled fix, nonlinear deflation, is sketched and left to future work. The second wall is the *marginal* form of the objective: every strategy here scores one direction at a time, so none can see dependence living in the recovered subspace as a whole, and taking the columns all at once rather than one after another does not change it. The deflation problem would persist even under a multivariate objective, because it concerns what a linear operation can remove rather than how the removal is scored; the marginal-sum problem would not.

Predictors that are neither normal nor continuous. Nothing in the two experiments above turns on the predictor law, and nothing in the objective could: (4.1) is a function of the distances between observations, which a discrete predictor supplies as readily as a continuous one. Sheng and Yin (2016) make the same point in their own design by running model A with Poisson(1) predictors. Their solver has a particular advantage in that setting, and it is worth naming because it is a property of the initialisation rather than of the objective: a predictor taking few distinct values makes the slices of an inverse regression nearly exact, so the inverse-regression seed their search starts from already sits close to the solution, where a search started at random has the whole sphere to cross.

CHAPTER 5

Unsupervised Projection Pursuit via Distance Variance

Chapter 4 studies a *supervised* method: given predictors X and a response Y, distance correlation identifies the directions of X carrying the most dependence on Y. The response supplies the meaning of "interesting".

This chapter asks what remains when there is no response. An unsupervised index has to score one projected sample with nothing but the predictors to go on. Most projections of high-dimensional data are close to Gaussian (Diaconis and Freedman, 1984), so whatever structure exists shows itself in the few that are not. Maximising distance variance over the unit sphere is the method studied here, and is called dVar-PP throughout. The optimisation itself, the analytical gradient, Riemannian Adam, the sequential and joint variants and the robust biloop index are developed in Chapter 3; this chapter specialises that machinery to distance variance and measures what it recovers.

What is offered here is not a new unsupervised dimension-reduction method, but a systematic study of one index. It asks where the maximisation of distance variance recovers a subspace, where it does not, and which of the two ingredients of the index, the scale of a projection or its shape, is responsible in either case. Since the index is scale-equivariant, the second question has to be settled before any recovery result can be read. Therefore Section 5.1 states the index, and Section 5.2 separates its two ingredients; what follows in Section 5.3 is the simulator, the recovery metric and the six questions which the experiments of Section 5.4 then take one at a time.

5.1 Distance variance as a projection index

Let $z = Xw$ be the projection of the data onto a unit vector w. A projection index is a scalar functional of that univariate sample, and projection pursuit searches for the w

maximising it. The index used here is the distance variance $\mathrm{dVar}(z)$, the square root of the diagonal case $\mathrm{dVar}^2(z) = \mathrm{dCov}^2(z, z)$ of the sample distance covariance (2.4). Where the variance reads the sample through one second moment, distance variance reads it through all $\binom{n}{2}$ pairwise distances. Section 2.1 sets out the ordering this produces at fixed variance, and two points from there govern every result below. The index rises as mass moves away from the centre and falls as mass concentrates in a peak, so two samples of equal variance are separated by it whenever their pairwise-distance profiles differ: a balanced bimodal sample scores 0.754 against a Gaussian's 0.635 (Table 2.1). Furthermore, the Gaussian sits in the *middle* of the distributions compared there rather than at one end, peaked and heavy-tailed shapes falling below it, so this is a shape index and not a measure of non-Gaussianity.

The method maximises it over the sphere,

$$w^\star = \arg\max_{\|w\|=1} \mathrm{dVar}(Xw), \tag{5.1}$$

which places it in the Friedman–Tukey and Huber tradition of maximising a shape functional over projections (Friedman and Tukey, 1974; Huber, 1985), with distance variance as the particular functional. Principal component analysis (PCA) is the same problem with a different index. It maximises $\mathrm{Var}(Xw)$ over the same sphere, ranking directions by a spread determined through second moments alone where (5.1) ranks them by the full pairwise-distance profile, and that substitution is the whole of the method.

5.2 Scale and shape in the index

The objective (5.1) combines two sources of information, which are worth separating before any recovery result is read. The distance variance is scale-equivariant, $\mathrm{dVar}(aZ) = |a|\,\mathrm{dVar}(Z)$, and is in this respect homogeneous of degree one, as a standard deviation is. A direction may therefore attain a large index value either because the projection along it has a large spread, or because its shape induces a large distance variance at the spread it has. The constraint $\|w\| = 1$ does not separate the two, since $\mathrm{Var}(Xw)$ still depends on w unless the covariance matrix of X is a multiple of the identity. Hereby the raw objective reads the scale of a projection first and its shape only afterwards.

Two devices remove the scale. The first is whitening, the classical projection pursuit recipe, which is taken up at the end of this section. The second compares the distance variance of a candidate projection with a second dispersion measure of the same projection,

$$\mathcal{K}(w) = \frac{\mathrm{dVar}(Xw)}{\sqrt{\mathrm{Var}(Xw)}} = \mathrm{dVar}\left(\frac{Xw}{\sqrt{\mathrm{Var}(Xw)}}\right), \tag{5.2}$$

where the second equality is the scale equivariance stated above. Since numerator and denominator are both homogeneous of degree one, $\mathcal{K}(aw) = \mathcal{K}(w)$ holds for every $a \neq 0$, so that the ratio is scale-invariant. It is the distance variance of a projection which has first been standardised to unit variance.

Whitening. In classical projection pursuit the data is whitened first, so that the index reacts to shape and not to spread (Huber, 1985). That step treats the spread as a nuisance, whereas in the factor model of Section 5.3 the spread is the signal itself, since there the signal direction is marked by its larger variance. Whitening then removes the information which marks the direction, and the recovery is worse than without it. Which of the two variants is the more informative depends on whether the structure of interest is expected to be marked by its scale. FastICA whitens as the first step of its own procedure and is affected in the same way, which may be one reason why it does not improve on PCA in this model. The variance gap which marks the signal direction is equalised away before the contrast function is ever evaluated.

Of these two options, studied here is the raw objective (5.1), the proposal of Nordhausen and Radojičić (2026), and neither the whitened data nor the ratio (5.2). The distinction is nevertheless needed for what follows, since wherever a recovery is reported below, the question whether the scale or the shape produced it has to be asked separately. Section 5.5 returns to (5.2) as the first of two open questions.

5.3 Experimental design

Six questions are put to the index, in the order of the subsections of Section 5.4. The first two follow from the previous section and are the halves of the main one: whether the maximisation recovers a signal direction which the variance of the data already marks, and whether it recovers one which nothing but the shape of its projection marks. Since the ordering of Section 2.1 places the Gaussian in the middle of the distributions compared there, a third question asks which shapes a maximiser reaches at all, and this setting is then extended to several directions, asking in addition whether a set of them reads more than one feature of the latent structure. The index is built from distances which a single remote observation can inflate, so contamination through gross errors is examined next; at last, whether the recovered directions are independent and not merely orthogonal is put to the test.

The controlled experiments use a non-Gaussian *factor model*, a standard generative model for the situation projection pursuit addresses: a few structured drivers, observed only through many noisy variables,

$$X = Z\,W_{\text{true}}^{\top}\,\sigma_{\text{signal}} + E. \tag{5.3}$$

Each observation is built from k latent factors, the columns of $Z \in \mathbb{R}^{n\times k}$, each a draw from a chosen non-Gaussian distribution standardised to unit variance. They are not observed directly: the mixing matrix $W_{\text{true}} \in \mathbb{R}^{p\times k}$, random orthonormal, spreads them across the p measured variables, its columns spanning the true signal subspace. The scalar σ_{signal} scales their contribution, and independent Gaussian noise $E \sim \mathcal{N}(0, I_p)$ is added to each variable. The task is to estimate the k-dimensional subspace spanned by the columns of W_{true} from X alone.

Table 5.1: The six latent distributions of the factor model (5.3), each standardised to unit variance before the signal scaling σ_{signal} is applied. The last column is the distance variance at unit variance (Table 2.1); the Gaussian noise the factors are buried in has dVar = 0.635, so `gaussian_mix` and `uniform` sit above the noise baseline and `t3`, `skewed` and `chisq` below it.

Key	Construction	Shape	dVar
`gaussian_mix`	$\pm 2 + \mathcal{N}(0,1)$, equal weights	balanced bimodal	0.754
`t3`	Student t_3	heavy-tailed	0.465
`uniform`	Uniform$(-\sqrt{3}, \sqrt{3})$	light-tailed	0.729
`skewed`	$\text{Exp}(1) - 1$	right-skewed	0.575
`chisq`	$\chi^2_3 - 3$	skewed	0.593
`gaussian`	$\mathcal{N}(0,1)$	control, no shape gap	0.635

The six latent distributions used for Z are listed in Table 5.1. Five span distinct departures from Gaussianity that a shape-based index might exploit: bimodality, heavy and light tails, and two forms of skewness. `gaussian_mix` is the bimodal one, an equally weighted two-component mixture with component means at ± 2 and unit-variance Gaussian components, standardised afterwards, so its two modes are four component standard deviations apart. The sixth, `gaussian`, is a control: its distance variance at unit variance is that of the noise the factors are buried in, so it is the one case in which the index has no shape difference to find.

Evaluation metric and baselines. Recovery is scored by the principal angles between the estimated and the true subspace, defined in Section 4.2.2. Where $k = 1$ the angle in degrees is reported directly. Where $k > 1$ the summary used is the mean squared sine (MSS), $\text{MSS}(W_{\text{true}}, \hat{W}) = \text{mean}(1 - \varsigma^2)$, with ς the singular values of $W_{\text{true}}^{\top} \hat{W}$, that is the cosines of the principal angles. Each term $1 - \varsigma_m^2 = \sin^2 \theta_m$ is 0 when a recovered direction lies in the true subspace and 1 when it is orthogonal to it, so MSS runs from 0 for perfect overlap to 1 for orthogonal subspaces, lower being better. It is the same object Chapter 4 reports as a mean principal angle, on a squared-sine rather than a degree scale. PCA is the principal classical baseline, for the reason given above, and a random direction provides a lower reference. Independent component analysis via FastICA (Hyvärinen and Oja, 2000; Pedregosa et al., 2011) is the shape-based competitor, and is reported in the two constructions of Sections 5.4.2 and 5.4.3 in which it is informative. It is run with full whitening, that is on all p principal components, so that its contrast function and not the variance ordering decides which directions it returns; both constructions have $p = k = 2$, where no dimension is discarded at all.

Table 5.2: Angle in degrees between the recovered direction and the true one, for dVar-PP on the factor model (5.3) at $p = 20$, $n = 200$, $k = 1$, across the latent distribution and the signal strength σ_{signal}; median over five seeds, lower being better. The signal strength is the factor which matters. The `gaussian` row is the control in which the index has no shape difference to exploit, and it recovers the direction as well as the shaped latents do.

Distribution	σ=0.5	σ=1.0	σ=2.0	σ=4.0
`gaussian_mix`	77.2	27.7	10.3	4.9
`t3`	62.6	46.2	18.0	7.4
`uniform`	82.3	25.5	11.9	6.1
`skewed`	61.6	55.8	14.6	6.1
`chisq`	78.1	29.8	11.9	6.0
`gaussian`	70.4	29.1	11.4	5.1

5.4 Experiments

5.4.1 A variance-marked signal

The first question is how well maximising distance variance recovers the signal subspace of (5.3). Because each latent factor is standardised to unit variance, the signal contributes variance σ^2_{signal} along its direction, so the signal direction in X has total variance $\sigma^2_{\text{signal}}+1$ while every pure-noise direction has variance 1. For $\sigma_{\text{signal}} \in \{2, 4\}$ that is a variance ratio of $5:1$ or $17:1$. The signal direction is therefore also the maximum-variance direction, which is exactly what PCA is defined to recover. In this model PCA is optimal by construction and is the natural benchmark. Section 5.4.2 turns to the opposite case, where variance carries no signal at all.

The comparison is a grid over the six latent distributions, four signal levels $\sigma_{\text{signal}} \in \{0.5, 1, 2, 4\}$ and three dimensions $p \in \{5, 20, 50\}$, each configuration replicated over five data seeds, at $n = 200$ with twenty restarts. Table 5.2 shows the recovered angle at $p = 20$; the same ordering holds at $p = 5$ and $p = 50$.

Recovery is governed almost entirely by the signal strength. At $\sigma_{\text{signal}} = 0.5$ the direction is essentially unrecovered, between 62° and 82°; at $\sigma_{\text{signal}} = 2$ the angle falls to 10–18° and at $\sigma_{\text{signal}} = 4$ to 5–7°. The latent distribution has a secondary effect that follows no simple ordering. At $\sigma_{\text{signal}} = 1$ the light-tailed `uniform` converges fastest at 25.5° and the right-skewed `skewed` slowest at 55.8°. PCA remains ahead of dVar-PP in the great majority of the configurations. The `gaussian` control, whose distance variance is that of the noise it is buried in, so that the index has no shape difference anywhere to climb towards, recovers at 70.4, 29.1, 11.4 and 5.1 degrees, inside the range the five shaped latents span. Therefore the shape information is not what finds the direction here. What the objective reads is the scale of the projections, the first of the two ingredients separated in Section 5.2.

The index is in addition the more expensive of the two. PCA returns its directions from a single eigenvalue decomposition, whereas every evaluation of the distance variance and of its gradient sums over all pairs of observations at a cost of $O(n^2)$, repeated over the iterations of twenty restarts (Chapter 3). What would put the shape contribution to the test in this model is the scale-free ratio (5.2), for which no run is reported in this work.

5.4.2 A shape-marked signal without a variance gap

The complementary setting removes the variance gap entirely, and with it the signal-plus-noise form. The question it puts is which structure distance variance is able to identify that the established projection pursuit and independent component criteria do not already reach. Against PCA the answer below is unambiguous, against a shape-based competitor it is not. Two latent variables of *equal* unit variance, one Gaussian and one bimodal, are rotated by 35° into the observed plane, with no additional noise dimensions; this is the second motivating example of Chapter 1, drawn in the right panel of Figure 1.1. Because both components have the same variance the observed covariance matrix is close to the identity: over five seeds its two eigenvalues differ by a median of 0.06 $[0.04, 0.10]$, so nothing distinguishes the bimodal axis except the shape of its projection.

PCA has no eigenvalue gap to lock onto, so its direction is fixed by whatever sampling fluctuation breaks the near-tie. Its angle to the bimodal axis has a median of 25.8° ranging from 7.6° to 89.3°, the spread of a direction drawn at random rather than of an estimate with a bias. Distance variance reads the shape difference directly and recovers the bimodal axis at a median of 1.7° $[0.1, 2.7]$ raw and 0.5° $[0.2, 2.1]$ after whitening. FastICA, whose own criterion is non-Gaussianity, does as well at 2.0° $[0.9, 3.5]$. The two read the shape differently, FastICA through third and fourth moments of the projection and distance variance through the whole pairwise-distance profile, and on this construction either suffices. Therefore the construction separates the shape-based indices from the variance-based ones. An advantage of distance variance over the classical shape-based indices is not established by it.

5.4.3 The direction of the search

Every result so far comes from *maximising* the index, and the ordering of Section 5.1 places the Gaussian in the middle of the distributions compared there rather than at one end. A maximiser can therefore only search towards the shapes above the Gaussian baseline of 0.635; the peaked and heavy-tailed shapes below it lie in the opposite direction, and minimising the same index searches that half. This is the argument of invariant coordinate selection (Tyler et al., 2009), that the interesting directions are to be looked for on both sides of such an ordering. In independent component analysis the sign of the departure separates two kinds of structure. Minimising the kurtosis brings out clusters, while maximising it brings out outlying observations (Radojičić et al., 2020).

The same construction tests this with the bimodal axis replaced by a Student t_3 one, at 0.469 below the baseline. Table 5.3 runs the optimiser in both directions on both constructions, and the result is nearly a mirror. Maximising finds the bimodal axis at

Table 5.3: Angle in degrees between the recovered direction and the interesting axis, for two equal-variance constructions with no variance gap, at $n = 1200$ over five seeds, median with $[\min, \max]$, lower being better. The interesting axis is bimodal in the first construction, with a distance variance above the Gaussian baseline, and Student t_3 in the second, below it. PCA and FastICA do not depend on the direction of search and are listed once.

Construction	Method	Angle	$[\min, \max]$
bimodal axis, dVar = 0.753	dVar-PP, maximised	1.7	$[0.1, 2.6]$
	dVar-PP, minimised	75.1	$[56.1, 89.4]$
	PCA	25.8	$[7.6, 89.3]$
	FastICA	2.0	$[0.9, 3.5]$
t_3 axis, dVar = 0.469	dVar-PP, maximised	75.0	$[68.2, 87.4]$
	dVar-PP, minimised	1.7	$[1.0, 4.2]$
	PCA	80.5	$[0.9, 83.6]$
	FastICA	2.9	$[0.5, 5.1]$

1.7° and misses the heavy-tailed one by 75.0°; minimising finds the heavy-tailed axis at 1.7° and misses the bimodal one by 75.1°. Neither search fails in the case it misses. The maximiser on the heavy-tailed construction reaches an index value of 0.629, above both the 0.469 carried by the axis it was meant to find and the 0.622 of the Gaussian axis, so it found what it was asked for. FastICA, whose criterion is symmetric about the Gaussian, finds both axes with one search.

The direction of the search is therefore a modelling choice. Where the structure of interest is expected to be peaked or heavy-tailed, the index is minimised, and nothing in the machinery of Chapter 3 changes. A criterion which reaches both halves with a single search would measure the deviation of the scale-free ratio (5.2) from its Gaussian reference value, as $|\mathcal{K}(w) - c_0|$, in the way in which the kurtosis-based indices measure the deviation from their own; Section 5.5 takes this up.

5.4.4 More than one direction

Both experiments of this subsection return to the factor model (5.3), where the signal is once more marked by its variance and PCA is once more optimal by construction, so that what is measured is not whether the index improves on PCA but whether the extraction of several directions works at all and what the two available strategies cost. Extending to $k > 1$ does not change the standing of the method. Table 5.4 reports four multi-direction scenarios, each run with both sequential deflation and joint Stiefel optimisation, the two strategies of Section 3.5, over five seeds. Three of them ask for two bimodal factors and vary one thing each: a baseline at $p = 50$, a high-dimensional case at $p = 100$ and a weak-signal case at $\sigma_{\text{signal}} = 0.8$. The fourth asks for four heavy-tailed factors at $p = 80$, and is the only one in which more than two directions are extracted. dVar-PP tracks

PCA within a factor of 1.2 to 2.4 in MSS, with PCA remaining slightly ahead for the same variance-gap reason as at $k = 1$. Sequential and joint optimisation are operationally identical here. Their medians differ by at most 0.003 and their seed ranges overlap in every scenario. Joint reaches an orthogonality defect of order 10^{-16} against sequential's drift of order 10^{-3}, but it is 1.5 times slower at $k = 4$ (1.89 against 1.26 seconds) and offers no recovery advantage. The supervised comparison of Section 4.4.5 reaches the same conclusion on its own model, that the two strategies separate on orthogonality precision and cost and not on recovery.

Those scenarios give each latent factor one interesting axis, which says nothing about a latent plane whose structure is a set of groups. For this the latent factors of (5.3) are replaced by a mixture of G equally weighted components whose centres lie on a circle of radius 2 in the $k = 2$ latent plane. The circle is used rather than a product of two bimodal coordinates because such a product places four groups at the corners of an axis-aligned square, where the structure factorises and each latent axis separates the same pair of groups twice over.

Table 5.5 reports the same subspace metric over $G \in \{3, 5\}$, $p \in \{10, 50\}$ and $\sigma_{\text{signal}} \in \{1, 2, 4\}$ at $n = 400$, five seeds. Group structure leaves the standing of the method unchanged. The groups are still lifted above the noise by their variance, the median ratio of dVar-PP's score to PCA's runs between 1.40 and 1.50, and sequential and joint extraction differ by at most 0.004. The last column answers what the subspace metric cannot. Each unordered pair of centres defines a separating axis in the latent plane, which W_{true} carries into observed space, and each recovered direction is assigned to the pair whose axis it aligns with most closely. In all 120 runs the two directions are assigned to *different* pairs, with median alignments of 0.95 and 0.96, at both group counts and every signal strength, so that the second direction is not another view of the first group split. Figure 5.1 shows one such run at each group count, projected onto the true signal plane.

Those scenarios give each latent factor one interesting axis, which says nothing about a latent plane whose structure is a set of groups. For this the latent factors of (5.3) are replaced by a mixture of G equally weighted components whose centres lie on a circle of radius 2 in the $k = 2$ latent plane. The circle is used rather than a product of two bimodal coordinates because such a product places four groups at the corners of an axis-aligned square, where the structure factorises and each latent axis separates the same pair of groups twice over.

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Table 5.4: Mean squared sine against the true signal subspace of the factor model (5.3), for dVar-PP (sequential and joint) against PCA; median over five seeds, with [min, max] for dVar-PP, lower being better. The scenarios are, as $(n, p, k, \sigma_{\text{signal}})$, baseline $(200, 50, 2, 2)$, high dimension $(200, 100, 2, 3)$ and weak signal $(200, 50, 2, 0.8)$, all three with bimodal factors, and four factors $(300, 80, 4, 2)$ with t_3 factors. In the weak-signal scenario no method recovers the subspace.

Scenario	Strategy	dVar-PP	[min, max]	PCA
baseline	sequential	0.088	[0.079, 0.102]	0.072
baseline	joint	0.086	[0.076, 0.106]	0.072
high dimension	sequential	0.090	[0.066, 0.095]	0.060
high dimension	joint	0.090	[0.066, 0.096]	0.060
weak signal	sequential	0.786	[0.692, 0.829]	0.637
weak signal	joint	0.784	[0.691, 0.839]	0.637
four factors	sequential	0.182	[0.168, 0.186]	0.075
four factors	joint	0.179	[0.154, 0.195]	0.075

Table 5.5: Mean squared sine against the true signal plane for a latent mixture of G equally weighted components on a circle, at $n = 400$, $k = 2$, over $p \in \{10, 50\}$ and $\sigma_{\text{signal}} \in \{1, 2, 4\}$; median over five seeds, [min, max] for dVar-PP, lower being better. The last column is the share of runs in which the two recovered directions separate different pairs of group centres.

Groups	Strategy	dVar-PP	[min, max]	PCA	distinct pairs
$G = 3$	sequential	0.018	[0.001, 0.435]	0.015	100%
$G = 3$	joint	0.017	[0.001, 0.429]	0.015	100%
$G = 5$	sequential	0.017	[0.001, 0.378]	0.012	100%
$G = 5$	joint	0.021	[0.001, 0.373]	0.012	100%

to *different* pairs, with median alignments of 0.95 and 0.96, at both group counts and every signal strength, so that the second direction is not another view of the first group split. Figure 5.1 shows one such run at each group count, projected onto the true signal plane.

5.4.5 Contaminated observations

The last comparison in the factor model contaminates it. A fraction of the rows is replaced by large random spikes of magnitude $10\sqrt{p}$, standing in for gross measurement errors. Table 5.6 and Figure 5.2 sweep that fraction for plain dVar-PP, for a variant built on the biloop transform of Leyder et al. (2026) whose redescending weight bounds the finite-sample sensitivity the raw index leaves unbounded (Section 2.5), and for PCA.

The two families deteriorate at different rates, and the power at which a spike enters the

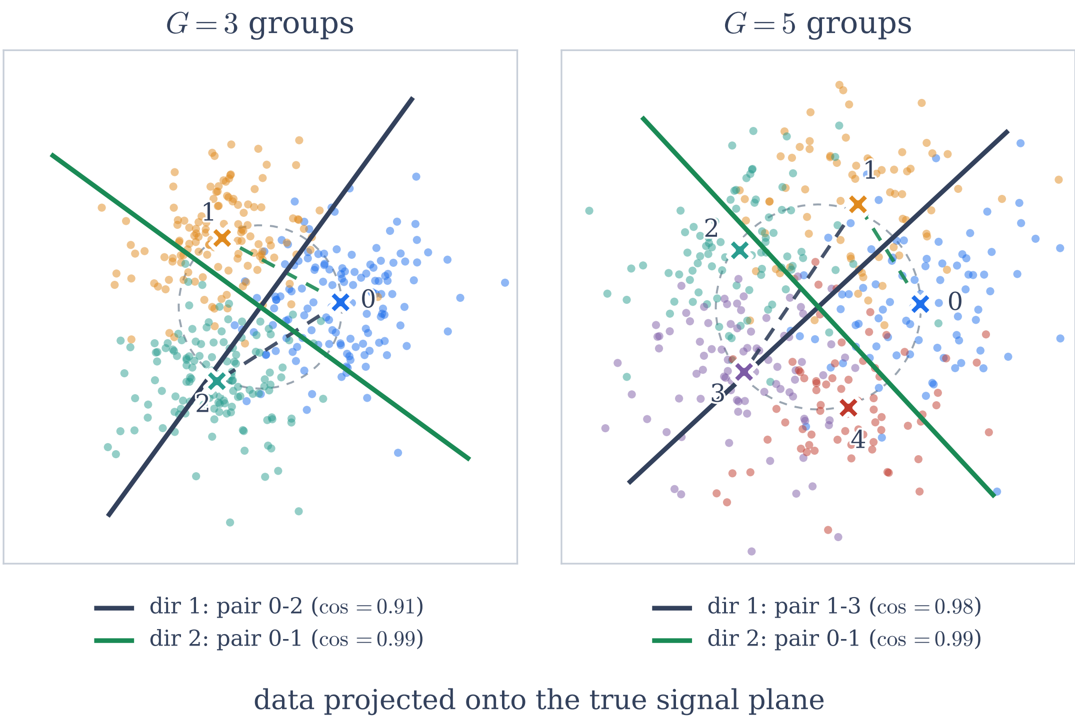

Figure 5.1: The observed data projected onto the true signal plane, for a latent mixture of G equally weighted components with centres on a circle (grey dashed), at $n = 400$, $p = 10$, $\sigma_{\text{signal}} = 4$ with sequential extraction; one seed of the five. Points are coloured by the component they were drawn from and the crosses mark the component centres. The two solid lines are the directions dVar-PP recovers, each drawn beside the dashed segment joining the pair of centres it is assigned to; the alignment between the two is the cosine in the legend. The two directions separate different pairs in both panels, which Table 5.5 reports for all 120 runs.

objective is the reason. A spike of magnitude M enters a sum of squared projections through M^2, so that a few observations at $M \approx 10\sqrt{p}$ outweigh the entire clean sample, whereas a distance-based index sees the same spike through its linear distance. At 5% contamination the mean squared sine of PCA moves from 0.032 to 0.830, that of the plain index from 0.046 to 0.268. The window in which this matters is narrow. From 10% contamination onwards no method in the sweep recovers the subspace, and every value from there on lies between 0.658 and 0.929. What the table records is a breakdown-point comparison, how large a contaminated fraction each estimate survives, and not the influence function. Of the three criteria only the biloop variant has a breakdown point strictly above zero, since it is computed on a bounded transform (Leyder et al., 2026). For the other two the table gives a rate of deterioration and not a guarantee.

The biloop variant is reported for its cost. On clean data it reaches 0.763 against the

Table 5.6: Mean squared sine against the true signal subspace of the factor model (5.3) at $n = 150$, $p = 30$, $k = 2$, $\sigma_{\text{signal}} = 2.5$ with bimodal factors, versus the fraction of rows replaced by spikes of magnitude $10\sqrt{p}$; median over five seeds, lower being better. At 5% contamination the plain index deteriorates less than PCA, and from 10% onwards no method in the sweep recovers the subspace.

Outlier fraction	dVar-PP (plain)	dVar-PP (biloop)	PCA
0.00	0.046	0.763	0.032
0.05	0.268	0.738	0.830
0.10	0.733	0.658	0.927
0.20	0.868	0.736	0.929

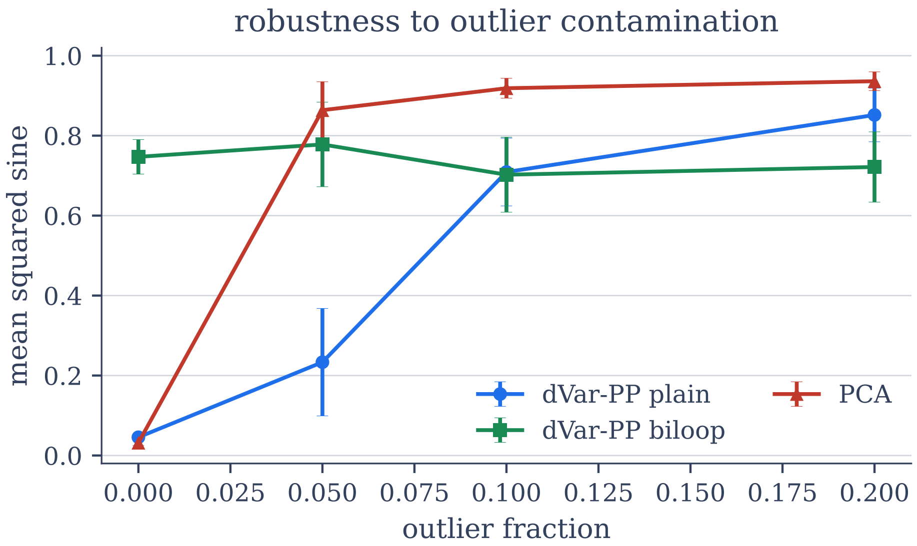

Figure 5.2: Mean squared sine against outlier fraction for plain dVar-PP, biloop dVar-PP and PCA, in the setting of Table 5.6. The biloop variant starts far worse on clean data and is the flattest of the three under heavy contamination.

plain index's 0.046, because its redescending saturation flattens the objective surface and suppresses legitimate structure; that price has not, as far as we are aware, been quantified before in a dimension-reduction setting. Its lower values from 10% contamination onwards, 0.658 against 0.733 and 0.736 against 0.868 at 20%, are not a recovery, since both numbers describe a subspace which has been lost. Plain dVar-PP is therefore the default, and it is also the cheaper of the two, having an analytical gradient where the biloop transform has none (Chapter 3). The natural multivariate successor of the transformation is the robust independent component analysis of Leyder et al. (2025).

Table 5.7: Mean squared sine against the true signal subspace of the factor model (5.3) at $n = 200$, $p = 50$, $k = 2$, $\sigma_{\text{signal}} = 2$ with bimodal factors, for independent latent factors against factors under the heteroscedastic coupling $z_j = |z_1|\varepsilon_j$; median over five seeds, $[\min, \max]$ in brackets, lower being better. dVar-PP degrades under the coupling and PCA barely moves; joint optimisation does not rescue dVar-PP.

Factors	dVar-PP (seq.)	dVar-PP (joint)	PCA
independent	0.088 $[0.079, 0.102]$	0.086 $[0.076, 0.106]$	0.072
dependent	0.105 $[0.097, 0.129]$	0.105 $[0.097, 0.128]$	0.065

5.4.6 Uncorrelated but dependent factors

A final experiment records a structural limitation of the method as constructed; as Chapter 7 argues, it is the unsupervised twin of the supervised deflation problem. Sequential deflation enforces orthogonality of the recovered directions in weight space, and the question is whether those directions are also genuinely *independent* when the latent factors are uncorrelated but higher-order dependent.

For this the independent latent columns of (5.3) are replaced by a *heteroscedastic coupling*: z_1 is drawn from the chosen distribution and $z_j = |z_1|\,\varepsilon_j$ for $j \geq 2$, with ε_j independent draws. Since $\mathbb{E}[\varepsilon_j] = 0$ the factors stay uncorrelated, $\mathrm{Cov}(z_1, z_j) = 0$, yet the spread of each z_j is governed by $|z_1|$, so they share higher-order dependence. At $n = 2000$ over five seeds the squared distance correlation between two columns is 0.021 $[0.021, 0.025]$ under the coupling against 0.0004 $[0.0000, 0.0013]$ when they are drawn independently, while the squared Pearson correlation stays below 4×10^{-3} in both cases and does not separate them at all. The multiplicative form keeps the latent columns full rank; an additive coupling $z_j = z_1 +$ small noise would collapse them onto a single direction.

Under this coupling, Table 5.7 shows dVar-PP degrading from 0.088 to 0.105 in MSS, a factor of 1.19, while PCA barely moves, from 0.072 to 0.065. The coupling leaves the second moments essentially untouched, which is all PCA reads. That is not a point in its favour, since PCA never claimed to deliver independent directions, and here it is only insensitive to the dependence; that is not evidence against it.

The two dVar-PP ranges meet only at their edges, so the shift is modest but consistent across seeds, whereas PCA's two ranges overlap. Joint Stiefel optimisation degrades by the same amount, its median differing from the sequential one only in the fourth decimal. This places the cause in the objective rather than in the search. Both strategies maximise a *sum of marginal* distance variances, $\sum_j \mathrm{dVar}(XB_j)$, which scores each candidate direction in isolation and never inspects whether the projections are jointly structured. An independent pair of directions and a higher-order-dependent pair with the same one-dimensional shapes therefore look identical to it, and heteroscedastic coupling exploits exactly that, since it inflates the marginal distance variance along *any* direction in the span of the dependent factors, while the constraint $B^\top B = I$ does nothing to stop the optimiser from trading subspace accuracy for objective value.

The remedy this diagnosis points to is a genuinely *multivariate* distance measure of the recovered subspace XB, scoring the projections jointly instead of one at a time, whose closest existing form is the distance-covariance independent component analysis of Matteson and Tsay (2017). One exploratory run with such an objective did not remove the degradation at $n = 200$, and since a single replicate settles nothing, it remains open whether the marginal form of the objective is the difficulty or whether the multivariate index is hard to estimate at this sample size. The supervised method meets a related obstacle when asked for several directions (Section 4.4), although there it is sequential extraction rather than the marginal form of the objective; Chapter 7 separates the two.

5.5 Summary and open questions

Where the signal is lifted above the noise by its own variance, PCA is optimal by construction, and the search recovers the same subspace once the signal is present, without reaching PCA and at a higher cost. The Gaussian control recovers as well as the shaped latents do, so in this regime the objective is reading the scale of the projections. Neither several directions nor a latent plane of three or five groups alters that, and sequential and joint extraction are indistinguishable in recovery throughout, separating only on orthogonality precision and cost.

Where variance does not carry the signal, the shape sensitivity becomes useful. With two equal-variance components the index recovers the bimodal axis which PCA can only pick at random. FastICA reads the same shape and does as well, so the advantage holds against the variance-based criteria; against the shape-based criteria, no advantage is established. The heavy-tailed counterpart of that axis is found by minimising the same index, since the Gaussian sits in the middle of the distributions compared. Under outlier contamination the index deteriorates more slowly than the criteria built on squared deviations, which opens a narrow window at around 5%; from 10% onwards no method in the sweep recovers the subspace, the bounded biloop variant included, which in addition costs a great deal on clean data.

Two questions are left open. The first concerns the criterion itself. The ratio (5.2) of the distance variance of a projection to its standard deviation is scale-invariant, so that it isolates the shape contribution which the raw objective mixes with the scale, and measured against its Gaussian reference value as $|\mathcal{K}(w) - c_0|$ it would reach both halves of the ordering with one maximisation in place of the two searches of Section 5.4.3. Whether it recovers the equal-variance axis as reliably as the raw index does is not examined here. The second is the marginal form of the objective, which Section 5.4.6 sets out and for which the remedy is a multivariate distance measure of the recovered subspace in the spirit of Matteson and Tsay (2017). That limitation is the unsupervised face of the marginal-sum problem which the supervised method shows under deflation, and Chapter 7 treats the two as one.

CHAPTER 6

Real Data Case Study: Excess Bond Returns

The two preceding chapters evaluated the distance-based methods of this thesis on simulators in which the truth was known by construction. This chapter evaluates both methods on real data, using a published out-of-sample forecasting exercise from empirical finance for which both a linear and a nonlinear reference model exist in the literature. Sections 6.1 to 6.3 set out the forecasting task, the two published solutions, and the experimental design; Sections 6.4 and 6.5 report the results for the supervised and the unsupervised method, and Section 6.6 summarises them.

6.1 The forecasting problem

Buy a five-year US Treasury bond, hold it for one year, and sell it. Whether this earns more than simply holding a one-year bond over the same year is the *excess return*, the compensation for bearing interest-rate risk. Its expectation is what the literature calls the *risk premium*: premium and excess return refer to the same quantity, the first to its expected value and the second to the value actually realised. The empirical question studied for decades is whether that premium is *predictable* twelve months in advance. The classical answer uses only the yield curve itself: Cochrane and Piazzesi (2005) showed that a single combination of forward rates predicts a substantial share of next year's excess returns. Ludvigson and Ng (2009) then showed that the *macroeconomy* adds predictive power beyond the yield curve. They compress a panel of 131 monthly macroeconomic series covering 1964–2007 into eight principal-component factors and use those factors to forecast the excess returns on two- to five-year bonds. It is this second, macroeconomic reduction step that the present chapter is about, so the forecasts below use the macro factors alone and leave the yield curve out.

In the notation of the earlier chapters, the predictors X are the eight factor scores summarising the state of the macroeconomy at month t; the response Y is the excess return realised over the year from t to $t+12$, one series per bond maturity, where the maturity is the lifetime of the bond whose return is being forecast (two to five years); and the task is genuine out-of-sample forecasting. The sample of 480 months (1964–2003) is split at the predictor date into a training half (1964–1983, 240 observations) and a test half (1984 onwards, 228 observations); the twelve months that separate $240+228$ from 480 are the final months of the sample, for which the return realised twelve months later falls outside the data and the observation therefore has no response. Every quantity is fit on the training half and *frozen.*[1]

What is measured, and how it is reported. Every method in this chapter does the same two things: it compresses the eight factors into d directions, and it then regresses next year's excess return on those d directions. How many directions to keep is not fixed by any of the methods, so d is a tuning choice, and every result is reported as a function of it, from a single direction up to all eight. This is also how Gunsilius and Schennach (2023) present their results.

Forecast accuracy on the test half is the mean squared prediction error (MSPE), the average squared difference between the forecast and the return that followed. A raw MSPE cannot be compared across bond maturities: a five-year bond moves more than a two-year bond, so its squared errors are an order of magnitude larger, and an equally good model looks worse on the longer bond. The remedy is to divide by the squared error of the crudest forecast available, which is to always predict the average excess return of the training years. One minus that ratio is the Campbell–Thompson out-of-sample R^2,

$$R^2_{\mathrm{OS}} \;=\; 1 - \frac{\mathrm{MSPE}}{\mathrm{MSPE}_{\mathrm{mean}}}, \tag{6.1}$$

of Campbell and Thompson (2008): it is 0 for a model no better than the historical mean and 0.25 for one that removes a quarter of the mean's squared error. Within a single maturity $\mathrm{MSPE}_{\mathrm{mean}}$ is a fixed number, so (6.1) is only a rescaling of the MSPE and ranks models in exactly the same order; it is used so that the four maturities can be shown in one table. The figure keeps the raw MSPE, because that is the unit in which the published figure it reproduces is drawn.

Significance. Whether the difference between two models is more than noise is decided by the Diebold–Mariano test (Diebold and Mariano, 1995) throughout. Write $e_{A,t}$ and $e_{B,t}$ for the forecast errors of two competing models in test month $t = 1, \ldots, T$ with

[1] The twelve-month alignment between predictors and response follows the convention of Cochrane and Piazzesi (2005); as a correctness check on the return construction, the implementation reproduces their canonical regression of returns on forward rates with $R^2 = 0.314$, the value reported in the original study.

$T = 228$, and form the loss differential and its mean

$$\delta_t \;=\; e_{A,t}^2 - e_{B,t}^2, \qquad \bar{\delta} \;=\; \frac{1}{T}\sum_{t=1}^{T} \delta_t. \tag{6.2}$$

The null hypothesis is $\mathbb{E}[\delta_t] = 0$, that is, equal expected squared error. The test statistic is

$$\mathrm{DM} \;=\; \frac{\bar{\delta}}{\sqrt{\hat{\sigma}^2/T}}, \qquad \hat{\sigma}^2 \;=\; \hat{\gamma}_0 + 2\sum_{j=1}^{L}\Big(1 - \tfrac{j}{L+1}\Big)\hat{\gamma}_j, \qquad \hat{\gamma}_j = \frac{1}{T}\sum_{t=j+1}^{T}(\delta_t - \bar{\delta})(\delta_{t-j} - \bar{\delta}), \tag{6.3}$$

and under the null DM is asymptotically standard normal, so $|\mathrm{DM}| > 1.645$ corresponds to a p-value below 10%. The variance in the denominator is not the naive one: consecutive twelve-month returns overlap in eleven of their twelve months, so the δ_t are strongly autocorrelated and a naive variance would be far too small. Equation (6.3) therefore uses the autocorrelation-robust Newey–West estimator (Newey and West, 1987) at lag $L = 18$, which widens the confidence bands to account for the overlap. This is not an addition of the present work: it is the test Gunsilius and Schennach (2023) themselves use to certify their result, at the same lag, so applying it here keeps the comparison like-for-like. Its power is limited by the same overlap, since 228 overlapping monthly observations carry the information of roughly nineteen independent ones, and Section 6.4 returns to that point where it matters.

6.2 Two published solutions

The relationship between the factors and the premium is *nonlinear*. The two published solutions handle that nonlinearity in two different places, and the contrast between them is what makes the benchmark a natural test bed for this thesis.

Ludvigson and Ng (2009) keep the reduction linear and put the nonlinearity in the *regression stage*: the specification they select by BIC adds the *cube of the first factor* to the linear model. We denote it `PCA+poly`. It works, but it requires a model-selection step over polynomial terms.

Gunsilius and Schennach (2023) argue the nonlinearity belongs in the *reduction stage* instead. Their independent nonlinear component analysis (INPCA) replaces linear PCA by a nonlinear analogue built on optimal transport, which ranks directions by an entropy criterion rather than by variance. Their headline result, obtained on exactly this bond dataset, is that the INPCA directions used as predictors in an ordinary *linear regression* match `PCA+poly`: once the reduction itself is nonlinear, no polynomial term has to be selected.

The question of this chapter is whether the two methods developed in Chapters 4 and 5, the supervised dCor-SDR and the unsupervised dVar-PP, can play the same role on the identical data and protocol: absorb the nonlinearity into the reduction step and reach the level of the two published models.

6.3 Experimental design

Methods under comparison. All methods work on the same eight-factor block and differ in two respects only: which d directions they extract from it, and which regression is run on those directions.

The *direction-finders* are: PCA, INPCA, the supervised dCor-SDR of Chapter 4 (sequential X-deflation, plus the Y-residual and joint variants), the unsupervised dVar-PP of Chapter 5 (sequential and joint, plain and robust), and three classical supervised peers from the sufficient-dimension-reduction lineage described in Section 2.2, namely partial least squares (PLS) (Wold et al., 2001), sliced inverse regression (SIR) (Li, 1991), and sliced average variance estimation (SAVE) (Cook and Weisberg, 1991). PCA is a direction-finder only in a nominal sense and is the baseline the two published models are built on: the eight factors *are* already principal components, so extracting d directions means keeping the first d of them, with nothing searched for.

The *second stage* is the regression run on the extracted directions, and it is one of two. Either `lin`, an ordinary linear regression on the d directions, or `poly`, a polynomial regression, the same linear model with the cube of the leading direction added, which is the form Ludvigson and Ng (2009) select by BIC.

Every combination of direction-finder, number of directions $d = 1, \ldots, 8$, second stage and bond maturity is computed within one common implementation. Each method is then scored at its own *best over* d: for a given maturity, the d at which its out-of-sample R^2 is highest. Because d is picked by looking at the test half, this is an optimistic score for every method; it is the way both published papers report their results, and it is applied identically to all methods here, so the comparison remains like-for-like. One consequence of the design is worth stating in advance: at $d = 8$ any set of eight linear directions spans the whole factor block, so `lin` on eight directions is the same model whatever found them, namely least squares on all eight factors ($R^2_{\text{OS}} = 0.184$ at two years). A linear reduction can only earn its place by beating that with fewer directions; a nonlinear one, such as INPCA, is not bound by it.

Since the distance-based methods are the object of interest, dCor-SDR is reported throughout with the `lin` second stage. The question is whether a supervised reduction can absorb the nonlinearity by itself, and a polynomial term placed on top of it would hide exactly that.

No public implementation of INPCA exists in either Python or R, so the method was reimplemented for this comparison. All INPCA numbers below are therefore this thesis's re-run of the method on the same factors, not the published numbers of Gunsilius and Schennach (2023); where a single INPCA curve or row is shown, it uses the hybrid the source paper favours, reshaping the leading $\min(d, 4)$ factors nonlinearly and keeping the remainder linear.

The factor block, and its look-ahead. The eight factors used throughout are the ones distributed with Ludvigson and Ng (2009). Their macroeconomic panel runs to 2007, but the forecasting sample used here ends in 2003, and the factors were extracted by PCA on the whole of that sample at once, before any train/test split. A method consuming them therefore sees eigenvectors computed in part from the test years. They are used here regardless, and deliberately: they are the input on which both published models were tuned and reported, and the question of this chapter is whether a supervised reduction can replace theirs on their own benchmark, which requires handing every method the same factors. What the look-ahead does affect is the *level* of every number below, inflating out-of-sample R^2 by roughly 0.07 across the board.[2] The results below are therefore comparisons between reductions on a shared input, not estimates of the forecast accuracy a practitioner could have achieved in 1984.

6.4 Results for the supervised method

This section runs the Gunsilius and Schennach (2023) experiment as closely as the published material allows: the same macro-factor block, the same out-of-sample MSPE reported against the number of directions, and the same equal-MSPE Newey–West test. The one substitution is the reduction itself, where the supervised dCor-SDR direction of Chapter 4 takes the place of the transport-based INPCA direction. Since both reductions are followed by ordinary least squares, the comparison is between two ways of choosing directions and nothing else.

Figure 6.1 reproduces the two panels of the source paper's central figure for the two-year bond and adds the dCor-SDR curve. Table 6.1 gives the same comparison in normalised units and extends it to all four maturities, which is this thesis's addition rather than the source paper's. Table 6.2 then places all methods in the study side by side.

Levels. Scored at its own best d, a supervised linear reduction followed by ordinary least squares is above the transport-based reduction at every maturity, by 0.026 to 0.054 in out-of-sample R^2, and below the polynomial benchmark by 0.033 to 0.006, the shortfall shrinking as the maturity lengthens. Where each method reaches its best is as informative as the level it reaches. `PCA+poly` and `INPCA+lin` both need the full eight-factor block at every maturity, whereas dCor-SDR is best with two directions, three at four years. The -0.03 in Table 6.1 is thus the price of a much smaller model: `PCA+poly` attains 0.269 at two years with nine regressors, eight factors and a cube, and dCor-SDR attains 0.242 with two, and no nonlinear term anywhere. That the benchmark's advantage rests on the largest model it is allowed to fit is visible in the left panel of Figure 6.1: its MSPE

[2] Measured by re-extracting the factors from the underlying 131-series panel with mean, scale and eigenvectors fit on the training months only. On the full sample that code reproduces the authors' own archived factors with |corr| = 1.00 on all eight, so it is a de-leaked version of the same construction rather than a different pipeline. On the train-only factors every reduction loses about 0.07 and the three fall within roughly 0.02 of one another, a spread the Newey–West test cannot resolve on this sample.

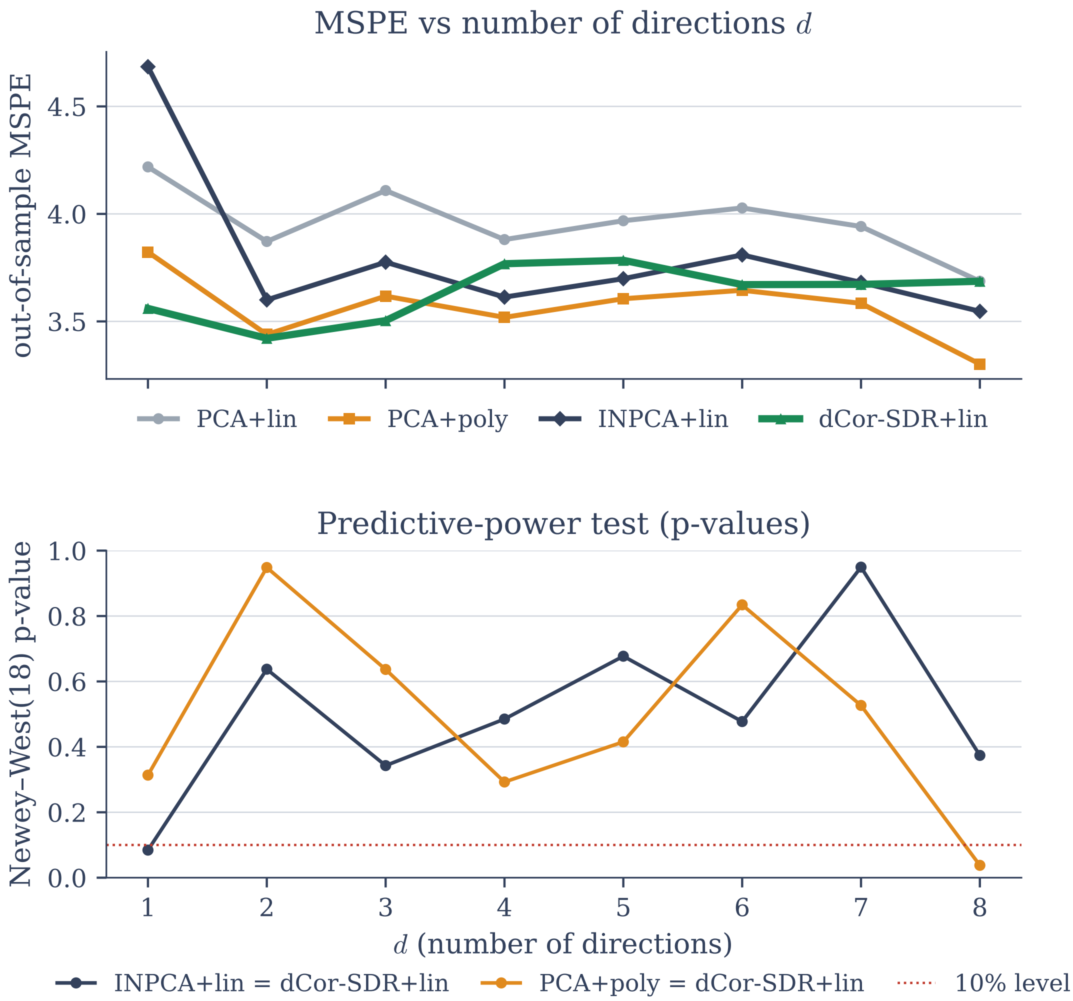

Figure 6.1: Left: out-of-sample MSPE for the two-year bond against the number of directions d. Right: the equal-MSPE Newey–West(18) p-value against d for the two comparisons (`INPCA+lin`) = (dCor-SDR+`lin`) and (`PCA+poly`) = (dCor-SDR+`lin`). Each model is scored at the minimum of its own curve, and the three minima sit at different places: $d = 2$ for dCor-SDR+`lin` and $d = 8$ for both `PCA+poly` and `INPCA+lin`.

curve stays between 3.44 and 3.65 for $d = 1, \ldots, 7$ and drops to 3.30 only at $d = 8$, whereas dCor-SDR's best, 3.42 at $d = 2$, is below every one of those seven points.

Significance. The ordering in Table 6.1 is largely a numerical one. Applying the source paper's own test to the two methods at their respective best d, dCor-SDR+`lin` is ahead of `INPCA+lin` at all four maturities but reaches the 10% level only at five years ($p = 0.068$, against p between 0.12 and 0.22 elsewhere), and the shortfall against `PCA+poly` is nowhere significant, the smallest p-value over the four maturities being

Table 6.1: Best-over-d Campbell–Thompson out-of-sample R^2, and the Newey–West(18) p-value for the difference, each method taken at its own best d.

Maturity (years)	2	3	4	5
`PCA+poly`	0.269	0.286	0.288	0.274
`INPCA+lin`	0.215	0.227	0.224	0.214
dCor-SDR + `lin`	0.242	0.253	0.260	0.268
dCor-SDR − `INPCA+lin`	+0.028	+0.026	+0.036	+0.054
p	0.223	0.155	0.120	0.068
dCor-SDR − `PCA+poly`	−0.027	−0.033	−0.028	−0.006
p	0.287	0.232	0.247	0.746

Table 6.2: Best-over-d Campbell–Thompson out-of-sample R^2 for every method in the study, with the mean over maturities in the last column. Apart from the `PCA+poly` benchmark, whose cube is the published model, every method is shown with the linear second stage.

Method	Maturity (years)				mean
	2	3	4	5	
`PCA+poly`	0.269	0.286	0.288	0.274	0.279
dCor-SDR + `lin`	0.242	0.253	0.260	0.268	0.256
SIR + `lin`	0.195	0.221	0.229	0.241	0.222
`INPCA+lin`	0.215	0.227	0.224	0.214	0.220
dVar-PP (sequential) + `lin`	0.198	0.213	0.226	0.228	0.216
SAVE + `lin`	0.184	0.224	0.210	0.224	0.210
PLS + `lin`	0.189	0.210	0.220	0.221	0.210
`PCA+lin`	0.184	0.204	0.210	0.206	0.201

0.23. Against the linear benchmark `PCA+lin` the advantage does become significant, at three of the four maturities ($p = 0.058$ to 0.090, and $p = 0.111$ at two years). This is what the overlapping annual returns permit: with an effective nineteen independent observations, differences of a few hundredths in out-of-sample R^2 are not resolvable, which is also the limitation that lets Gunsilius and Schennach (2023) report their own reduction as being as predictive as the polynomial benchmark. What the comparison establishes is accordingly narrower than a ranking, and it concerns where the nonlinearity has to be modelled: on this benchmark a supervised linear direction, with no nonlinear term anywhere in the model, is at least as predictive as an optimal-transport reduction and close to a polynomial specification found by model selection.

Comparison with the classical supervised peers. The like-for-like comparison for a supervised direction-finder is against other supervised direction-finders, and there

Table 6.3: dCor-SDR + `lin` against each classical supervised peer, both methods at their own best d. Entries are the R^2 difference in dCor-SDR's favour; the last column is the smallest Newey–West(18) p-value over the four maturities. `PCA+lin`, the unsupervised baseline, is shown for reference.

Peer	ΔR^2 (dCor-SDR − peer)				smallest p
	2	3	4	5	
SIR + `lin`	+0.047	+0.032	+0.031	+0.027	0.099
PLS + `lin`	+0.053	+0.043	+0.040	+0.047	0.098
SAVE + `lin`	+0.059	+0.029	+0.050	+0.044	0.058
`PCA+lin`	+0.059	+0.048	+0.050	+0.062	0.058

dCor-SDR is ahead across the board. Table 6.3 reports the difference at each method's own best d: dCor-SDR+`lin` has the higher out-of-sample R^2 against every peer at every maturity, by 0.027 to 0.059, and the Newey–West test reaches the 10% level in three of those twelve comparisons. The peers are also the methods most exposed to the $d = 8$ collapse noted in Section 6.3: SAVE's best at two years *is* the full eight-direction fit, which is why it coincides with `PCA+lin` there. Comparing instead at each fixed $d \leq 3$, where the reductions are genuinely different models, sharpens the same picture: dCor-SDR has the lower MSPE at all 36 combinations of peer, maturity and number of directions, and is significantly better in 8 of 12 against SIR (p down to 0.017), 8 of 12 against SAVE (p down to 0.018) and 1 of 12 against PLS, and is never significantly worse than any of them.

Interpretation. Both reductions set out to remove the need for the polynomial term, and both largely do. They differ in what they use to choose the direction. INPCA reshapes the factor distribution towards a Gaussian without ever consulting the response, and ranks the resulting directions by entropy; dCor-SDR looks for the direction whose dependence with the realised excess return is largest. On these factors the supervised criterion extracts more of the return-relevant structure, which is the outcome one would expect if the informative directions are not also the ones the shape criteria single out. It does so, moreover, with a purely linear model: without the cube, the factors reach 0.184 at two years, and choosing the directions with reference to the response recovers most of what the cube adds. The claim to take from this section is one of possibility rather than of superiority: the nonlinearity in this benchmark can be moved into the reduction step by dependence maximisation alone, with no optimal transport and no model-selection search over polynomial terms.

6.5 Results for the unsupervised method

Moving the nonlinearity into the reduction works less well without supervision. The unsupervised dVar-PP has no access to the returns, and the result is the one that ordering predicts: it does not fall behind PCA, the unsupervised baseline it should be measured against, but it stays well short of the polynomial benchmark and of the supervised direction of the previous section. A second, unplanned finding sits alongside it. Because the factors are built from every month in the sample, the criterion is free to point at whatever carries the most distance variance in the macroeconomic record, and what it points at is interpretable: the direction it selects isolates two months of the test period almost perfectly. That was not what the direction was asked to do, and it is of no use for forecasting, but it says something about the data and about what the index responds to.

Levels. Table 6.4 places dVar-PP+`lin` against every other method in the study, each at its own best d. Three groups emerge. Against the unsupervised baseline `PCA+lin` the distance-variance criterion is a small but consistent improvement, +0.009 to +0.022, and this is the only comparison in the table that is significant at all four maturities ($p = 0.019$ to 0.051). Against the three classical *supervised* peers it is level: the differences run from −0.013 to +0.017 with no clear sign. Three of those twelve comparisons fall below the five per cent level, all of them in dVar-PP's favour, but they are scattered across references and maturities rather than forming a pattern, and the differences behind them are at most 0.017 in R^2. That an unsupervised reduction matches SIR, PLS and SAVE, all of which see the response, is the most favourable thing that can be said for it here. Against the two published models and against dCor-SDR it falls short. The gap to `PCA+poly` is the largest in the table, −0.046 to −0.072, and is significant at three of the four maturities; the gap to dCor-SDR+`lin` is −0.034 to −0.045 and is not significant at any of them. Only against `INPCA+lin` is the comparison a genuine draw, with differences between −0.017 and +0.014 and p-values above 0.6 throughout.[3]

The two months the criterion singles out. The direction that maximises distance variance on the training half is not a copy of a principal component: it loads on the first factor (−0.62), the third (+0.50) and the sixth (−0.55), a combination no variance criterion would form. Figure 6.2 shows what the projection on that direction looks like. The direction shown is the first column of the *joint* solution rather than the sequential one reported in the tables above, because that is the configuration in which the failure was most pronounced and therefore easiest to read. Two months of the test half sit far outside everything the training half contains, at −3.6 and +8.2 training standard deviations from the training mean against a training range of roughly $[-1.7, 3.5]$; the next most extreme test month is at 1.7. Those two months are September and October

[3] `INPCA+lin` is reported here, as everywhere in this chapter, as the hybrid the source paper favours. The strongest of the three INPCA variants implemented, the one reshaping two factors, reaches 0.227 to 0.238 across maturities and is above dVar-PP+`lin` at every maturity; it remains below dCor-SDR+`lin` at every maturity.

Table 6.4: dVar-PP (sequential) + `lin` against every other method, each at its own best d. Entries are the R^2 difference in dVar-PP's favour, with the Newey–West(18) p-value beneath. Only the comparison against the unsupervised baseline `PCA+lin` is significant throughout.

Reference	ΔR^2 (dVar-PP − reference), p beneath			
Maturity (years)	2	3	4	5
`PCA+poly`	−0.071	−0.072	−0.062	−0.046
p	0.049	0.071	0.096	0.168
dCor-SDR + `lin`	−0.045	−0.039	−0.034	−0.040
p	0.185	0.146	0.168	0.230
`INPCA+lin`	−0.017	−0.014	+0.002	+0.014
p	0.617	0.689	0.949	0.708
SIR + `lin`	+0.003	−0.007	−0.003	−0.013
p	0.726	0.330	0.706	0.337
SAVE + `lin`	+0.014	−0.010	+0.017	+0.004
p	0.019	0.615	0.042	0.962
PLS + `lin`	+0.009	+0.003	+0.006	+0.006
p	0.019	0.228	0.194	0.178
`PCA+lin`	+0.014	+0.009	+0.017	+0.022
p	0.019	0.031	0.042	0.051

2001, the months of the September 11 terrorist attacks, which halted air traffic, closed the US equity market for four sessions and produced a simultaneous shock to confidence, employment and output indicators worldwide. The same two months on the first PCA factor are unexceptional: they lie at 1.2 and 1.1 standard deviations, comfortably inside the training range, and they are not among the four most extreme test months on that axis.

The criterion never sees a single excess return and is estimated on 1964–1983 alone, yet the axis it commits to is one along which the largest macroeconomic disturbance of the following twenty years stands out, while the leading principal component absorbs the same shock into its bulk. An index built to find directions along which the projected mass lies far from its centre has found one along which a macroeconomic shock is visible, without being told what to look for.

Three qualifications belong with that. The method detected nothing: the direction is fixed on the training half and frozen, so the two months had no influence on the choice of axis. The behaviour is not a stable property of the criterion but of the factor block, which was extracted over the full 1964–2003 sample and has therefore already seen 2001; on a train-only re-extraction the same criterion selects a direction dominated by the stable leading factor and singles out no test month. And for forecasting the property

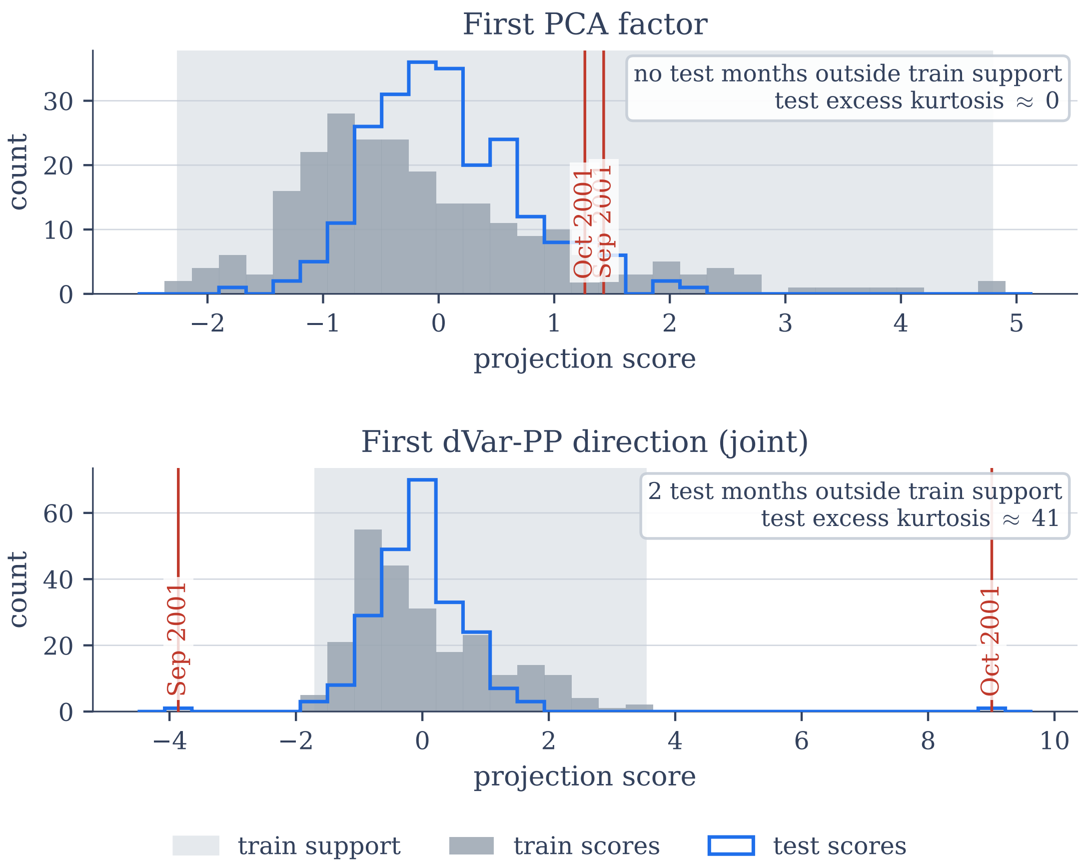

Figure 6.2: Distribution of the projection scores on the first PCA factor (left) and the first joint dVar-PP direction (right): training scores (grey), test scores (blue outline), the range of the training scores (shaded band), and the two months of the September 2001 shock (red). On the dVar-PP axis they are the extremes of the test half and lie outside the training range; on the PCA axis they are ordinary months well inside it. The sign of each axis is arbitrary and is fixed here so that the larger deviation is positive.

is a liability: the projection has an excess kurtosis of about 41 out of sample against approximately 0 in training, which is precisely a direction whose shape does not survive the split.

Interpretation. Chapter 5 evaluated dVar-PP by how well it recovered a known subspace *in sample*, and by that measure it performed well. The present chapter adds a criterion the simulator never tested: whether the *shape* that made a direction interesting persists out of sample. It need not, and here it does not. The objective is doing what it was designed to do, namely finding the projection of the training sample with the largest distance variance, and the September 2001 months are the clearest possible demonstration that it succeeds at that question. It is simply a different question from which projection predicts next year's returns, and the two answers need not coincide. There is also a purely statistical reason to expect the first to transfer less well than the second: the index is driven by the observations furthest from the centre, and which observations those are does not carry across the split, whereas the second moments PCA reads do. The kurtosis of 41 is that instability made visible. The supervised method is not exposed to this failure mode, not because its optimiser or its index is better, but because its objective ties the chosen direction to the quantity being forecast. On this evidence the missing dVar-PP gain is not repairable from within the unsupervised method; what is missing is supervision.

6.6 Summary

On the published bond-return benchmark, scored the way its authors score it, the supervised dCor-SDR directions entered linearly have a higher out-of-sample R^2 than the optimal-transport reduction at every maturity and than every classical supervised peer, and fall short of the hand-tuned polynomial benchmark by at most 0.033 while using two directions, three at four years, instead of eight factors and a cube. It reaches that level with no optimal transport and no model-selection search. Two limits have to be carried with the claim: the factors contain a full-sample look-ahead that lifts the level of every method by about 0.07, and the overlapping annual returns leave the test able to certify only the largest of these differences. The result is therefore about where the nonlinearity can be handled, not about which model is best.

The unsupervised dVar-PP is the counterpart. Entered linearly it improves on `PCA+lin` at every maturity, significantly so, and it holds level with the classical supervised peers and with the optimal-transport reduction, but it does not approach `PCA+poly` and remains behind the supervised direction throughout. What it does deliver is a by-product: the direction it selects makes September and October 2001 stand out as the extremes of the test period, where the leading principal component does not, so the criterion has located a direction along which a macroeconomic shock is visible. That is the objective working as designed, on a question other than the forecasting one, and the shortfall behind it is not repairable by a better optimiser. The projection of the training sample with the largest distance variance need not be the projection that predicts, and a shape set by the most extreme observations transfers across a sample split less reliably than a second-order one.

Both findings, and their reconciliation with the synthetic chapters, are taken up in the comparative discussion of Chapter 7. They agree on one point that the discussion returns to: on this benchmark the objective's connection to the forecast target does more for out-of-sample performance than its flexibility does.

CHAPTER 7

Discussion and Conclusion

The preceding chapters developed and tested the two reduction methods. This chapter states what they contribute, where they fall short, and what remains open.

7.1 Contributions

The two established foundations on which this thesis rests, Riemannian optimisation on matrix manifolds (Absil et al., 2008) and sequential projection pursuit (Huber, 1985), are not new here. What the present work contributes is their combination with objectives built from the distance correlation and the distance variance, the closed-form derivatives those objectives require, the search strategies suited to them, and a systematic account of what the resulting reductions recover and where they fail.

Closed-form Riemannian gradients. The distance-covariance literature on sufficient dimension reduction avoids the derivative of the nonsmooth objective. Sheng and Yin (2013) and Sheng and Yin (2016) employ sequential quadratic programming; Wu and Chen (2021) use a difference-of-convex reformulation. The present work takes the derivative where it exists, which is everywhere outside a set of probability zero, and verifies it against central differences to a relative error below 10^{-9} (Table 3.1). For both indices the gradient is derived on the sphere and on the Stiefel manifold. Two derivative-free solvers confirm the result independently: COBYLA matches or exceeds the attained objective in all 36 runs and returns the same direction to within 2° in eleven of the twelve configurations, at a median cost of 13 times the gradient optimiser's on the unsupervised index and 43 times on the supervised one (Table 3.2). The analytical gradient enables a standard adaptive Riemannian optimiser to be applied directly, at the cost of one index evaluation per step, and it reduces the per-step complexity from $O(n^2p)$ for central differences to $O(n^2 + np)$.

A systematic study of distance variance as a projection index. Distance variance is well established as a dependence measure and has been little examined as a dimension-reduction criterion. Since it is scale-equivariant, an index built on it reads the scale of a projection as well as its shape, and the two contributions have to be separated before any recovery result can be read (Section 5.2). Where the construction separates them, the index recovers an axis distinguished by nothing but the shape of its projection, and there PCA has no eigenvalue gap to work with (median 1.7° against 25.8°; Section 5.4.2). Nevertheless FastICA reads the same shape and recovers the axis at 2.0°, so that the advantage established there is one over the variance-based criteria and not over the classical shape-based ones. Under outlier contamination the index deteriorates more slowly than the criteria built on squared deviations, which opens a window at around 5% in which its recovery is the better one (mean squared sine 0.268 against 0.830; Table 5.6), while from 10% onwards no method in the sweep recovers the subspace. In the non-Gaussian factor model, where the signal direction is also the maximum-variance direction and PCA is thus optimal by construction, dVar-PP stays behind PCA but recovers the same subspace once the signal is present.

Real-data validation on a published benchmark. On the bond-return benchmark of Ludvigson and Ng (2009), the supervised reduction dCor-SDR followed by an ordinary linear regression attains a higher out-of-sample R^2 than the optimal-transport reduction of Gunsilius and Schennach (2023) at every maturity and than every classical supervised peer (SIR, SAVE, PLS), and comes within 0.033 of the published polynomial benchmark at its worst maturity, with no transport computation, no polynomial term and no model-selection search (Section 6.4). The unsupervised dVar-PP improves on `PCA+lin` at every maturity, significantly so, and matches the classical supervised peers without ever seeing a return, but it does not approach `PCA+poly` and stays behind the supervised direction throughout (Section 6.5). On this evidence the objective's connection to the forecast target does more for out-of-sample performance than the flexibility of the criterion does.

The robustness trade-off of the biloop transform. The bounded biloop transform of Leyder et al. (2026) is examined here as a projection index. On clean data it is much worse than the plain index (mean squared sine 0.763 against 0.046), because its redescending saturation flattens the objective surface, while from about 10% contamination onwards it is the less damaged of two failed recoveries (Section 5.4.5). Therefore nothing in the sweep recommends it as a default. As far as we are aware, this cost on clean data has not been quantified previously in a dimension-reduction setting.

7.2 Limitations

Two principal limitations run across the thesis. Both were encountered empirically and reported in the chapter in which they arose; this section adds that the first belongs to the objective and the second to the procedure built on top of it.

The marginal-sum objective. When more than one direction is sought, the objective scores each direction on its own and adds up the scores. It never inspects whether the recovered projections are jointly structured. Under the heteroscedastic coupling of Section 5.4.6, where the latent factors are uncorrelated but share higher-order dependence, recovery worsens from 0.088 to 0.105 in mean squared sine while PCA barely moves, from 0.072 to 0.065, because the coupling leaves the second moments it reads essentially untouched (Table 5.7). That is insensitivity to this particular dependence rather than a guarantee, since PCA never undertook to return independent directions. Optimising all directions jointly on the Stiefel manifold returns the same degraded value, which places the ceiling in the objective rather than in the search. The natural candidate remedy is a genuinely multivariate distance measure in the spirit of Matteson and Tsay (2017); one exploratory run with such an objective did not help at $n = 200$ (Section 5.4.6), and whether the difficulty resides in the marginal form or in the estimation of the multivariate index at moderate sample sizes remains open.

Sequential deflation. None of the three available deflation strategies for extracting a second direction is sound. Y-residual deflation subtracts a correlation from the response while the objective measures dependence; X-deflation makes the weight vectors orthogonal but not the projections they induce, so that with correlated predictors two perpendicular weights can still produce scores that are correlated; and Σ-orthogonal deflation reaches uncorrelated projections and stops there, short of the independence the objective asks for (Section 4.4.4). Σ-orthogonal deflation is the recommended default because its failure is one of degree rather than of kind, but recommending it is choosing the least flawed of three flawed options, and every multi-direction result in the thesis carries that flaw. In the unsupervised case, this same limitation appears as a degradation when latent factors share higher-order dependence (Section 5.4.6).

Further qualifications. Five narrower qualifications accompany the results. The joint supervised optimiser projects the gradient onto the horizontal space rather than the full tangent space, so it searches over subspaces and holds the frame within the recovered subspace at its initialisation, which the marginal-sum objective does not reward (Section 3.4); the subspace accuracies are unaffected, the attained objective values are a lower bound. The $O(n^2)$ gradient imposes a ceiling near $n = 2000$ (Section 3.3), so nothing here speaks to large samples. The number of directions k is treated as known throughout the simulation chapters, which suits a method study and not an application. The biloop index has no analytical gradient and falls back on finite differences, at a median of 54 times the cost of one plain fit at $p = 30$ (Section 3.6). And the real-data study works with a factor block extracted over the full sample, so the comparison inherits a look-ahead that lifts the level of every method by about 0.07; it measures relative rankings under a shared protocol, not absolute predictive power (Section 6.6).

7.3 Future work

Each item follows from one of the limitations stated above, ordered by how much it would change.

Nonlinear deflation. The deflation obstacle is the one limitation with a principled fix already in view. Replacing the rank-one linear residual by a kernel-ridge one,

$$\tilde{e} = Y - K(X\hat{\beta}_1)\big(K(X\hat{\beta}_1) + \lambda I\big)^{-1} Y, \tag{7.1}$$

for a universal kernel K, removes from the response the projection onto the whole space of smooth functions of the found direction, which is the condition the objective requires (Section 4.4.4). This is the machinery of kernel dimension reduction (Fukumizu et al., 2009), and it composes coherently with the objective because distance covariance and the kernel dependence statistics are equivalent (Sejdinovic et al., 2013). It costs an $O(n^3)$ solve and two hyperparameters, and the open question is whether it recovers second directions where both linear strategies fail.

A scale-free unsupervised index. The distance variance is homogeneous of degree one, so the raw objective of Chapter 5 reads the scale of a projection as well as its shape (Section 5.2). Dividing it by the standard deviation of the same projection gives a scale-invariant ratio, which is a generalised kurtosis in the sense in which invariant coordinate selection compares two scatter functionals (Tyler et al., 2009; Radojičić et al., 2020), and measuring the deviation of that ratio from its Gaussian reference value would in addition search both sides of the ordering with one maximisation instead of the two searches of Section 5.4.3. Neither variant is evaluated here.

A multivariate index for the unsupervised case. Scoring the whole projection with one multivariate distance-variance index instead of summing marginal ones, in the spirit of Matteson and Tsay (2017), would in principle address the marginal-sum limitation. One exploratory run at $n = 200$ did not remove the degradation, and a single replicate settles nothing, so at what sample size, if any, the multivariate form overtakes the marginal one is an open empirical question.

Choosing the number of directions. A permutation or parametric-null stopping rule, testing whether the next direction's attained index exceeds what independent data would produce, would remove the assumption that k is known, at the cost of one resampling loop per direction.

Scaling past the dense gradient. Subsampled or randomised gradient approximations, or a low-rank factorisation of the distance matrix, would move the $n \approx 2000$ ceiling. They were avoided here so that the reported behaviour is that of the exact objective, which gives them a baseline to be measured against.

A leak-free and wider real-data study. Chapter 6 reports all comparisons on the published factor block, so that the benchmark is the one its authors defined. Running the entire comparison on train-only factors, widening the panel, and splitting the sample by volatility regime would settle both the ranking under a clean protocol and the one claim the present study could not test: that the robustness advantage observed under synthetic contamination also appears when real data are heavy-tailed.

The methods developed in this thesis show that it is practicable to maximise distance correlation and distance variance over orthonormal frames by means of analytical Riemannian gradients, and the simulation studies delineate the conditions under which the resulting reductions recover structure that second-order criteria miss. Where the methods remain limited, the limitation lies in the objective (the marginal sum and the estimator variance) rather than in the search, which is what a gradient-based optimiser can be expected to settle. Whether the remedies outlined above close the remaining gaps is the natural next step.

Overview of Generative AI Tools Used

The generative AI tool Claude (Anthropic) was used with the models Sonnet and Opus as an aid in the preparation of this thesis, in the following ways:

- **Language polishing:** Claude was used to refine the wording of drafted text, tightening formulations and shortening overly verbose passages. The entire content originated from the author; every LLM-suggested rephrasing was critically scrutinized for accuracy and fidelity to the intended meaning before acceptance, only a subset was incorporated, and no text was copy-pasted directly from model output.
- **Code generation and debugging:** Claude was used as an aid during implementation and debugging. The design of all algorithms, the structure of the code, and its overall organization are the author's own work; the AI was used only to assist with specific, delimited sub-tasks. All AI-assisted code (with the exception of plotting code) was read and tested line by line before use, and the author takes full responsibility for its correctness.

None of the results reported in this thesis are AI-generated outputs; all results were produced by the author's own implementation and are reproducible.

Reflection. Working with this tool made clear that it is most reliably used where its output can be checked quickly and unambiguously. For code in particular, subtle design or logic errors are easy to overlook, so its suggestions could not be trusted without close, deliberate review, and were treated accordingly.

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